Lowcountry Shrimp Dataset Deep Dive

September 11, 2025

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Summary

About the Project

Funded by the NERRS Science Collaborative - see the project page here.

Collaborators

Project PIs were Robert Dunn[1,2] and Joshua Stone[1]. The work group that guided the compilation of this document included Matt Kimball[1], Michael Kendrick[3], Liam Batchelder[1], Graham Wagner[1], and Maggie[1]. The R code developed to work with the data and produce this document was written by Kim Cressman[4].

[1] University of South Carolina, Baruch Marine Lab
[2] Ecological Dynamics, LLC
[3] South Carolina Department of Natural Resources
[4] Catbird Stats, LLC

Part I

Definitions and Structure

1 Shrimp Year Abundance Index Explanation

The concept of **shrimp year** is somewhat complicated - a detailed explanation is below. If you prefer, jump straight to Table 1.1 and/or Figure 1.1.

1.1 Shrimp Year

When examining long-term datasets, it is convenient, if not necessary, to perform some aggregation of data to means or totals over some representative length of time (e.g. monthly mean temperature, annual precipitation total, monthly or annual totals of money spent on [anything related to budgets]).

In this project, we want to be able to link different life stages of shrimp to each other, and ask questions like "if postlarvae did well this year, did juveniles also do well? does that hold all the way through adults, and fisheries landings?"

Animal life cycles, incoveniently for us, do not necessarily align with calendar years. It may be (and for brown shrimp generally is the case) that young are spawned early in a calendar year and are adults late in that calendar year. However, white shrimp (and many other species) may be spawned in the middle or late part of a calendar year, and not be adults until the next calendar year. So, in order to relate those adults to the earlier life stages, we cannot use calendar year to represent the cohort of shrimp.

We use **shrimp year** to represent a single cohort of shrimp, and assign the calendar year in which that group of shrimp will live most of their lives. For brown shrimp, the shrimp year is the calendar year in which the individuals will show up in the fishery. For white shrimp, it is the year in which they go through most life stages; though adults will bleed into (and spawn in) the following calendar year.

Each survey in this project, based on its location of sampling, is expected to primarily catch a single life stage of each shrimp. Based on the month of sampling, the targeted life stage, and the shrimp species, we assigned shrimp year to each sampling event based on the following:

1.1.1 Table based on sample date and life stage

need to give it some formatting love, but this aligns with assignment

Table 1.1: Shrimp year assignments, based on month and calendar year of sampling, by species and life stage. 'year' represents the calendar year of sample date. 'year - 1' represents the calendar year prior to the sample date (e.g. 2004 when sample date is 2005). 'year + 1' represents the calendar year after the sample date (e.g. 2006 when sample date is 2005).

	Brown Shrimp				White Shrimp			
Month	Postlarvae	Juvenile	Subadult	Adult	Postlarvae	Juvenile	Subadult	Adult
1 - Jan	year	year	year	year	year	year	year - 1	year - 1
2 - Feb	year	year	year	year	year	year	year - 1	year - 1
3 - Mar	year	year	year	year	year	year	year - 1	year - 1
4 - Apr	year	year	year	year	year	year	year - 1	year - 1
5 - May	year	year	year	year	year	year	year - 1	year - 1
6 - Jun	removed	year	year	year	year	year	year - 1	year - 1
7 - Jul	removed	year	year	year	year	year	year	year
8 - Aug	removed	year	year	year	year	year	year	year
9 - Sep	removed	year + 1	year + 1	year	year	year	year	year
10 - Oct	removed	year + 1	year + 1	year	year	year	year	year
11 - Nov	removed	year + 1	year + 1	year	year	year	year	year
12 - Dec	removed	year + 1	year + 1	year	year	year	year	year

1.1.2 Environmental data

Winter temperatures and cold spells can impact shrimp growth and/or survival, and understanding these dynamics is important to shrimp fishery managers. Salinity during early life stages may also impact shrimp, and these effects extend to later stages of the life cycle.

Again, we need to aggregate data. **Winter** here is defined as December through February. For brown shrimp, the shrimp year impacted by winter temperatures is the calendar year containing that January and February. For white shrimp, we have assigned each winter metric separately to two life stages, as they occur in different years: postlarval winters and adult winters. Subadults and adults will be impacted by winter temperatures occurring a year after the winter temperatures that affect the postlarval and juvenile stages. See Figure 1.1. **Nursery period** is defined separately for each species, based on Batchelder et al. 2024*: April through July for brown shrimp, and June through October for white shrimp.

*Batchelder, L., Stone, J., Kimball, M., Pfirrmann, B., & Dunn, R. (2024). Phenology of penaeid shrimp nursery habitat use: Trends and environmental drivers over four decades. Marine Ecology Progress Series, 751, 79–95. https://doi.org/10.3354/meps14741

1.2 Graphic of abundances, temperatures, and assigned shrimp years

this will be what Kim showed the group at some point in the past, with 2 or 3 calendar years worth of data

need to actually process the data though and use quarto's `embed` functionality to show it here

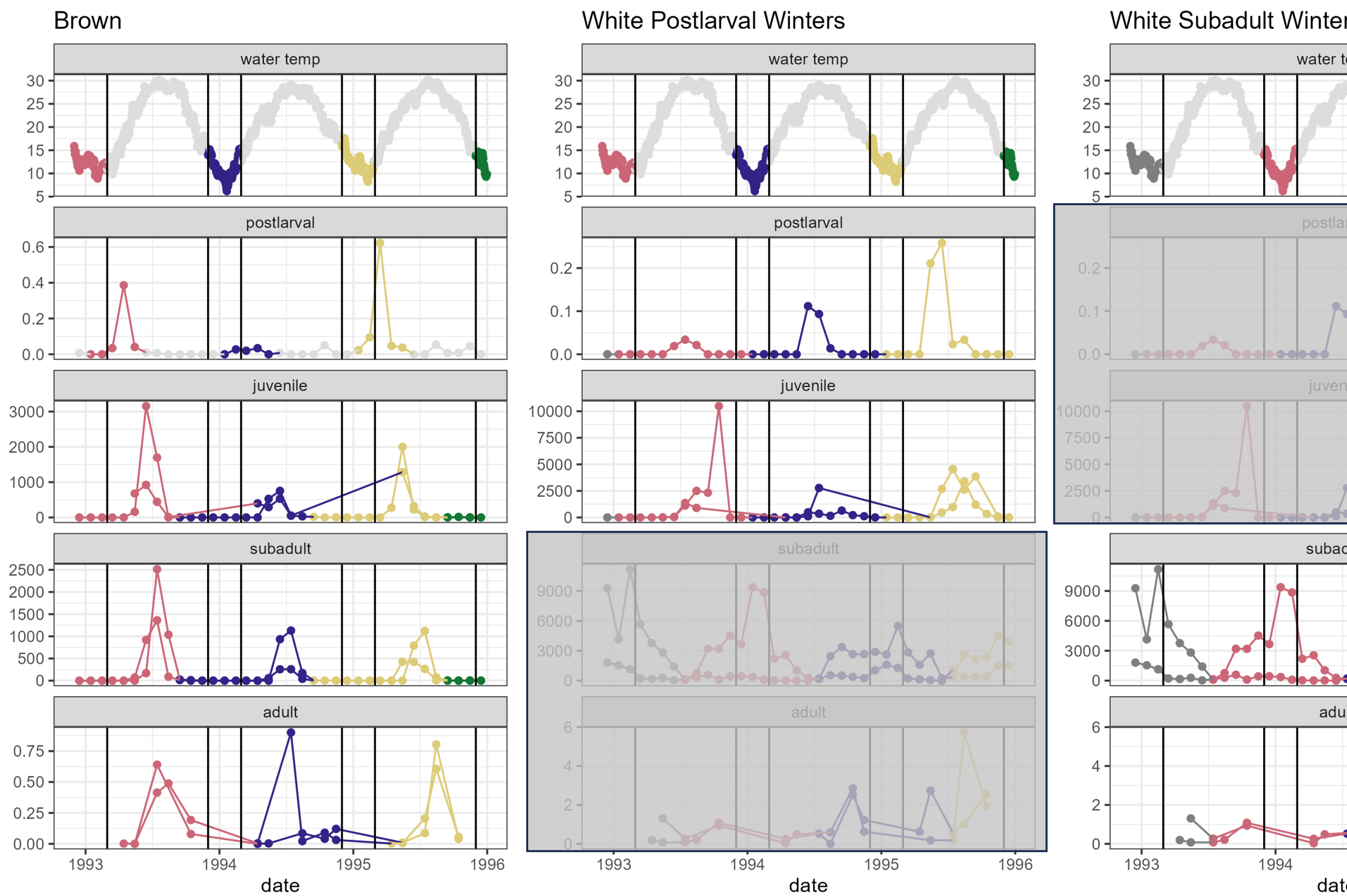


Figure 1.1: Temperature and abundance data, colored by shrimp year, with irrelevant life stages grayed out.

group may want to insert conceptual diagram too when that is completed

1.3 Abundance Index

This project brought together a number of surveys by different organizations, with different methods, and different ways of quantifying shrimp abundance. In order to make these numbers comparable, while accounting for possible variation in the number of sampling stations and sample dates over time, we have adjusted abundance metrics to an index using the following steps. More details for individual datasets will be provided in that life stage's chapter in the "Summarizing to Abundance Index" portion of this book.

For each survey:

1. Assign shrimp year to each sample, based on month and species.

2. If a sampling event (site/station on a date) had replicates, average them.

3. Average data across all sites/stations on a sample day.

4. Average data across all sampling dates within each month = average abundance for the month.

5. Add the monthly averages, to get a total abundance for each shrimp year. This absorbs variability in timing for a year.

6. From total abundance per shrimp year, derive an index:

    a. Take the square root of total abundance (deals some with extreme high numbers).

    b. Mean-center the result from (a) across all shrimp years for the dataset.

    c. Divide by the standard deviation across all shrimp years for the datset.

talk about multiple datasets and how they were combined

1.4 About

Input `.qmd` file for this section was: **shrimp_year_explanations.qmd** and can be found in the main directory.

2 Data Processing and Directory Structure

This chapter primarily exists to guide other members of the working group in modifying, updating, or extracting information from this book.

2.1 Directory organization

- `_quarto.yml` is the main 'table of contents' of all .qmd files in order. Each chapter also has an 'About' section at the bottom identifying the exact file used to generate it, as well as details about the input and output datasets for the chapter.

- `_book/` is the subfolder with output files. Open any .html in here and you'll have access to the full structure. The pdf output is also in this directory.

2.2 Data, data, data

- `data/raw` = time series, unmodified

- `data/intermediate` = time series with some simplifications for reporting and summarizing; generally outputs from the "Data Exploration" chapters of this book

- `data/processed` - as of this writing, I haven't processed data yet. This folder will contain data that are aggregated to different levels:

- month or season

- shrimp year (based on month or season); these files will also contain a column for the abundance index

The older version of the processed data folder has the following subfolders:

- *01 Survey level* - averaged to month and then shrimp year, per survey. ALL files have the same columns: survey, species, life_stage, shrimp_year, abundance_measure, abundance, sqrt_abundance

- *02 Life Stage level* - averaged survey results to shrimp year x life stage, if relevant Again all files have same columns. Survey, species, life_stage, shrimp_year, abundance, abundance_measure (if average of two surveys, will be told here), abundance, sqrt_abundance

- *03 Combined abundance* - combined all abundance files and calculated abundance index

- *04 Environmental Survey level*

- *05 Combined environmental*

2.3 About

Input `.qmd` file for this section was: **directory_structure.qmd** and can be found in the main directory.

Part II

Environmental Data

3 Environmental Data

3.1 Winter Temperatures at Charleston Harbor

winter = dec, jan, feb.

for brown shrimp, this is all one 'shrimp year'. white shrimp go through 2 winters: postlarval winter and subadult winter. subadult winter is the one managers focus on but I will calculate metrics for both.

See Chapter 1 for more on how these fit into the 'shrimp year' definitions used in this document.

for shrimp year 1993, postlarval winter is calendar year 1993 and subadult winter is calendar year 1994

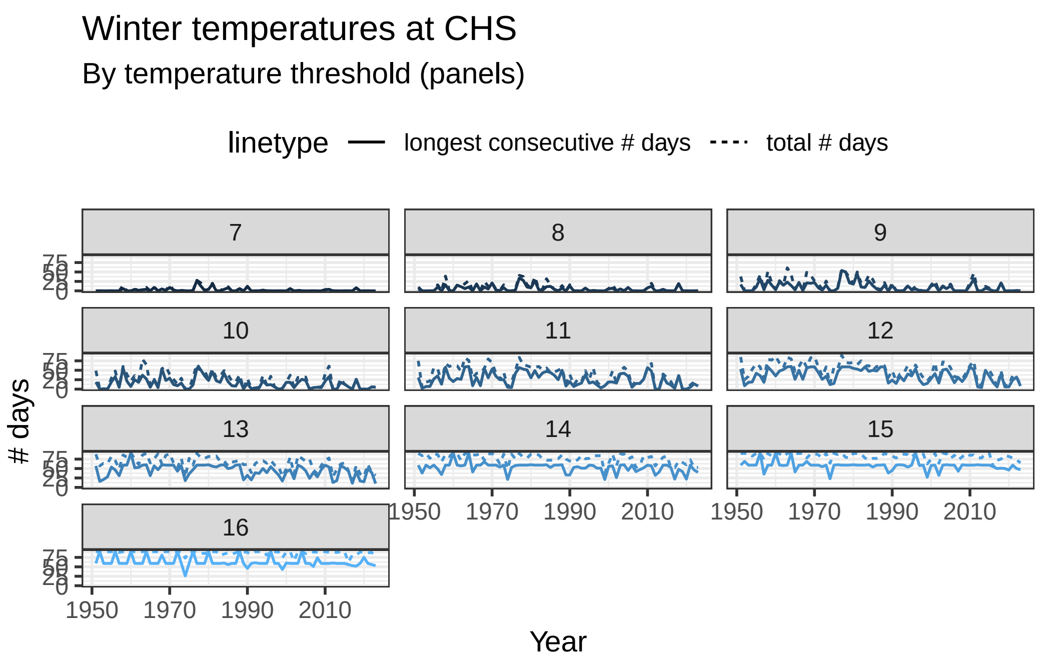
want min winter temp, mean winter temp, and # days temp < 11C

nursery periods, per Batchelder et al. 2024, are Apr-Jul for brown shrimp and Jun-Oct for white shrimp

nursery period salinity was important to phenology in that pub so I'm now pulling out some nursery season info, 2/26/25 based on our definitions, for nursery periods, shrimp year is current year for both species

3.1.1 Consecutive days below thresholds

There is some debate over which temperature threshold is most important. Here, we compare to each temperature from 7-16C.



3.1.2 Temperature Distribution

3.1.3 Apply to different species

3.2 Salinity

Nursery period salinity was important to phenology in Batchelder et al. 2024. We define nursery periods as per that manuscript: Apr-Jul for brown shrimp and Jun-Oct for white shrimp. I use both 'calendar_year' and 'shrimp_year' here because even though they are the same by our definitions (see Table 1.1), it can be a little difficult to keep track of - so I'm being as explicit as possible.

Here we pull data from the stations CH07, CH09, CH10, and CH18.

Salinity at each station was averaged to the day and month. There was at least one anomalously high value, so a line of code removed any values above 50.

3.2.1 Brown Shrimp Nursery Period

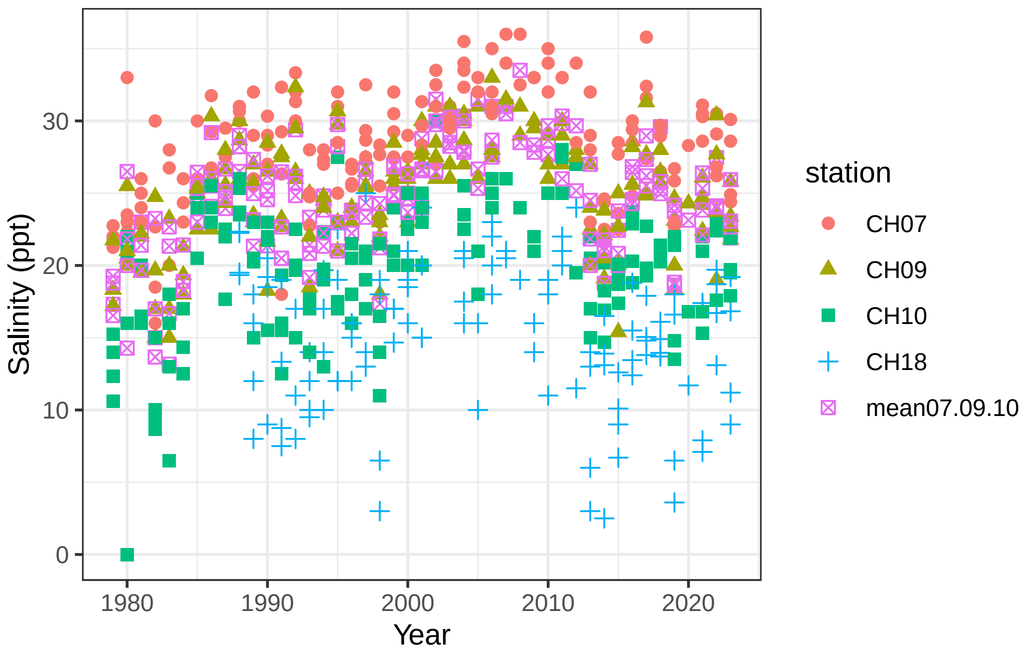


Figure 3.1: All brown shrimp nursery period salinity values by shrimp year

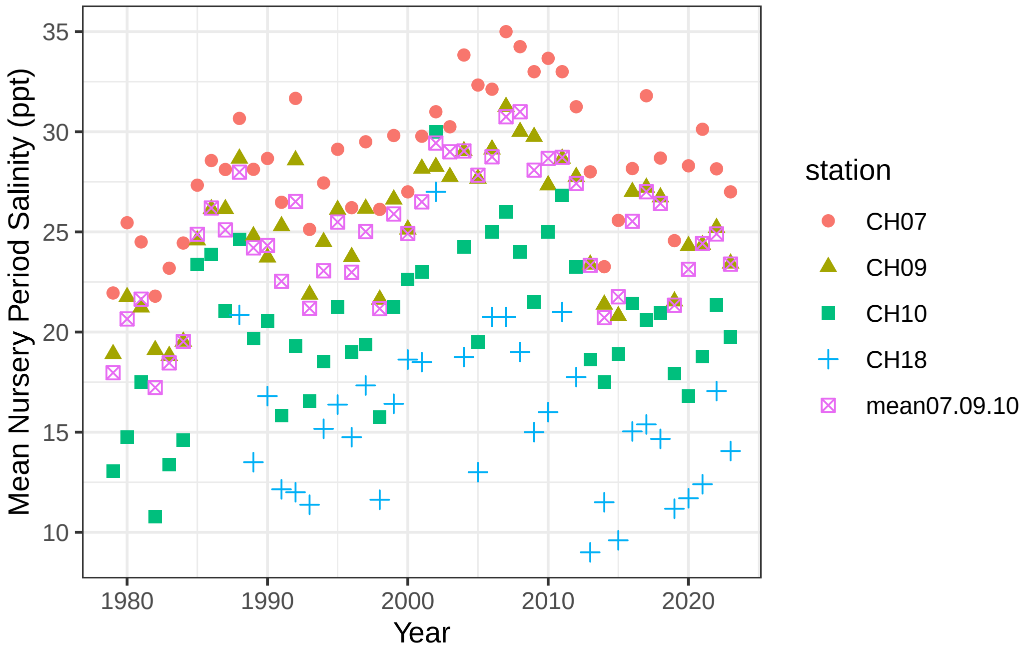


Figure 3.2: Mean brown shrimp nursery period salinity values by shrimp year

Correlations between stations

Mean winter salinity for each year - we can see that mean nursery period salinity is generally well correlated between stations.

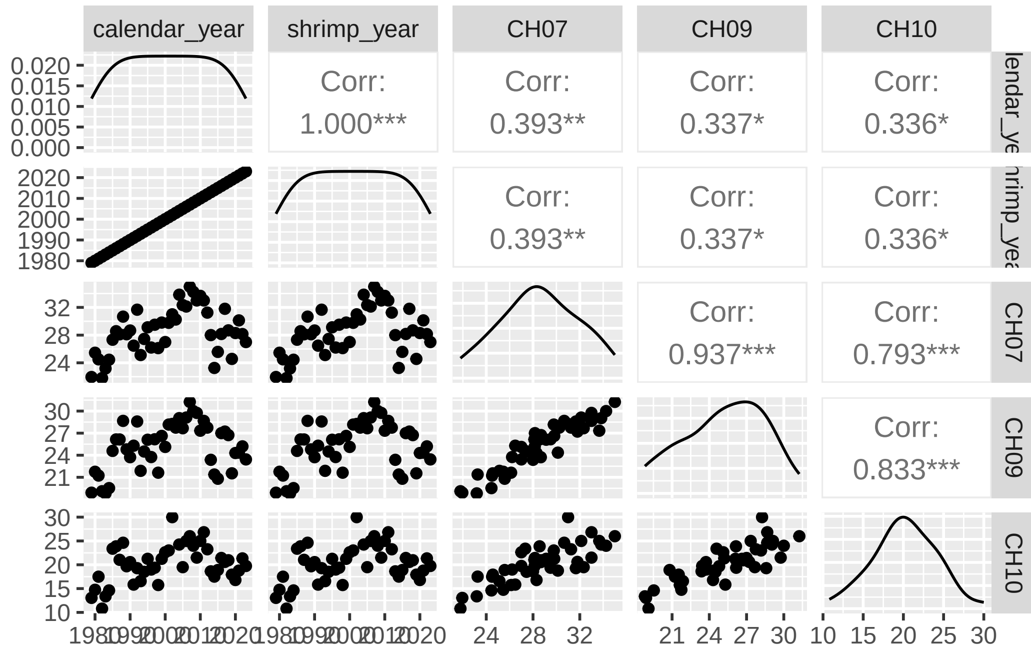


Figure 3.3: Pairs plot of brown shrimp nursery period salinities (mean for each year) at the different stations

3.2.2 White Shrimp Nursery Period

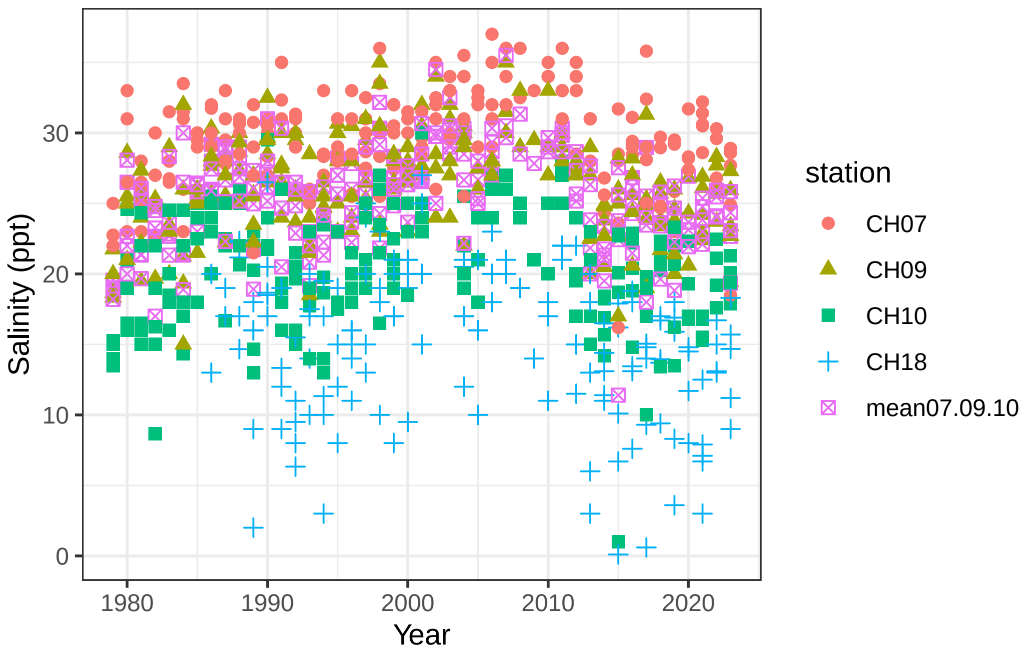


Figure 3.4: All white shrimp nursery period salinity values by shrimp year

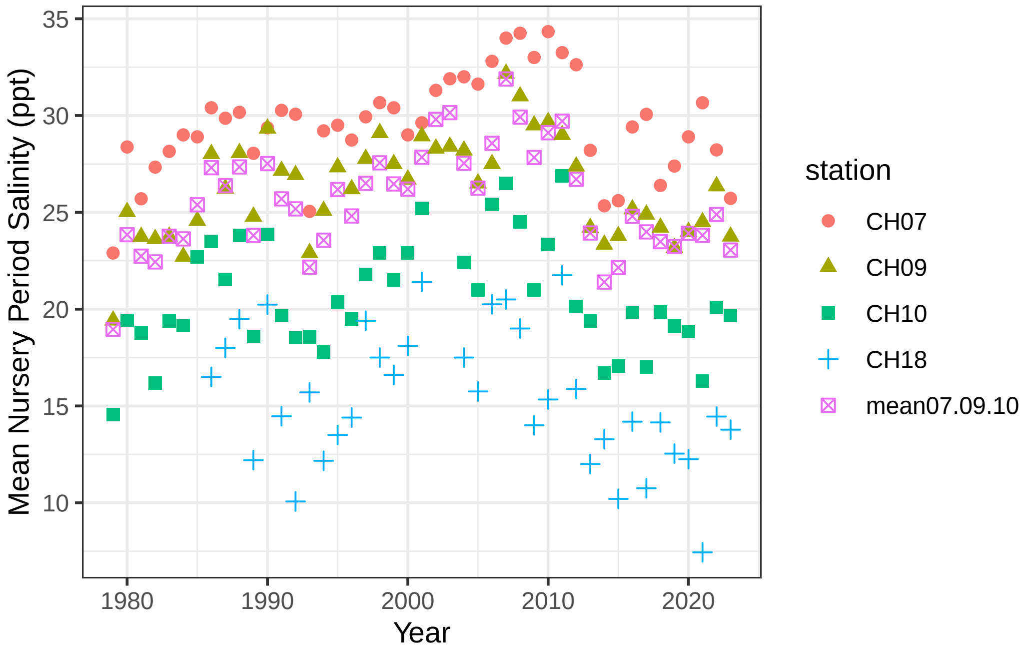


Figure 3.5: Mean white shrimp nursery period salinity values by shrimp year

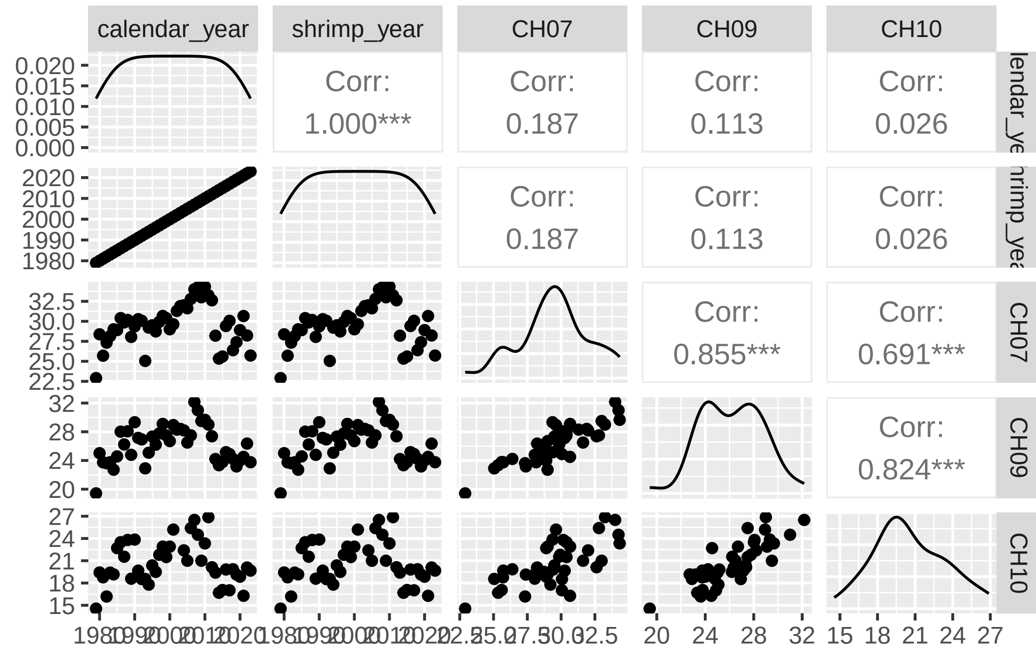


Figure 3.6: Pairs plot of white shrimp nursery period salinities (mean for each year) at the different stations

3.3 Combine all environmental data

Write out the intermediate data frames - yearly summaries.

3.4 About

Input `.qmd` file for this section was: **process_environmental.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/environmental`:

- `SCDNR_CHS_TempData_ 1950_2024.xlsx` for temperature
- `SCDNR_EstuarineTrawlShrimpSize.csv` for salinity from trawls

Modifications to data:

Summary calculations, described below for each output data frame.

Generated data file(s):

Data frames were modified as described above, and written to `data/intermediate/environmental_dfs.RData` (the single `.RData` file contains multiple data frames):

- `tmp_thresh` - counts of days below temperature thresholds for each year.
- `calendar year` represents the year containing January and February. Because we defined winter as December-February, calendar_year 2004 summarizes the time period December 2003 - February 2004.

- `threshold` column contains each integer from 7-16, representing degrees C.
- `totalDays` is the total number of days in that time period below the threshold in the `threshold` column.

- `totalSpells` represents how many times there were at least 7 consecutive days below the given threshold.

- `longestSpell` is the longest stretch of consecutive days (with a minimum of at least 7) within the time period.

- `tmp_distn` follows the same `calendar_year` scheme as `tmp_thresh`. In this data frame, there is only one column for each 'calendar_year'. Columns represent which 'shrimp year' the calendar year would fall into for each species/life stage, followed by summary statistics for water temperature over Dec-Feb: min, median, max, mean, sd, iqr, etc.

- `sal_means` contains one row for each 'shrimp year' (matches calendar year) and species. Columns are average nursery period salinity from each of 4 stations, and one column containing the mean from 3 of the four stations (one station was less like the others). Nursery periods were defined as in Batchelder et al. 2024; see above for more.

The data files to be used further in the workflow combined temperature and salinity information for each year x species, and were separated by species. These are both saved in `data/processed/environmental`. Each has a single row per shrimp year. For white shrimp, there are columns for each of 'postlarval winter' and 'subadult winter', along with columns identifying which calendar year was used for the calculations, because the white shrimp life cycle is much trickier than brown:

- `brn_environmental.csv`

- `wht_environmental.csv`

Other notes about data:

Part III

Shrimp Data Exploration

Here, we go through each file to see the distribution of data and to make some graphics illustrating what is present in each dataset.

4 Postlarvae

4.1 Dataset

The only dataset representing postlarvae is from North Inlet-Winyah Bay.

4.2 Graphics - Size Distributions

In these plots, there is a panel for each shrimp species. Size is represented on the y-axis, and a cloud of points called a “beeswarm” has representation for every data point. Each point is colored by the numeric month in which the individual was caught, allowing us to see differences in size at different parts of the year and life-cycle.

There are three different measures of size in this dataset: Rostrum length, Telson length, and Uropod length. A graph is below for each.

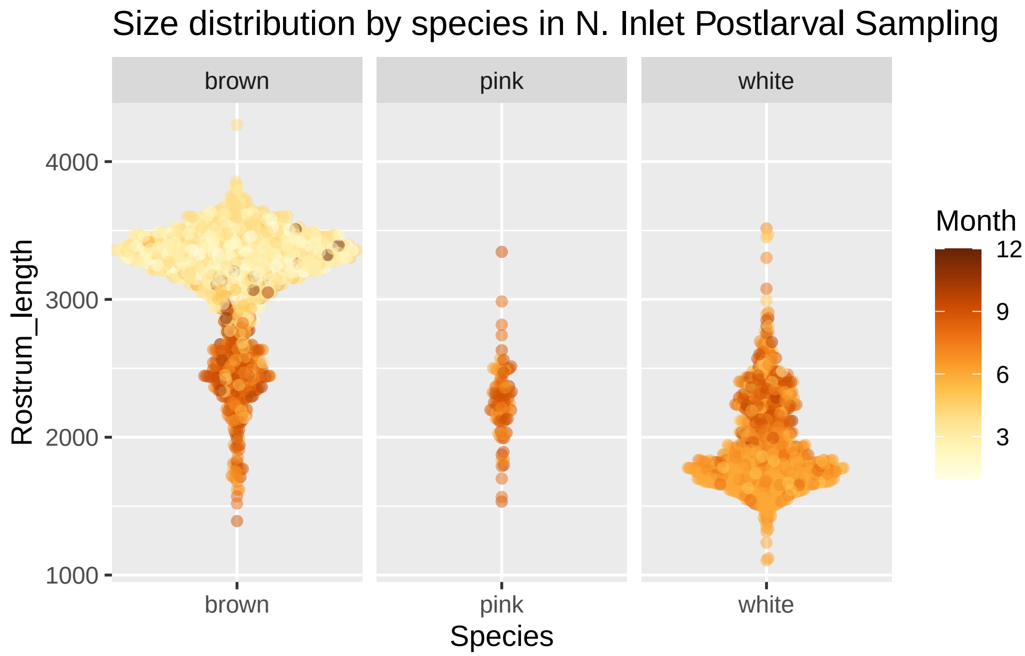


Figure 4.1: Beeswarm plot of rostrum length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

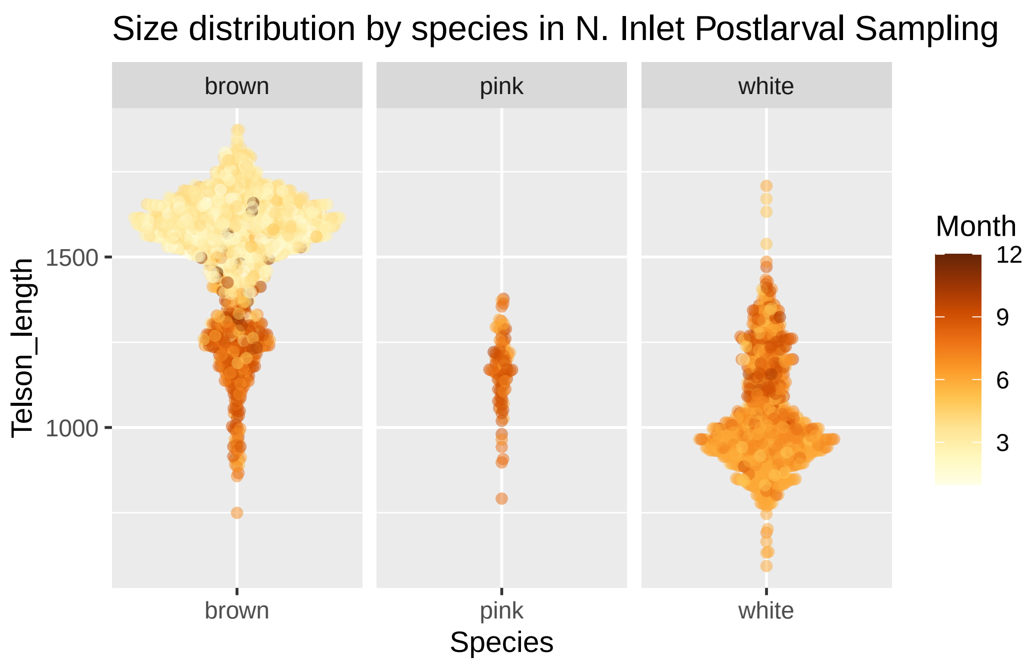


Figure 4.2: Beeswarm plot of telson length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

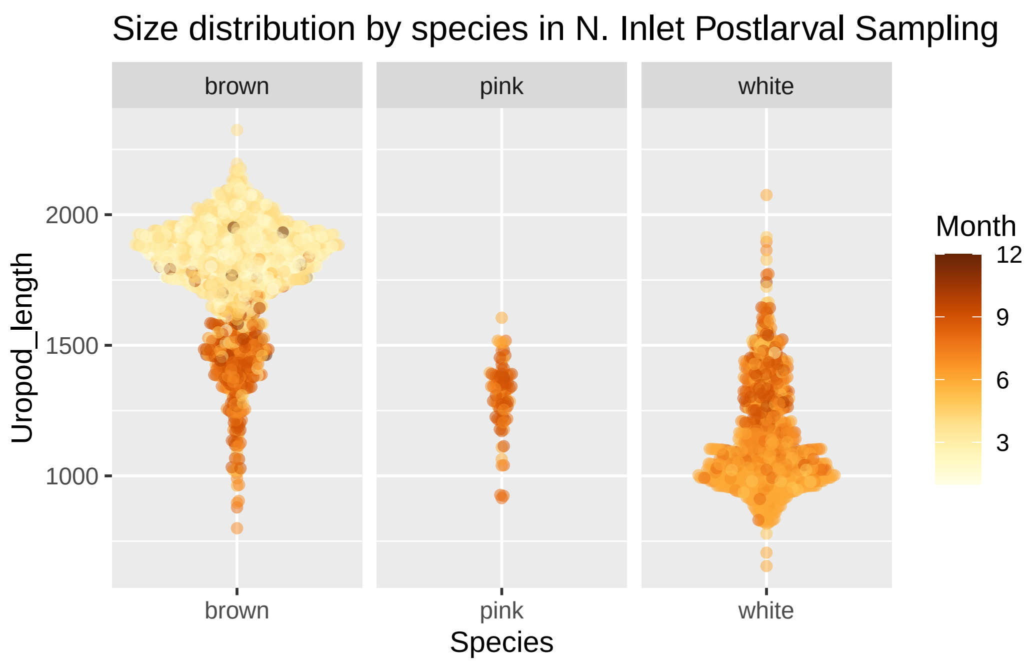


Figure 4.3: Beeswarm plot of uropod length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

With brown shrimp, looks like we've got a couple of size classes in here - bigger shrimp earlier in the year, and smaller ones that are mostly later months.

The plot below shows the entire time series, with date (as year-month) along the x-axis and uropod length on the y-axis. Points are again colored by month.

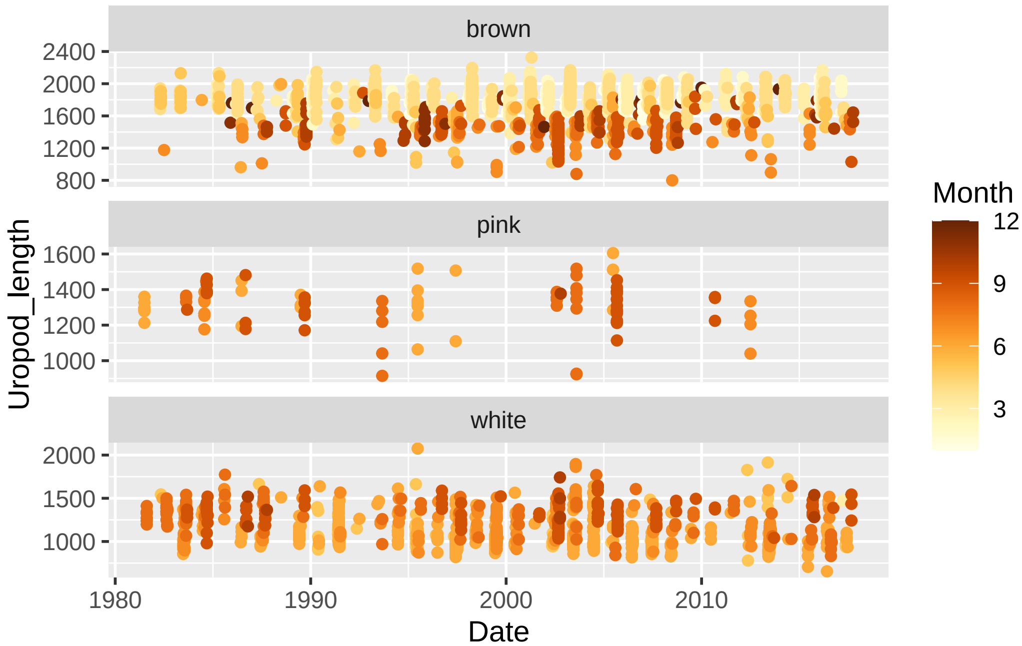


Figure 4.4: Time series plot of uropod length in postlarval shrimp. Points represent measurements of individual shrimp, and are colored by the month in which they were captured.

4.3 Graphics - Abundance and Size Boxplots

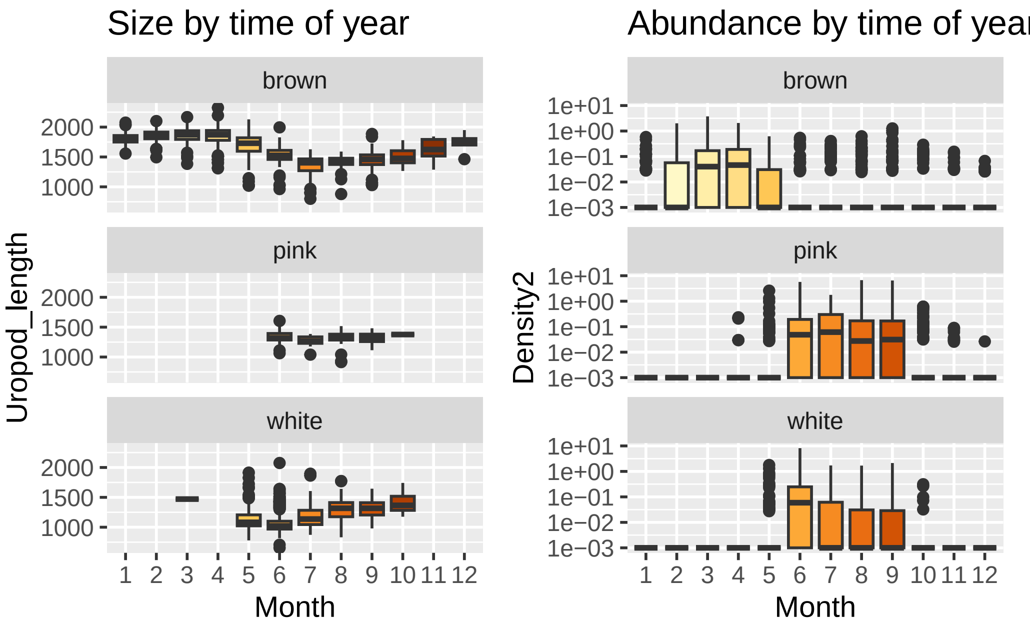


Figure 4.5: Boxplots representing shrimp size and abundance Note the log10-scaled y-axes.

So even though we're seeing that pattern in brown shrimp of being so much bigger early in the year and smaller later, they're just really not all that common later in the year.

4.4 Tabular summary of abundance

Table 4.1: Abundance data frame summary

Table 4.1: Data summary

Name	postlarv_abund
Number of rows	1830
Number of columns	21
Column type frequency:	
character	6
Date	1

numeric	14
Group variables	None

Variable type: character

Table 4.2: Abundance data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
replicate	0	1	1	10	0	13	0
date	0	1	0	10	6	915	0
season	0	1	4	6	0	4	0
ol_sal	0	1	0	7	2	181	0
ol_temp	0	1	0	7	2	241	0
notes	0	1	0	29	1746	22	0

Variable type: Date

Table 4.3: Abundance data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
date2	6	1	1981-01-20	2017-12-30	1999-06-17	914

Variable type: numeric

Table 4.4: Abundance data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
sample	0	1	458.00	264.21	1.0	229.25	458.00	686.75	915.00
year	0	1	1998.67	12.07	1900.0	1990.00	1999.00	2008.00	2017.00
month	0	1	6.48	3.45	1.0	3.00	6.00	9.00	12.00
day	0	1	15.58	8.83	0.0	8.00	16.00	23.00	31.00
dayofyear	0	1	181.63	105.57	0.0	90.25	181.00	271.75	365.00
week	0	1	26.96	15.08	1.0	14.00	27.00	40.00	53.00
bb_surface_salinity	0	1	31.59	5.08	5.9	30.83	33.20	34.50	38.60
bb_surface_temp	0	1	19.28	7.15	3.3	13.20	19.50	26.37	33.00
total_153_zoop_ind_m.3	0	1	9946.45	10692.28	0.0	3440.03	6531.50	12306.58	119401.90
vol.filt.0.13	7	1	32.36	4.76	0.0	29.36	32.47	35.74	45.68
total_ppl_density	9	1	0.20	0.63	0.0	0.00	0.00	0.13	8.40

Table 4.4: Abundance data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
brown_density	9	1	0.05	0.19	0.0	0.00	0.00	0.03	3.77
white_density	9	1	0.05	0.29	0.0	0.00	0.00	0.00	8.19
pink_density	9	1	0.09	0.42	0.0	0.00	0.00	0.00	6.71

4.5 Tabular summaries of size

4.5.1 Brown Shrimp

Table 4.5: Brown shrimp size summary

Table 4.5: Data summary

Name	postlarv_size_brown
Number of rows	1952
Number of columns	16
Column type frequency:	
character	5
numeric	10
POSIXct	1
Group variables	None

Variable type: character

Table 4.6: Brown shrimp size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Shrimp_ID	0	1.00	6	8	0	1949	0
TowID	0	1.00	3	4	0	414	0
Replicate	0	1.00	1	1	0	2	0
Species	0	1.00	5	5	0	1	0
Notes	1830	0.06	1	34	0	33	0

Variable type: numeric

Table 4.7: Brown shrimp size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Cruise	0	1.00	510.58	192.75	32.00	388.00	521.00	606.00	909.00
Shrimp_ID_number	0	1.00	10.36	13.36	1.00	2.00	5.00	14.00	73.00
Surface_Salinity	2	1.00	26.58	7.84	5.90	23.10	29.60	32.38	38.60
Surface_Temperature	2	1.00	17.30	5.44	7.00	13.60	16.50	19.50	30.10
Rostrum_teeth	40	0.98	2.68	0.84	0.00	2.00	3.00	3.00	6.00
Rostrum_length	61	0.97	3141.90	415.89	1391.45	3018.81	3278.33	3415.37	4266.98
Telson_length	70	0.96	1516.31	179.36	749.62	1441.84	1567.86	1635.95	1872.39
Uropod_length	38	0.98	1768.37	214.82	799.73	1675.06	1822.97	1914.16	2324.15
Year	0	1.00	2001.33	7.81	1982.00	1996.00	2001.50	2005.00	2017.00
Month	0	1.00	4.26	2.31	1.00	3.00	4.00	4.00	12.00

Variable type: POSIXct

Table 4.8: Brown shrimp size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1982-04-20	2017-10-03	2002-01-03	302

4.5.2 White Shrimp

Table 4.9: White shrimp size summary

Table 4.9: Data summary

Name	postlarv_size_white
Number of rows	1096
Number of columns	16
Column type frequency:	
character	5
numeric	10
POSIXct	1
Group variables	None

Variable type: character

Table 4.10: White shrimp size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Shrimp_ID	0	1.00	7	8	0	1087	0
TowID	0	1.00	3	4	0	249	0
Replicate	0	1.00	1	1	0	2	0
Species	0	1.00	5	5	0	1	0
Notes	1029	0.06	5	35	0	29	0

Variable type: numeric

Table 4.11: White shrimp size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Cruise	0	1.00	487.77	228.39	15.00	356.00	482.00	632.50	907.00
Shrimp_ID_number	0	1.00	7.28	8.19	1.00	2.00	4.00	10.00	47.00
Surface_Salinity	0	1.00	31.57	4.82	6.90	31.40	33.20	34.40	38.60
Surface_Temperature	0	1.00	26.03	2.43	13.90	25.20	26.70	27.50	30.10
Rostrum_teeth	16	0.99	1.65	0.79	0.00	1.00	2.00	2.00	6.00
Rostrum_length	27	0.98	1940.35	321.57	1107.64	1710.19	1831.93	2179.20	3514.92
Telson_length	28	0.97	1033.78	155.01	593.65	928.25	990.63	1144.55	1708.58
Uropod_length	18	0.98	1142.02	197.06	654.82	995.32	1081.15	1281.64	2075.20
Year	0	1.00	2000.24	9.24	1981.00	1995.00	2000.00	2006.00	2017.00
Month	0	1.00	6.65	1.16	3.00	6.00	6.00	7.00	10.00

Variable type: POSIXct

Table 4.12: White shrimp size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1981-08-13	2017-09-01	2000-06-29	184

4.6 About

Input `.qmd` file for this section was: **raw_postlarv.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/postlarv`:

- `Penaeus_PostLarvae_NInlet_1981_2017_wide_kac.csv`
- `Penaeus_Postlarval_Lengths_NInlet_1981 - 2017.xlsx`

Modifications to data:

- minor re-naming: inserted underscores rather than spaces; changed `Shrimp Species` to `Species`.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/postlarval_dfs.RData` (the single `.RData` file contains multiple data frames):

- `postlarv_abund`
- `postlarv_size`

Other notes about data:

none

5 Juveniles

5.1 Datasets

There are two datasets that represent juveniles: SCDNR's Creek Trawls, and NIW NERR's Oyster Landing seines. We will explore each.

5.2 Graphics - Size Distributions

Beeswarm plots can be slow to render so I have subsetted both data frames here to 15,000 rows.

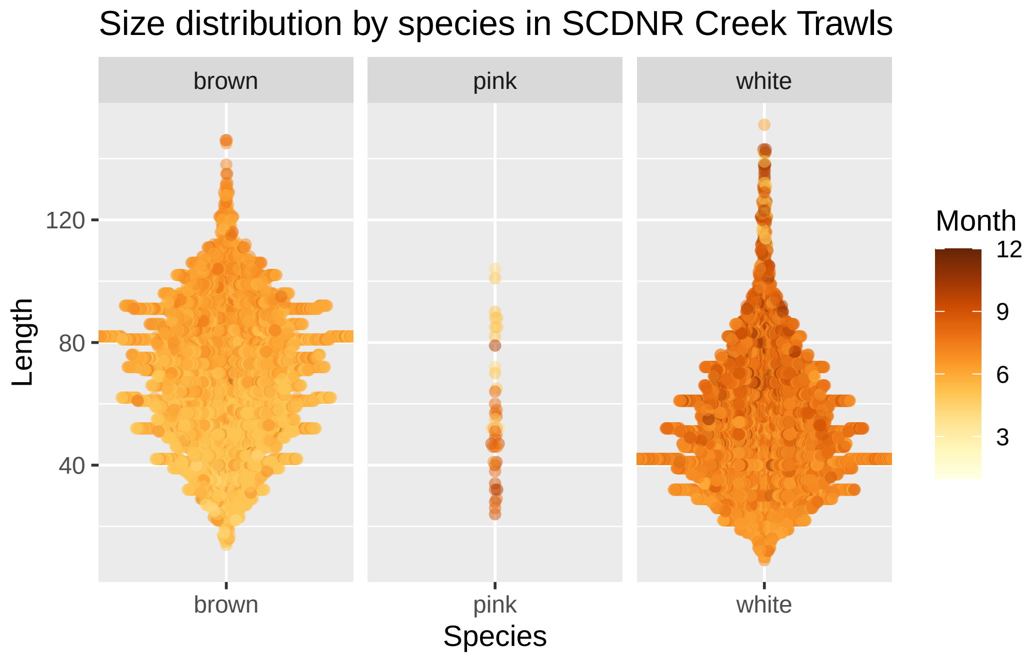


Figure 5.1: Beeswarm plot of juvenile shrimp length for the SCDNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

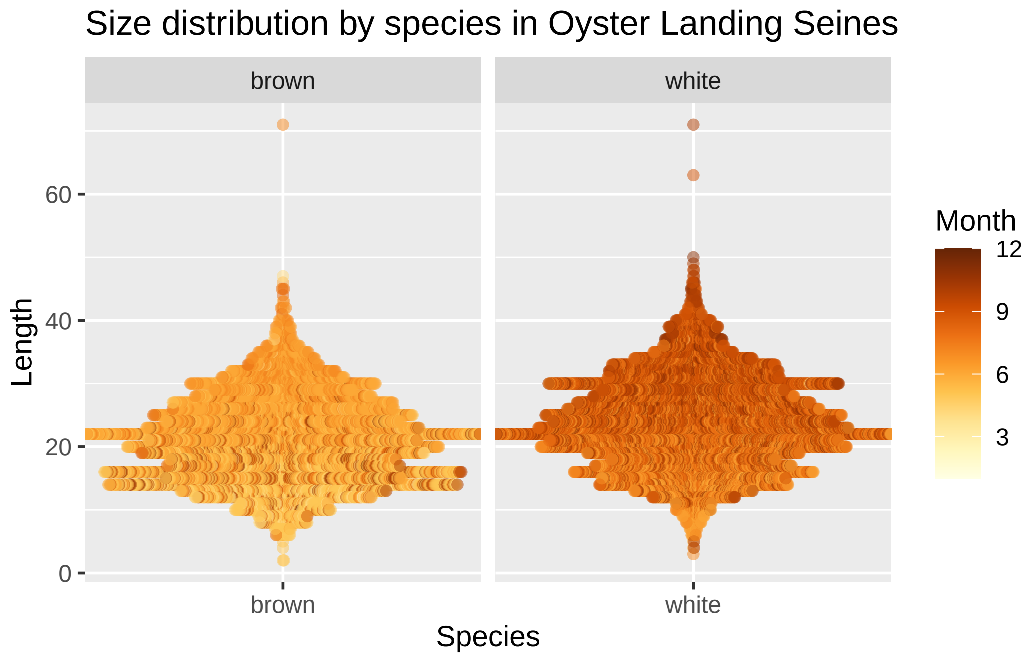


Figure 5.2: Beeswarm plot of juvenile shrimp length for the Oyster Landing seine data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

Looks like OL seines are generally catching smaller individuals of both species than the SCDNR creek trawls.

5.3 Graphics - Abundance/CPUE Boxplots

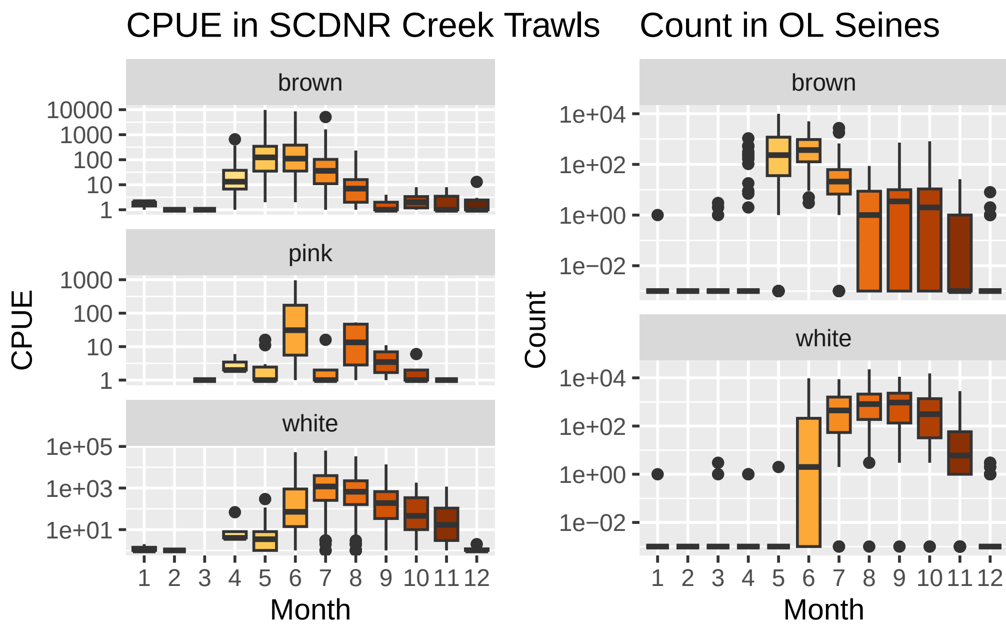


Figure 5.3: Boxplots representing shrimp abundance (as either CPUE or count, depending on the survey) by month. Note the log10-scaled y-axes.

Both surveys (unsurprisingly) show the same temporal pattern of abundance - aztecus in May/June, with setiferus later in the year.

5.4 Tabular summaries of abundance

SCDNR Creek Trawls

Table 5.1: Abundance (SC) data frame summary

Table 5.1: Data summary

Name	juv_sc_cpue
Number of rows	5619
Number of columns	13

Column type frequency:

character	5
numeric	8
Group variables	None

Variable type: character

Table 5.2: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
StationCode	0	1	4	4	0	7	0
EstuaryCode	0	1	2	2	0	1	0
DTStart	0	1	8	10	0	621	0
SpCode	0	1	4	4	0	3	0
Species	0	1	4	5	0	3	0

Variable type: numeric

Table 5.3: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1	2810.00	1622.21	1.00	1405.50	2810.00	4214.50	5619.00
Coll	0	1	20030686.43	133365.28	19801003.00	19911032.00	20031007.00	20161081.00	20231084.00
Year	0	1	2002.97	13.34	1980.00	1991.00	2003.00	2016.00	2023.00
Month	0	1	6.62	1.91	1.00	5.00	7.00	8.00	12.00
Day	0	1	15.29	8.38	1.00	9.00	15.00	22.00	31.00
CPUE	3	1	546.00	2690.49	0.00	0.00	0.00	82.00	62976.00
Lat	0	1	32.84	0.07	32.75	32.80	32.86	32.86	32.95
Long	0	1	-79.89	0.08	-79.99	-79.98	-79.88	-79.85	-79.76

Oyster Landing Seines

Table 5.4: Abundance (Oyster Landing) data frame summary

Table 5.4: Data summary

Name	juv_ol_count
Number of rows	1796
Number of columns	12

Column type frequency:	
character	4
Date	1
numeric	7
Group variables	None

Variable type: character

Table 5.5: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	2	1	8	19	0	4	0
WQ.Time	2	1	0	5	36	74	0
Species	0	1	5	5	0	2	0

Variable type: Date

Table 5.6: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-01-04	2023-12-21	2002-02-18	898

Variable type: numeric

Table 5.7: Abundance (Oyster Landing) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	449.76	259.63	1.0	225.0	449.5	675.00	899.0
Year	0	1.00	2002.63	11.71	1984.0	1993.0	2002.0	2012.00	2023.0
Month	0	1.00	6.60	3.26	1.0	4.0	7.0	9.00	12.0
Temp	94	0.95	20.47	6.91	3.8	14.5	21.6	26.90	31.4
Sal	106	0.94	31.82	4.78	2.3	30.0	33.3	35.00	38.7
Weight	489	0.73	625.27	2355.88	0.0	0.0	0.0	77.05	39196.1
Count	240	0.87	380.32	1326.23	0.0	0.0	0.0	62.00	22560.0

5.5 Tabular summaries of size

5.5.1 Brown Shrimp

SCDNR Creek Trawls

Table 5.8: Brown shrimp (SC) size summary

Table 5.8: Data summary

Name	sz_sc_brn
Number of rows	34818
Number of columns	14
Column type frequency:	
character	4
Date	1
numeric	9
Group variables	None

Variable type: character

Table 5.9: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
DTStart	0	1	8	10	0	489	0
StationCode	0	1	4	4	0	7	0
SpCode	0	1	4	4	0	1	0
Species	0	1	5	5	0	1	0

Variable type: Date

Table 5.10: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1980-05-08	2023-12-11	1996-06-12	489

Variable type: numeric

Table 5.11: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	35073.55	22862.74	1.00	15867.25	32821.50	52650.75	81763.00
Coll	0	1.00	19981663.41	119955.91	19801003.00	19891005.00	19961014.00	20061026.00	20231080.00
Length	0	1.00	69.54	22.67	12.00	52.00	71.00	86.00	151.00
Lat	0	1.00	32.83	0.07	32.75	32.80	32.81	32.86	32.95
Long	0	1.00	-79.91	0.08	-79.99	-79.98	-79.96	-79.85	-79.76
TempS	372	0.99	27.61	2.92	8.40	25.90	28.00	29.70	36.70
SalinityS	913	0.97	17.58	5.21	0.00	14.00	18.00	21.00	28.00
Year	0	1.00	1998.07	12.00	1980.00	1989.00	1996.00	2006.00	2023.00
Month	0	1.00	5.81	0.86	1.00	5.00	6.00	6.00	12.00

Oyster Landing Seines

Table 5.12: Brown shrimp (Oyster Landing) size summary

Table 5.12: Data summary

Name	sz_ol_brn
Number of rows	15212
Number of columns	12
Column type frequency:	
character	5
Date	1
numeric	6
Group variables	None

Variable type: character

Table 5.13: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	0	1	8	19	0	4	0
WQ.Time	0	1	0	5	409	48	0
Species	0	1	5	5	0	1	0
Rep	0	1	4	6	0	100	0

Variable type: Date

Table 5.14: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-05-14	2023-08-29	1999-06-11	410

Variable type: numeric

Table 5.15: Brown shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	384.06	230.34	10.0	185	383.0	543	891.0
Year	0	1.00	1999.55	10.18	1984.0	1991	1999.0	2006	2023.0
Month	0	1.00	6.27	1.50	1.0	5	6.0	7	12.0
Temp	895	0.94	25.64	2.82	6.6	24	26.0	28	31.4
Sal	903	0.94	32.54	3.53	7.7	31	33.8	35	38.7
Length	0	1.00	21.20	6.70	0.0	16	21.0	26	71.0

5.5.2 White Shrimp

SCDNR Creek Trawls

Table 5.16: White shrimp (SC) size summary

Table 5.16: Data summary

Name	sz_sc_wht
Number of rows	46792
Number of columns	14
Column type frequency:	
character	4
Date	1
numeric	9
Group variables	None

Variable type: character

Table 5.17: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
DTStart	0	1	8	10	0	467	0
StationCode	0	1	4	4	0	7	0
SpCode	0	1	4	4	0	1	0
Species	0	1	5	5	0	1	0

Variable type: Date

Table 5.18: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1980-05-08	2023-12-11	2003-07-29	467

Variable type: numeric

Table 5.19: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	45299.88	23170.58	7.00	25467.75	48030.50	65043.25	81764.00
Coll	0	1.00	20031152.32	124812.02	19801003.00	19921067.00	20031027.00	20151040.00	20231080.00
Length	1	1.00	53.26	21.32	8.00	37.00	51.00	66.00	152.00
Lat	0	1.00	32.84	0.07	32.75	32.80	32.81	32.86	32.95
Long	0	1.00	-79.90	0.08	-79.99	-79.98	-79.96	-79.85	-79.76
TempS	215	1.00	29.22	2.98	11.00	28.30	29.60	31.00	36.70
SalinityS	822	0.98	15.99	5.81	0.00	12.00	17.00	20.00	30.00
Year	0	1.00	2003.02	12.48	1980.00	1992.00	2003.00	2015.00	2023.00
Month	0	1.00	7.43	1.11	1.00	7.00	7.00	8.00	12.00

Oyster Landing Seines

Table 5.20: White shrimp (Oyster Landing) size summary

Table 5.20: Data summary

Name	sz_ol_wht
Number of rows	24387
Number of columns	12

Column type frequency:	
character	5
Date	1
numeric	6
Group variables	None

Variable type: character

Table 5.21: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Protocol	0	1	7	8	0	3	0
WQ.Source	0	1	8	19	0	4	0
WQ.Time	0	1	0	5	874	51	0
Species	0	1	5	5	0	1	0
Rep	0	1	4	6	0	100	0

Variable type: Date

Table 5.22: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1984-07-12	2023-12-11	2000-09-11	436

Variable type: numeric

Table 5.23: White shrimp (Oyster Landing) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Sample	0	1.00	421.82	222.99	14.0	236.0	414	592.0	898.0
Year	0	1.00	2001.01	9.94	1984.0	1993.0	2000	2008.0	2023.0
Month	0	1.00	8.45	1.39	1.0	7.0	8	10.0	12.0
Temp	2200	0.91	25.85	3.78	10.6	23.8	27	28.8	31.4
Sal	2305	0.91	32.81	3.96	5.4	31.8	34	35.1	38.7
Length	0	1.00	23.84	7.09	3.0	19.0	23	29.0	71.0

5.6 About

Input `.qmd` file for this section was: **raw_juv.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/juv`:

- `SCDNR_CreekTrawlShrimpCPUE.csv`
- `SCDNR_CreekTrawlShrimpSize.csv`
- `Oyster Landing Seine Shrimp Dataset - 1984-2023 (BWP, 2025-04-24)(kac 2025-06-02).csv`

Modifications to data:

- All files: changed `Species` from provided scientific names to 'brown', 'white', and 'pink'.

- Oyster Landing file: removed 'DataGap' from temp, salinity, and weight columns and turned these to numeric. Split data into counts data frame and sizes data frame. Pivoted sizes to long format.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/juvenile_dfs.RData` (the single `.RData` file contains multiple data frames):

- `juv_sc_cpue`
- `juv_sc_size`
- `juv_ol_count`
- `juv_ol_size`

Other notes about data:

- Oyster Landing file used had a manual correction from Kim; original file had a typo - Sample 748, from 2017-07-20, LEN21, value of 221. Changed to 21.

6 Subadults

6.1 Datasets

There are two datasets that represent subadults: GADNR's EMTS sampling and SCDNR's Estuarine Trawls. We will explore each.

6.2 Graphics - Size Distributions

South Carolina's file has over 300,000 points, which makes for a beeswarm plot that is very large and slow to render. Georgia's file is "only" 65k and even it is too slow to render beeswarm plots. So rather than graphing the full datasets, I have randomly sampled 10,000 rows from GA and 15,000 rows from SC (because SC has three species).

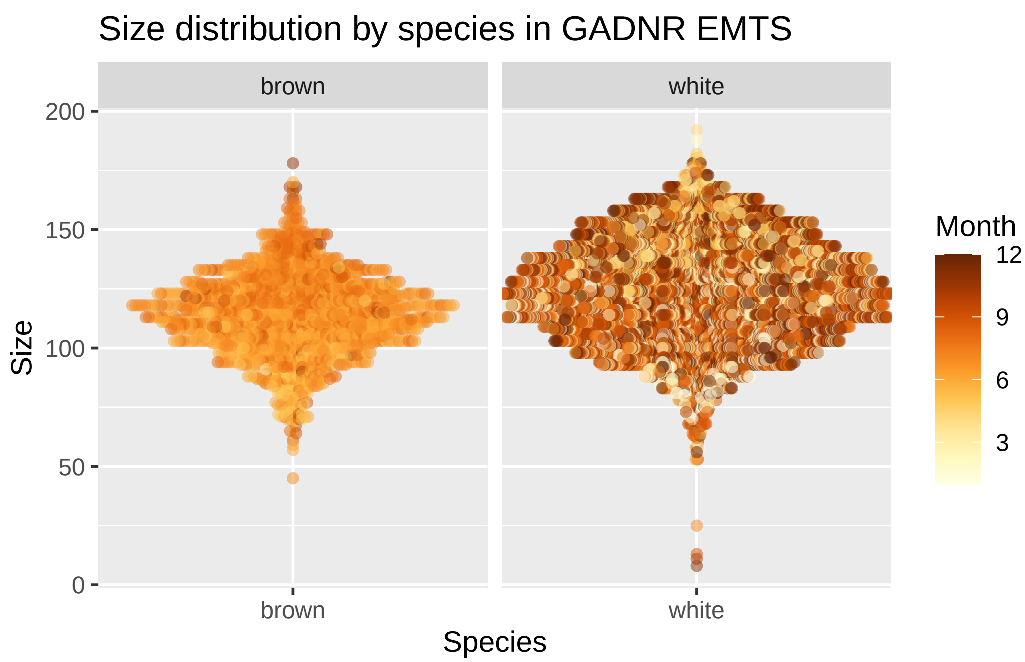


Figure 6.1: Beeswarm plot of subadult shrimp length for the GADNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

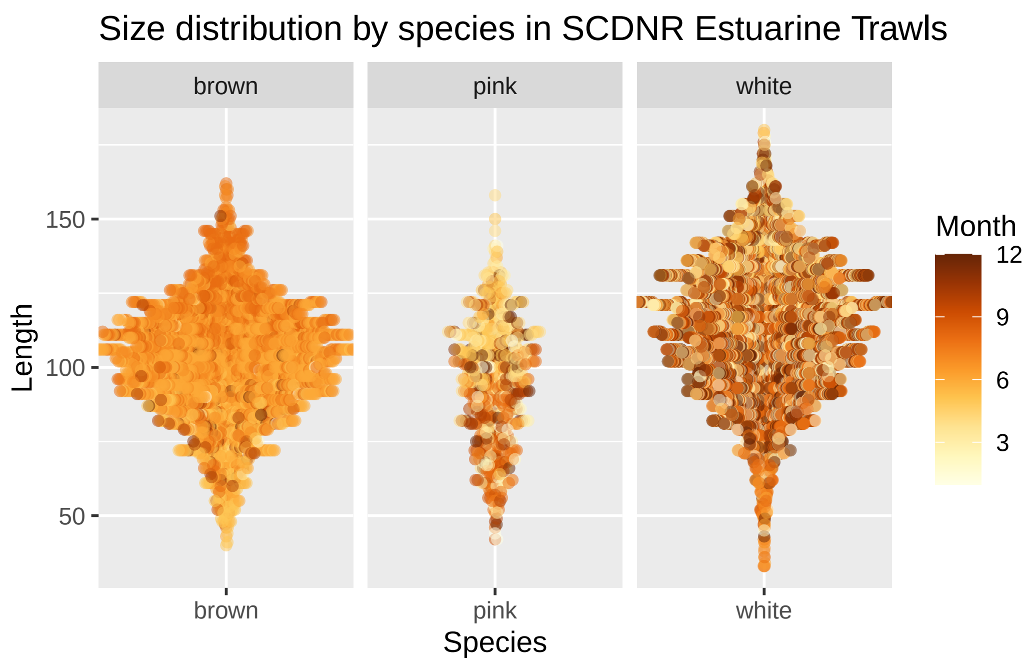


Figure 6.2: Beeswarm plot of subadult shrimp length for the SCDNR data. Points represent measurements of individual shrimp, and are colored by the month in which they were captured. The y-axis represents length. More points spread out along the x-axis for a given length means that there were more individuals of that length captured than regions where points remain closer to the center.

6.3 Graphics - Abundance/CPUE Boxplots

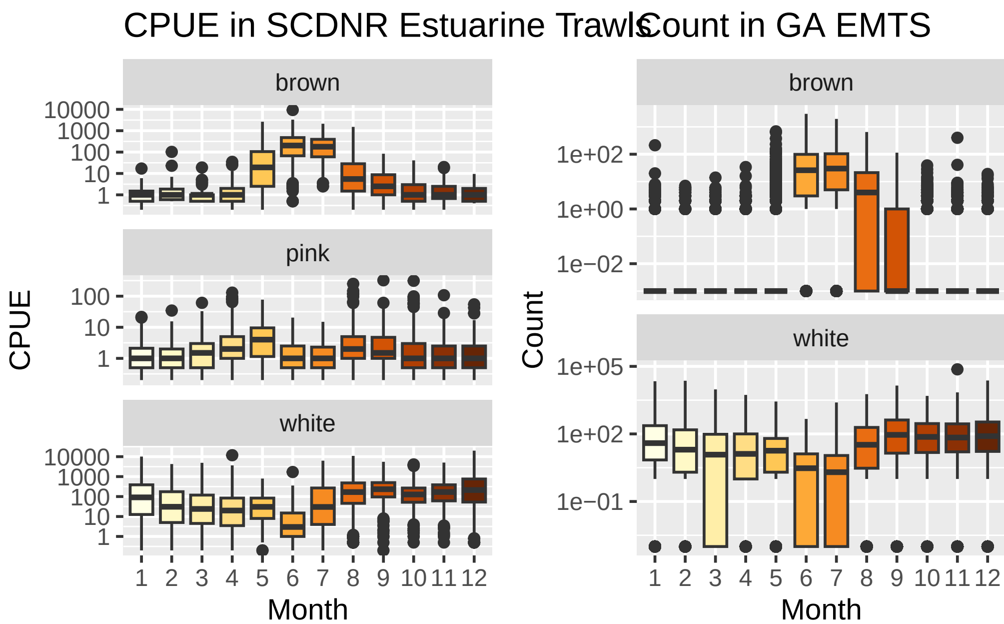


Figure 6.3: Boxplots representing shrimp abundance (as either CPUE or count, depending on the survey) by month. Note the log10-scaled y-axes.

Both surveys (unsurprisingly) show the same temporal pattern of abundance - brown shrimp most abundant in June and July, also high in August; and May in SC. For white shrimp, we see a dip in June and (less so) July.

6.4 Tabular summaries of abundance

GADNR EMTS

Table 6.1: Abundance (GA) data frame summary

Table 6.1: Data summary

Name	subad_ga_count
Number of rows	39669
Number of columns	14

Column type frequency:	
character	2
Date	1
numeric	11
Group variables	None

Variable type: character

Table 6.2: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	19828	0
Species	0	1	5	5	0	2	0

Variable type: Date

Table 6.3: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
GuessedDate	0	1	1975-12-19	2021-10-26	2000-01-10	3254

Variable type: numeric

Table 6.4: Abundance (GA) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
TotWt	3	1.00	1491.00	8839.01	0	0.00	27.22	680.39	1394789.25
TotNum	83	1.00	115.10	610.81	0	0.00	2.00	43.00	73800.00
SampleWt	13	1.00	382.37	716.10	0	0.00	27.22	680.39	65770.89
SampleNum	120	1.00	28.03	47.17	0	0.00	2.00	42.00	1220.00
NumMeas	21154	0.47	12.23	16.40	0	0.00	2.00	30.00	207.00
LbsperHr	12269	0.69	17.83	75.69	0	0.16	2.08	12.88	7060.50
NumperLb	12359	0.69	33.89	253.35	0	15.27	25.60	40.00	31751.50
NumFemales	2395	0.94	7.63	10.59	0	0.00	2.00	15.00	122.00
NumMales	2382	0.94	6.27	9.46	0	0.00	1.00	11.00	97.00
Year	0	1.00	1998.94	13.00	1975	1988.00	2000.00	2010.00	2021.00
Month	0	1.00	6.49	3.42	1	4.00	6.00	9.00	12.00

SCDNR Estuarine Trawls

Table 6.5: Abundance (SC) data frame summary

Table 6.5: Data summary

Name	subad_sc_cpue
Number of rows	21771
Number of columns	13
Column type frequency:	
character	5
numeric	8
Group variables	None

Variable type: character

Table 6.6: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
DTStart	0	1	8	10	0	1647	0
SpCode	0	1	4	4	0	3	0
Species	0	1	4	5	0	3	0

Variable type: numeric

Table 6.7: Abundance (SC) data frame summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1	10886.00	6284.89	1.000e+00	5443.50	10886.00	16328.50	21771.00
Coll	0	1	19986283.95	122808.61	1.979e+07	19890034.00	19970150.00	20090016.00	20230114.00
Year	0	1	1998.61	12.29	1.979e+03	1989.00	1997.00	2009.00	2023.00
Month	0	1	6.60	3.37	1.000e+00	4.00	6.00	9.00	12.00
Day	0	1	17.04	8.02	1.000e+00	11.00	18.00	23.00	31.00
CPUE	4	1	119.11	525.17	0.000e+00	0.00	0.50	24.00	20052.00
Latitude	0	1	32.57	0.23	3.215e+01	32.32	32.67	32.77	32.83
Longitude	0	1	-80.30	0.36	-8.085e+01	-80.65	-80.29	-79.92	-79.89

6.5 Tabular summaries of size

6.5.1 Brown Shrimp

GADNR EMTS

Table 6.8: Brown shrimp (GA) size summary

Table 6.8: Data summary

Name	subad_ga_size_brown
Number of rows	11029
Number of columns	4
Column type frequency:	
character	3
numeric	1
Group variables	None

Variable type: character

Table 6.9: Brown shrimp (GA) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	745	0
TowDate	0	1	8	10	0	308	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 6.10: Brown shrimp (GA) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Size	0	1	113.84	17.45	8	103	114	125	182

SCDNR Estuarine Trawls

Table 6.11: Brown shrimp (SC) size summary

Table 6.11: Data summary

Name	sz_sc_brn
Number of rows	66499
Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 6.12: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SpCode	0	1	4	4	0	1	0
Sex	0	1	0	1	61324	4	0
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 6.13: Brown shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	129622.34	80691.21	1.000e+00	59387.50	126106.00	187401.50	304572.00
Coll	0	1.00	19955732.85	112429.92	1.979e+07	19870163.00	19940120.00	20010143.00	20230106.00
Length	0	1.00	102.12	19.99	2.100e+01	90.00	103.00	115.00	177.00
Year	0	1.00	1995.56	11.25	1.979e+03	1987.00	1994.00	2001.00	2023.00
Month	0	1.00	6.66	1.22	1.000e+00	6.00	6.00	7.00	12.00
Day	0	1.00	17.19	7.95	1.000e+00	10.00	19.00	24.00	31.00
Latitude	0	1.00	32.68	0.19	3.215e+01	32.67	32.77	32.80	32.83
Longitude	0	1.00	-80.12	0.30	-8.085e+01	-80.29	-79.97	-79.92	-79.89
TempB	1640	0.98	27.81	2.84	0.000e+00	27.00	28.40	29.40	32.40
SalinityB	2759	0.96	24.83	6.13	0.000e+00	21.00	26.00	30.00	39.00

6.5.2 White Shrimp

GADNR EMTS

Table 6.14: White shrimp (GA) size summary

Table 6.14: Data summary

Name	subad_ga_size_white
Number of rows	54505
Number of columns	4
Column type frequency:	
character	3
numeric	1
Group variables	None

Variable type: character

Table 6.15: White shrimp (GA) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
RefNum	0	1	12	12	0	2231	0
TowDate	0	1	8	10	0	726	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 6.16: White shrimp (GA) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Size	0	1	124.02	21.61	2	108	124	140	212

SCDNR Estuarine Trawls

Table 6.17: White shrimp (SC) size summary

Table 6.17: Data summary

Name	sz_sc_wht
Number of rows	224834
Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 6.18: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
SpCode	0	1	4	4	0	1	0
Sex	0	1	0	1	173246	4	0
estuary	0	1	2	2	0	6	0
StationCode	0	1	4	4	0	24	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 6.19: White shrimp (SC) size summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
X	0	1.00	165616.57	89820.94	7.000e+00	89692.25	172903.50	244450.75	307960.00
Coll	0	1.00	19991381.12	123497.97	1.979e+07	19890293.00	19970176.00	20090085.00	20230114.00
Length	0	1.00	113.38	22.51	1.100e+01	98.00	114.00	130.00	207.00
Year	0	1.00	1999.12	12.36	1.979e+03	1989.00	1997.00	2009.00	2023.00
Month	0	1.00	7.49	3.49	1.000e+00	4.00	8.00	10.00	12.00
Day	0	1.00	16.44	8.29	1.000e+00	10.00	17.00	23.00	31.00
Latitude	0	1.00	32.59	0.22	3.215e+01	32.46	32.67	32.77	32.83
Longitude	0	1.00	-80.27	0.35	-8.085e+01	-80.54	-80.24	-79.92	-79.89
TempB	4077	0.98	20.86	6.76	0.000e+00	15.00	20.60	27.90	32.40
SalinityB	4665	0.98	25.93	7.06	0.000e+00	22.00	27.00	31.00	209.00

6.6 About

Input `.qmd` file for this section was: **raw_subad.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/subad`:

- `GADNR_EMTS_PenaeidShrimp_Brown_Count_thru2021.xlsx`
- `GADNR_EMTS_PenaeidShrimp_White_Count_thru2021.xlsx`
- `GADNR_EMTS_PenaeidShrimp_SIZE_thru2021.xlsx`, sheet = "Brown shrimp"
- `GADNR_EMTS_PenaeidShrimp_SIZE_thru2021.xlsx`, sheet = "White Shrimp"
- `SCDNR_EstuarineTrawlShrimpCPUE.csv`
- `SCDNR_EstuarineTrawlShrimpSize.csv`

Modifications to data:

- All files: changed `Species` from provided scientific names to 'brown', 'white', and 'pink'.
- GA counts file: extracted date from the Reference Number, as date was not its own field.
- SC size file: filtered data to remove lengths >1000. There were 3 such rows.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/subadult_dfs.RData` (the single `.RData` file contains multiple data frames):

- `subad_ga_count`
- `subad_ga_size`
- `subad_sc_cpue`
- `subad_sc_size`

Other notes about data:

- In GA count files, early exploration not shown here indicated that `NumperLb` was miscalculated when the sample weight wasn't the typical 1360.78 - calculations in the spreadsheet used that value anyway. This issue seemed to happen when catch was low.

7 Adults - fisheries-independent

SEAMAP info here

About the Coastal Trawl Survey: https://seamap.org/seamap-sa-coastal-trawl/

Some excerpts:

- The standardized gear used since 1987 is a pair of 75 ft (22.9 m) mongoose-type Falcon trawl nets (manufactured by Beaufort Marine Supply, Beaufort, SC) without Turtle Excluder Devices. The body of the trawl net was constructed of #15 net twine with 47 mm stretch mesh.

- Historically, the catch from each net was processed separately and assigned a unique collection number, with data from both nets at each station pooled for analysis to form a standard unit of effort (tow). Beginning in 2021, only the catch from one net was processed with the tail bag of the other net fished untied.

- three sampling seasons: Spring (April - May), Summer (July - August), and Fall (September - November).

So it will probably be best to average to the event, because we've got events in 2022 with only one collection (prior to 2022, it was 2 collections per event).

Starting in 1989, which is when this file starts, sampling was pretty consistent - seasonally rather than monthly; daytime; 75ft nets.

There's also some inner vs. outer depth zone stuff to check. See below - 14,576 rows are from "inner" and only 534 from "outer", so will remove "outer" completely.

More on depth zones, from the Figure 1 caption on the Coastal Trawl website linked above: "The first map shows what's going on, and has this in the caption:"....Inner (shallow) strata are sampled during all seasons (1990-present). Outer (deep) strata were sampled south in Spring (green) and north in Fall (yellow) from 1990-2000. In 1989, both depth zones were sampled over the entire latitudinal range in Spring and Fall."

NC, SC, GA, and FL are present in this dataset.

7.1 Datasets

There are two SEAMAP files: one of events, and one of catch data.

Looks like when no shrimp were caught in a net, the species is left blank and count and weight are both 0. So I'll widen the data frame, filling in 0s for NAs. Then lengthen back out so 0s are all accounted for.

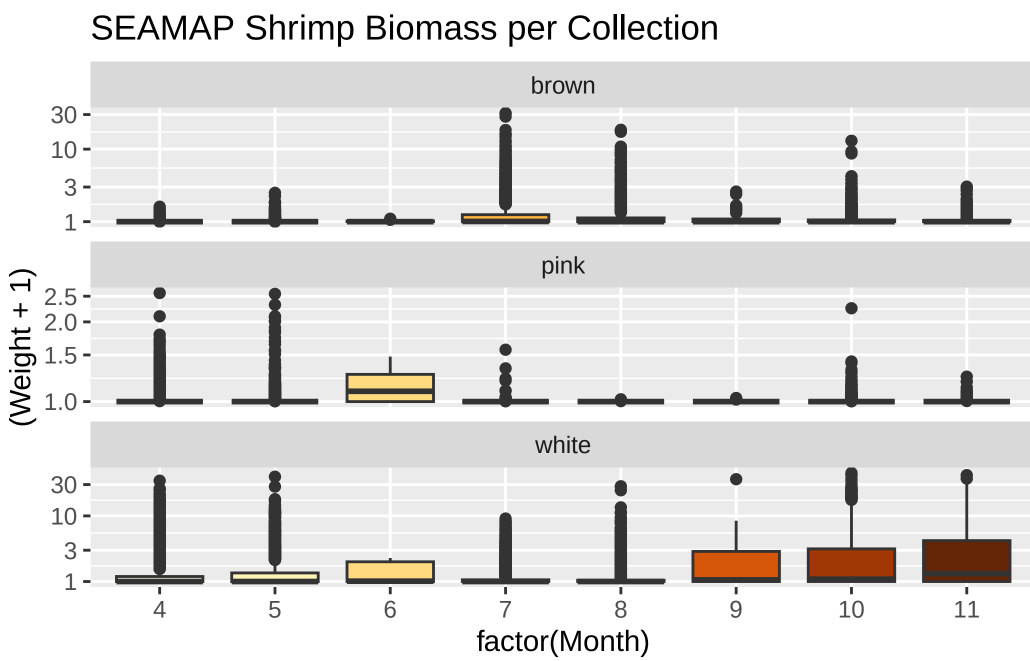
Also filtering to "INNER" depth zone.

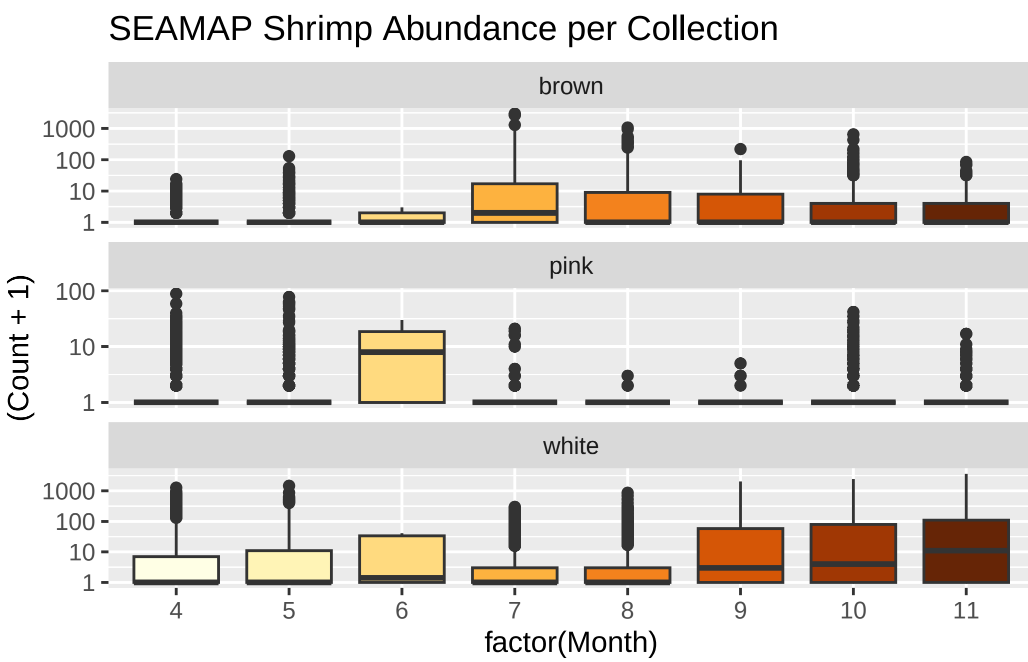
Weight and count were both recorded, even in subsamples, BUT "when large numbers of a species occurred in the processed catch", all individuals were weighed; a subsample was processed for abundance (typically 32-64 individuals); and count was estimated from the weight.

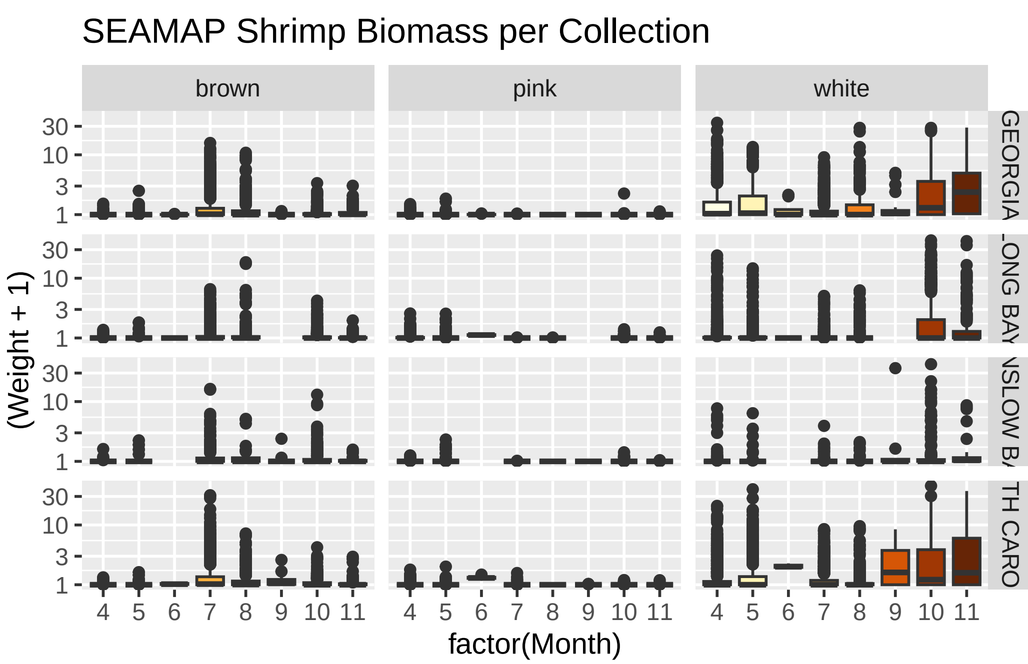
FOR THAT REASON I will use weights in my analyses, because it is the most likely piece to have been measured rather than estimated.

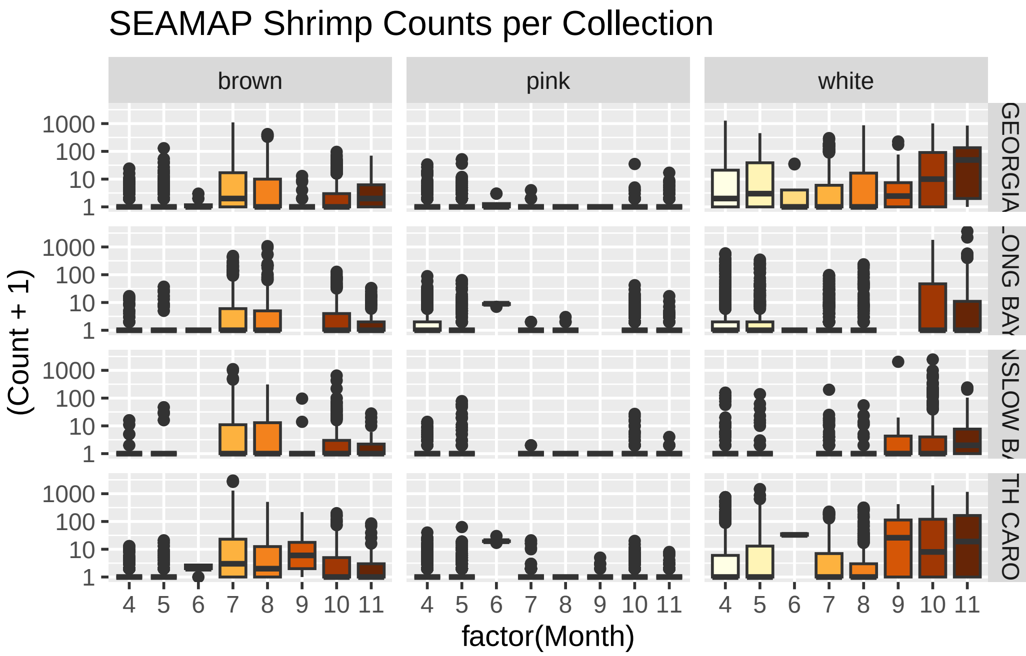
7.2 Graphics - Abundance/Landings/CPUE Boxplots

STILL IN PROGRESS




SEAMAP Shrimp Abundance per Collection
brown
pink
white
(Count + 1)
1000
100
10
1
4
5
6
7
8
9
10
11
factor(Month)


SEAMAP Shrimp Biomass per Collection
brown
pink
white
GEORGIA
(Weight + 1)
30
10
3
1
4
5
6
7
8
9
10
11
factor(Month)



7.3 Tabular summaries of abundance

Brown Shrimp

Table 7.1: Brown shrimp SEAMAP summary

Table 7.1: Data summary

Name	filter(adsm_long, species...
Number of rows	11310
Number of columns	9
Column type frequency:	
character	2
Date	1
numeric	6
Group variables	None

Variable type: character

Table 7.2: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Region	0	1	7	14	0	4	0
species	0	1	5	5	0	1	0

Variable type: Date

Table 7.3: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1989-04-12	2022-11-16	2005-07-19	994

Variable type: numeric

Table 7.4: Brown shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2004.80	9.08	1989	1997	2005	2012.00	2022.0
Month	0	1	7.30	2.41	4	5	7	10.00	11.0
Event	0	1	2005105.07	9090.43	1989001	1997302	2005309	2012542.50	2022559.0
Collection	0	1	20048332.69	90798.12	19890001	19970302	20050310	20120542.75	20220559.0
Weight	0	1	0.18	0.92	0	0	0	0.03	30.4
Count	0	1	11.64	69.76	0	0	0	2.00	2983.0

White Shrimp

Table 7.5: White shrimp SEAMAP summary

Table 7.5: Data summary

Name	filter(adsm_long, species…
Number of rows	11310
Number of columns	9
Column type frequency:	
character	2
Date	1
numeric	6

Group variables	None

Variable type: character

Table 7.6: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Region	0	1	7	14	0	4	0
species	0	1	5	5	0	1	0

Variable type: Date

Table 7.7: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	min	max	median	n_unique
Date	0	1	1989-04-12	2022-11-16	2005-07-19	994

Variable type: numeric

Table 7.8: White shrimp SEAMAP summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2004.80	9.08	1989	1997	2005	2012.00	2022.00
Month	0	1	7.30	2.41	4	5	7	10.00	11.00
Event	0	1	2005105.07	9090.43	1989001	1997302	2005309	2012542.50	2022559.00
Collection	0	1	20048332.60	90798.12	19890001	19970302	20050310	20120542.75	20220559.00
Weight	1	1	1.00	2.86	0	0	0	0.46	44.07
Count	0	1	36.95	116.85	0	0	0	15.00	3641.00

7.4 About

Input `.qmd` file for this section was: **raw_ad_seamap.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/ad_seamap`:

- `SEAMAP_ShrimpCatches_19892022.csv`

Modifications to data:

- Changed `Species` from provided names to 'brown', 'white', and 'pink'.

- For `adsm_long` and `adsm_wide` data frames, filtered to only the "INNER" depth zone and filled in NAs with 0s, because when nothing was caught in a net, both columns in the original data were left blank.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/adultSeamap_dfs.RData` (the single `.RData` file contains multiple data frames):

- `seamap` - the unfiltered data frame
- `adsm_wide`
- `adsm_long`

Other notes about data:

- Contains 4 states: GA, NC, SC, FL

8 Adults - landings and trip tickets

Commercial data here

8.1 Datasets

There are three files that represent commercial fisheries data for adult shrimp: Landings, from NMFS, at two different frequencies (annual for both species, and for white shrimp, fall vs. winter-spring-summer); and trip tickets, by 2-month blocks for both species.

For brown shrimp, calendar year is fine. White shrimp have different life cycles, and we are interested in spawning adults vs. non-spawning adults for each 'shrimp year'; so we obtained the head-off white shrimp data file, split into seasons.

8.2 Graphics - Abundance/Landings/CPUE Boxplots

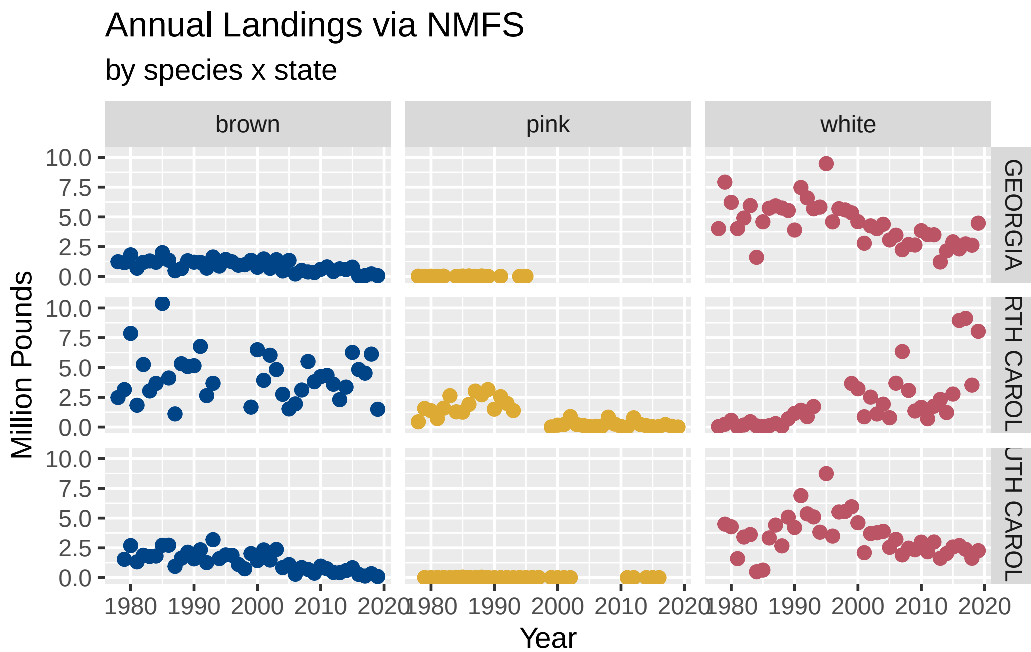


Figure 8.1: Time series plots of annual landings by species and state.

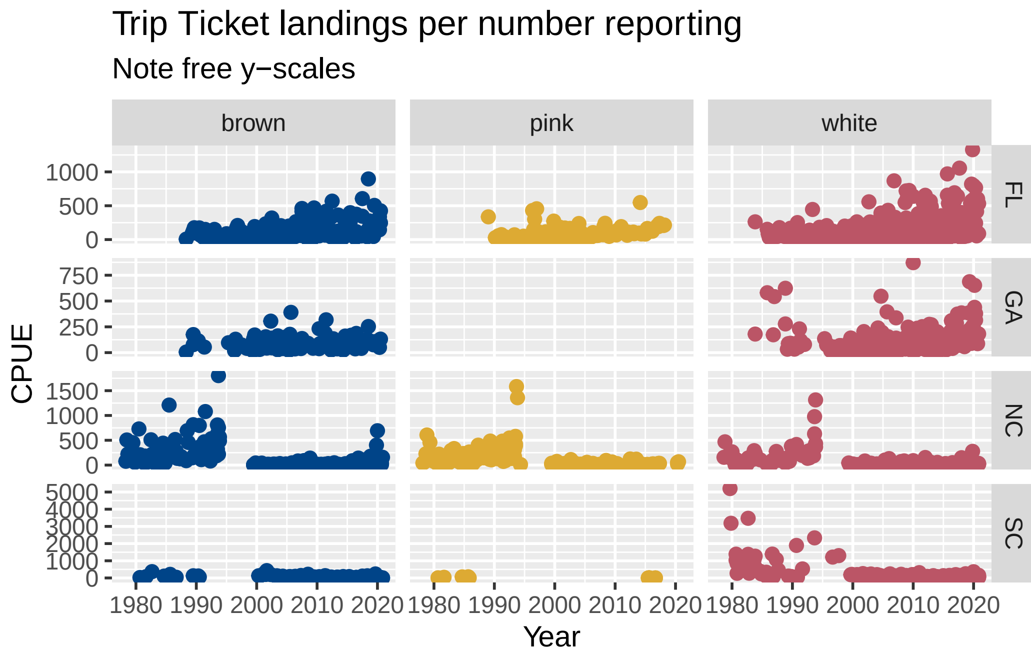


Figure 8.2: Time series plots of estimated CPUE by species and state, from trip ticket data. This plot uses free y-axes for overall context. Large differences probably signal changes in trip ticket collection and reporting, as states are in charge of their programs and may implement changes.

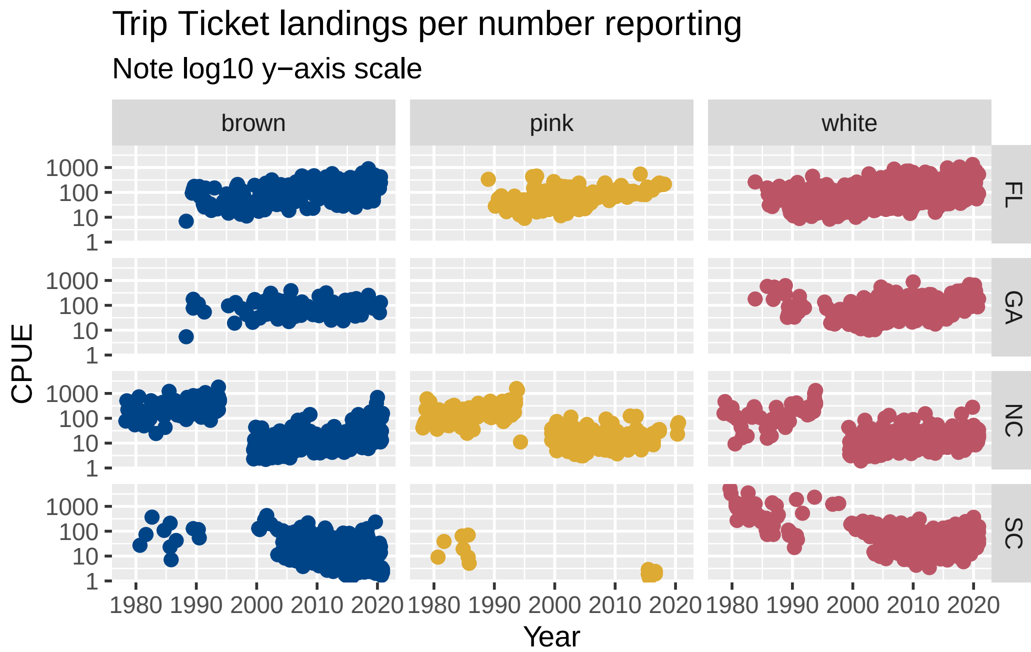


Figure 8.3: Time series plots of estimated CPUE by species and state, from trip ticket data. This plot is the same as that above, but it uses log-10 axes for easier comparison, especially at the low abundances. Large differences probably signal changes in trip ticket collection and reporting, as states are in charge of their programs and may implement changes.

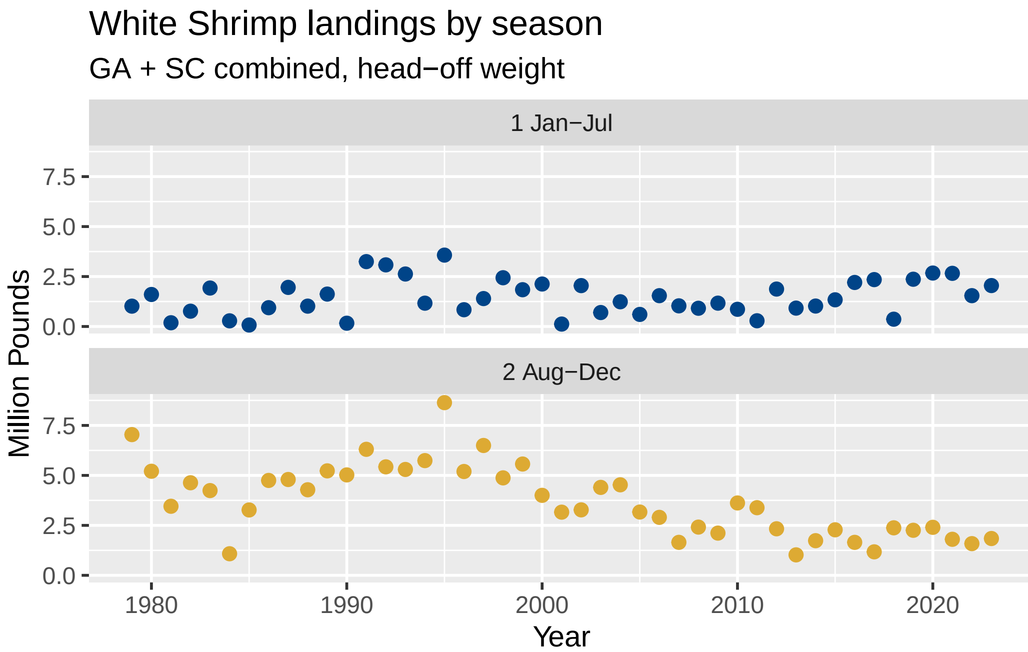


Figure 8.4: Time series plots of white shrimp landings, faceted by season (1 Jan-Jul represents spring spawning adults; 2 Aug-Dec represents fall non-spawning adults).

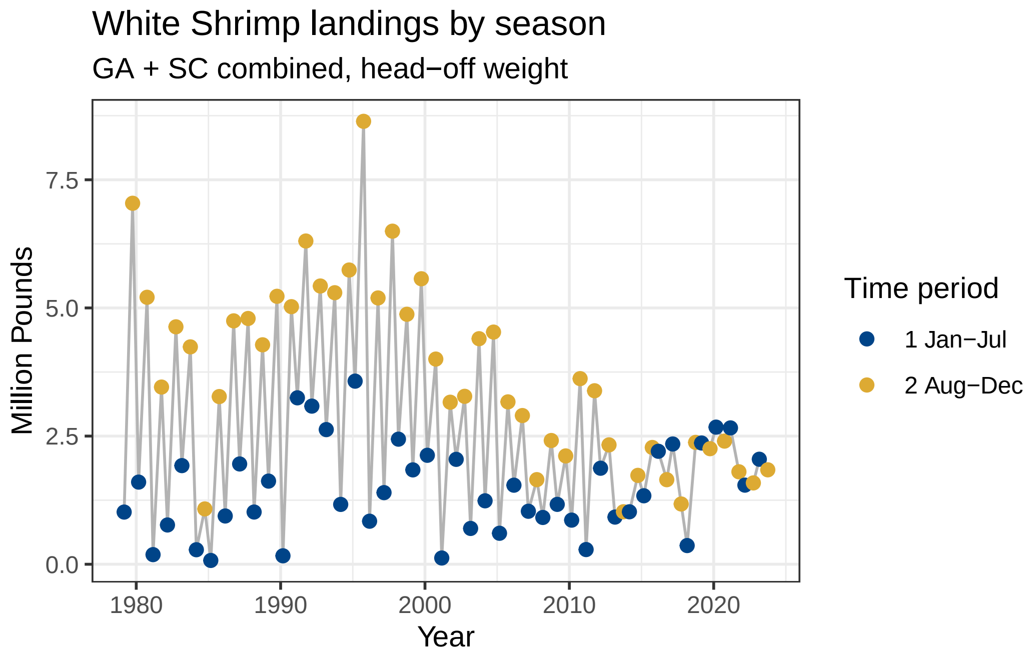


Figure 8.5: Time series plots of white shrimp landings, colored by season (1 Jan-Jul represents spring spawning adults; 2 Aug-Dec represents fall non-spawning adults) but in the same panel and connected with a line. Jan-Jul points have been placed at March 1st each year, and Aug-Dec points placed at October 1st of each year.

8.3 Tabular summaries of abundance

Annual Landings (NMFS)

Brown Shrimp

Table 8.1: Brown shrimp annual landings summary

Table 8.1: Data summary

Name	filter(landings_annual, S...
Number of rows	120
Number of columns	10
Column type frequency:	
character	5
numeric	5

Group variables	None

Variable type: character

Table 8.2: Brown shrimp annual landings summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
State	0	1	7	14	0	3	0
NMFS.Name	0	1	23	23	0	1	0
Collection	0	1	10	10	0	1	0
Confidentiality	0	1	6	6	0	1	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 8.3: Brown shrimp annual landings summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	1998.78	12.31	1978.00	1988.00	1999.50	2009.25	2019.00
Pounds	0	1	2059941.38	1872035.54	41907.00	765917.75	1427067.50	2720053.75	10378157.00
Dollars	0	1	4039632.02	3692856.60	59049.00	1461823.75	2876552.00	4983011.25	18847514.00
Millions_of_dollars	0	1	4.04	3.69	0.06	1.46	2.88	4.98	18.85
Millions_of_Pounds	0	1	2.06	1.87	0.04	0.77	1.43	2.72	10.38

White Shrimp

Table 8.4: White shrimp annual landings summary

Table 8.4: Data summary

Name	filter(landings_annual, S...
Number of rows	120
Number of columns	10
Column type frequency:	
character	5
numeric	5

Group variables	None

Variable type: character

Table 8.5: White shrimp annual landings summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
State	0	1	7	14	0	3	0
NMFS.Name	0	1	23	23	0	1	0
Collection	0	1	10	10	0	1	0
Confidentiality	0	1	6	6	0	1	0
Species	0	1	5	5	0	1	0

Variable type: numeric

Table 8.6: White shrimp annual landings summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	1998.78	12.31	1978.00	1988.00	1999.50	2009.25	2019.00
Pounds	0	1	3372708.22	2171286.61	11372.00	1743573.75	3144572.50	4590372.75	9472533.00
Dollars	0	1	8334883.54	5549245.49	25119.00	4402430.75	7460425.00	10911377.23	23688259.00
Millions_of_dollars	0	1	8.33	5.55	0.03	4.40	7.46	11.91	23.69
Millions_of_Pounds	0	1	3.37	2.17	0.01	1.74	3.14	4.59	9.47

Trip Ticket Landings

Brown Shrimp

Table 8.7: Brown shrimp trip tickets summary

Table 8.7: Data summary

Name	filter(tripTick_abund, Sp...
Number of rows	983
Number of columns	15
Column type frequency:	
character	5
numeric	10

Group variables	None

Variable type: character

Table 8.8: Brown shrimp trip tickets summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area.name	0	1	7	46	0	13	0
Common.Name	0	1	14	14	0	1	0
Species	0	1	5	5	0	1	0
Wave	0	1	9	9	0	6	0
State	0	1	0	2	237	5	0

Variable type: numeric

Table 8.9: Brown shrimp trip tickets summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
object	0	1	1883.32	1036.12	2.00	1032.00	1952.00	2785.00	3573.00
Year	0	1	2004.77	10.94	1978.00	1998.00	2007.00	2014.00	2020.00
WAVE	0	1	3.91	1.24	1.00	3.00	4.00	5.00	6.00
Lat	0	1	32.18	1.93	28.53	30.50	32.24	34.27	35.43
Long	0	1	-78.66	1.75	-81.21	-80.35	-78.25	-77.36	-75.68
annland	0	1	287816.95	768368.26	24.00	10378.01	41119.21	167878.98	8084396.00
numdeal	0	1	14777.14	67470.92	3.00	161.00	900.00	4312.00	919803.00
numfisher	0	1	14777.14	67470.92	3.00	161.00	900.00	4312.00	919803.00
CPUE_approx	0	1	140.18	294.14	0.52	13.27	42.87	149.40	4847.00
Wave_start	0	1	6.81	2.49	1.00	5.00	7.00	9.00	11.00

White Shrimp

Table 8.10: White shrimp trip tickets summary

Table 8.10: Data summary

Name	filter(tripTick_abund, Sp...
Number of rows	1636

Number of columns	15
Column type frequency:	
character	5
numeric	10
Group variables	None

Variable type: character

Table 8.11: White shrimp trip tickets summary

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
Area.name	0	1	7	46	0	14	0
Common.Name	0	1	14	14	0	1	0
Species	0	1	5	5	0	1	0
Wave	0	1	9	9	0	6	0
State	0	1	0	2	256	5	0

Variable type: numeric

Table 8.12: White shrimp trip tickets summary

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
object	0	1	1889.27	1018.10	6.00	1056.25	1938.50	2755.75	3575.00
Year	0	1	2004.90	10.65	1978.00	1998.00	2006.00	2014.00	2020.00
WAVE	0	1	3.87	1.68	1.00	3.00	4.00	5.00	6.00
Lat	0	1	31.77	1.92	25.25	30.50	31.33	33.19	35.43
Long	0	1	-79.26	1.76	-81.21	-80.50	-80.34	-78.25	-75.68
annland	0	1	349093.95	796531.34	32.24	29626.52	108025.30	314248.95	12391609.78
numdeal	0	1	15000.49	47907.43	16.00	309.75	1330.00	8266.50	648442.00
numfisher	0	1	15000.49	47907.43	16.00	309.75	1330.00	8266.50	648442.00
CPUE_approx	0	1	145.03	281.83	0.92	19.60	52.89	156.18	5202.00
Wave_start	0	1	6.75	3.35	1.00	5.00	7.00	9.00	11.00

Seasonal White Shrimp Landings

Table 8.13: White Shrimp (SC + GA) seasonal landings.

Table 8.13: Data summary

Name	landings_seas
Number of rows	90
Number of columns	6
Column type frequency:	
character	3
numeric	3
Group variables	None

Variable type: character

Table 8.14: White Shrimp (SC + GA) seasonal landings.

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
season	0	1	4	11	0	2	0
Species	0	1	5	5	0	1	0
Months	0	1	9	9	0	2	0

Variable type: numeric

Table 8.15: White Shrimp (SC + GA) seasonal landings.

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
Year	0	1	2001.0	13.06	1979	1990	2001.0	2012	2023
lbs_headOff	0	1	2593141.3	1799455.64	74022	1169931	2164551.5	3542747	8639831
Period	0	1	1.5	0.50	1	1	1.5	2	2

8.4 About

Input `.qmd` file for this section was: **raw_ad_commercial.qmd** and can be found in the main directory.

Input data file(s) - as-used; can be found in `data/raw/ad_commercial`:

- `Nonconfidential Shrimp landings by species area year two month block.xlsx` - Trip Tickets. Commercial shrimp landings in NC, SC, GA, and FL, by 2-month block, as reported by commercial fishers and seafood processors. From state trip ticket data. per the data provider, this dataset probably best if we're interested in estimating CPUE. " Trip tickets, which have area fished, came online at different times (GA 1999, SC 2003, NC 1994). That is causing the differences. Also NC didn't always differentiate among the species (I think 96 to 98). "
- `Shrimp Landings NC SC GA from NMFS.csv` - both species, calendar year level (okay for brown shrimp, not white). Annual commercial shrimp landings in NC, SC, and GA, as reported by commercial fishers and seafood processors, and downloaded from https://www.fisheries.noaa.gov/foss/f?p=215:200:::::: per the data provider, this dataset is probably the better one for total landings.
- `scga_landings_headOff_whiteShrimp_byseason.csv` - white shrimp, by Fall vs. Winter-Spring-Summer. Combined white shrimp landings for SC and GA by two 'seasons'. The Fall is defined as Aug-Dec and the Win_Spr_Sum is defined as Jan-July. As such, these two seasons can be combined to come up with 'Annual' landings.

Modifications to data:

- All files: changed `Species` from provided names to 'brown', 'white', and 'pink'.

- Trip Tickets file: Calculated `CPUE_approx` as landings / number of dealers (number of dealers and number of fishers were the same). Made two new columns for two-month `WAVE`: `Wave` to identify the months, e.g. `1 Jan-Feb`, `2 Mar-Apr`, etc.; and `Wave_start` to identify the first month numerically, for better plotting of time series, e.g. Wave 1 (Jan-Feb) is assigned 1; Wave 2 (Mar-Apr) is assigned 3; etc. Added a column for `State`, based on mapping the lat/longs of each named location in the file during earlier exploration.

- Annual Landings from NMFS file: both `Pounds` and `Dollars` were read in as character due to commas in the number formatting; parsed these to numeric. Then calculated versions where the numbers will be smaller: `Millions_of_Dollars` and `Millions_of_Pounds`.

- Seasonal head-off landings of white shrimp: added two columns for better ordering, because 'Win_Spr_Sum' is Jan-July, and 'Fall' is Aug-Dec of each year. Added `Months` to show this, and `Period` where Fall is 1 and Win_Spr_Sum is 2. Otherwise Fall appears first in the data frame due to alphabetical ordering; but chronologically it should be 2nd.

Generated data file(s): Data frames were modified as described above, and written to `data/intermediate/adultCommercial_dfs.RData` (the single `.RData` file contains multiple data frames):

- `landings_annual`

- `landings_seas`
- `tripTick_abund`

Other notes about data:

- `landings_seas` only represents two states: GA and SC. The annual landings and trip ticket data frames also include NC.

Part IV

Summarizing to Abundance Index

9 Postlarval Stage

Setup

Compiling to monthly

Compiling to shrimp year

Saving files out

9.1 Tabular summaries of output datasets

9.1.1 Monthly

Table 9.1: Summary of monthly postlarval abundance data

Table 9.1: Data summary

Name	postlarv_abund_monthly
Number of rows	888
Number of columns	8
Column type frequency:	
character	4
numeric	4
Group variables	None

Variable type: character

Table 9.2: Summary of monthly postlarval abundance data

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	6	6	0	1	0
species	0	1	5	5	0	2	0

Table 9.2: Summary of monthly postlarval abundance data

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
life_stage	0	1	10	10	0	1	0
abundance_measure	0	1	7	7	0	1	0

Variable type: numeric

Table 9.3: Summary of monthly postlarval abundance data

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
year	0	1	1999.00	10.68	1981	1990.00	1999.0	2008.00	2017.00
month	0	1	6.50	3.45	1	3.75	6.5	9.25	12.00
shrimp_year	0	1	4332.33	3638.30	1981	1994.00	2007.0	9999.00	9999.00
abundance	0	1	0.05	0.17	0	0.00	0.0	0.03	3.27

9.1.2 Yearly

Table 9.4: Summary of postlarval abundance data by shrimp year

Table 9.4: Data summary

Name	postlarv_yearly
Number of rows	74
Number of columns	9
Column type frequency:	
character	4
numeric	5
Group variables	None

Variable type: character

Table 9.5: Summary of postlarval abundance data by shrimp year

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	6	6	0	1	0
species	0	1	5	5	0	2	0

Table 9.5: Summary of postlarval abundance data by shrimp year

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
life_stage	0	1	10	10	0	1	0
abundance_measure	0	1	7	7	0	1	0

Variable type: numeric

Table 9.6: Summary of postlarval abundance data by shrimp year

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
shrimp_year	0	1.00	1999.00	10.75	1981.00	1990.00	1999.00	2008.00	2017.00
abundance	0	1.00	0.54	0.64	0.01	0.13	0.33	0.82	4.01
sqrt_abundance	0	1.00	0.64	0.37	0.11	0.36	0.57	0.91	2.00
abundanceIndex_fullTimePeriod	0	1.00	0.00	0.99	-1.42	-0.73	-0.14	0.57	3.32
abundanceIndex_commonTimePeriod	16	0.78	0.00	0.99	-1.49	-0.78	-0.17	0.48	3.25

9.2 About

Input `.qmd` file for this section was: **process_postlarv.qmd** and can be found in the main directory.

Input data file(s) - in `data/intermediate/postlarval_dfs.RData` (the single `.RData` file contains multiple data frames):

- `postlarv_abund`
- `postlarv_size` (this data frame was ignored)

Modifications to data:

- there were some 'year' entries of 1900. These were removed. No data or dates were associated with these rows; presumably they were extra rows in the spreadsheet to aid in data entry.

- removed data for pink shrimp.

Generated data file(s): - fully processed files are in the `data/processed` directory:

- `postlarval_byMonth.csv`, in the subfolder `moreDetailed`, is summarized to species and month within year.

- `postlarval_byShrimpYear.csv` is summarized to species and shrimp year.

Other notes about data:

- there were some 'year' entries of 1900. These were removed. No data or dates were associated with these rows; presumably they were extra rows in the spreadsheet to aid in data entry.

10 Juvenile

There are two datasets that represent juveniles: SCDNR's Creek Trawls, and NIW NERR's Oyster Landing seines.

Setup

SC Creek Trawls - compile to monthly

Oyster Landing - compile to monthly

Combine monthly data frames

Make sure numbers are generally on the same scale before averaging.

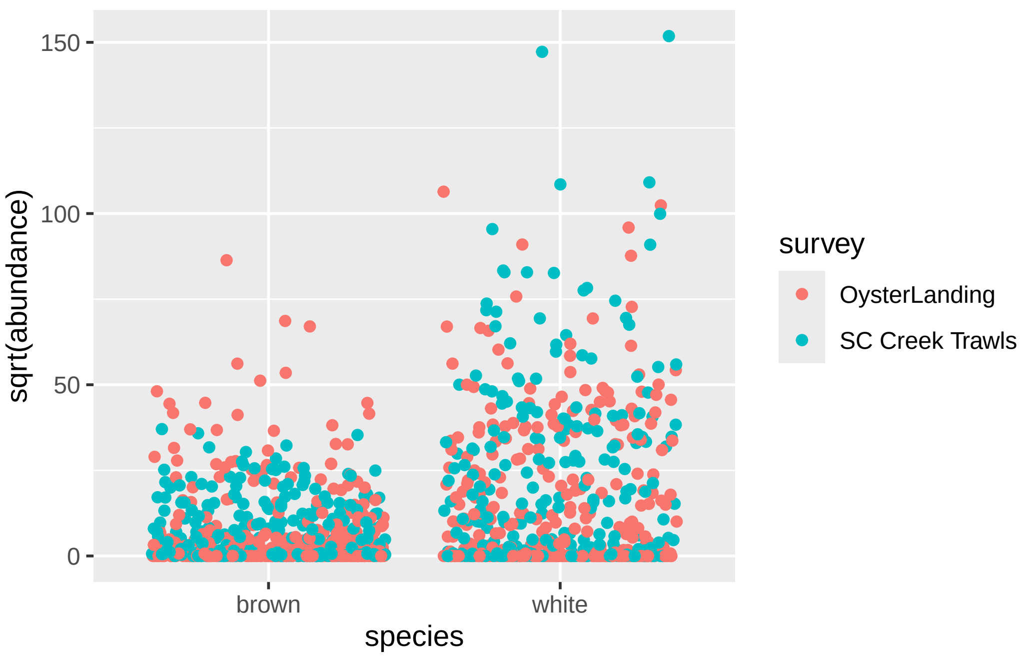


Figure 10.1: Jittered graphic of the square-root of abundance by species and survey. The different surveys used different measures of abundance (SC Creek Trawls used CPUE; Oyster Landing used count) but they appear to be on the same scale, so averaging them should be fine.

Compile to Shrimp Year

First, get annual totals by survey.

Now average the annual values across surveys.

Write out.

10.1 Tabular summaries of output datasets

10.1.1 Monthly

Table 10.1: Summary of monthly juvenile abundance data, SC Creek Trawls

Table 10.1: Data summary

Name	juv_cpue_monthly
Number of rows	472
Number of columns	8
Column type frequency:	
character	4
numeric	4
Group variables	None

Variable type: character

Table 10.2: Summary of monthly juvenile abundance data, SC Creek Trawls

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	15	15	0	1	0
species	0	1	5	5	0	2	0
life_stage	0	1	8	8	0	1	0
abundance_measure	0	1	4	4	0	1	0

Variable type: numeric

Table 10.3: Summary of monthly juvenile abundance data, SC Creek Trawls

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
year	0	1	2003.24	14.06	1980	1991.00	2003.0	2017.00	2023.00
month	0	1	6.85	2.11	1	5.00	7.0	8.00	12.00
shrimp_year	0	1	2003.34	14.08	1980	1991.00	2003.0	2017.00	2024.00
abundance	0	1	718.67	2074.63	0	0.43	56.2	481.33	23051.64

Table 10.4: Summary of monthly juvenile abundance data, Oyster Landing

Table 10.4: Data summary

Name	juv_ol_monthly
Number of rows	914

Number of columns	8
Column type frequency:	
character	4
numeric	4
Group variables	None

Variable type: character

Table 10.5: Summary of monthly juvenile abundance data, Oyster Landing

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	13	13	0	1	0
species	0	1	5	5	0	2	0
life_stage	0	1	8	8	0	1	0
abundance_measure	0	1	5	5	0	1	0

Variable type: numeric

Table 10.6: Summary of monthly juvenile abundance data, Oyster Landing

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
year	0	1.00	2002.96	11.55	1984	1993	2003.0	2012.00	2023
month	0	1.00	6.59	3.35	1	4	7.0	9.00	12
shrimp_year	0	1.00	2003.12	11.56	1984	1993	2003.0	2013.00	2024
abundance	130	0.86	363.37	1086.03	0	0	0.5	101.12	11318

10.1.2 Yearly

Table 10.7: Summary of juvenile abundance data by shrimp year, after averaging SC and OL surveys.

Table 10.7: Data summary

Name	juv_combined_byShrimpYear...
Number of rows	89
Number of columns	9

Column type frequency:	
character	4
numeric	5
Group variables	None

Variable type: character

Table 10.8: Summary of juvenile abundance data by shrimp year, after averaging SC and OL surveys.

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	29	29	0	1	0
species	0	1	5	5	0	2	0
life_stage	0	1	8	8	0	1	0
abundance_measure	0	1	52	52	0	1	0

Variable type: numeric

Table 10.9: Summary of juvenile abundance data by shrimp year, after averaging SC and OL surveys.

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
shrimp_year	0	1.00	2001.75	12.92	1980.00	1991.00	2002.00	2013.00	2024.00
abundance	0	1.00	3565.68	4478.64	0.43	594.59	1849.46	4813.75	24709.21
sqrt_abundance	0	1.00	50.12	32.65	0.65	24.38	43.01	69.38	157.19
abundanceIndex_fullTimePeriod	0	1.00	0.00	0.99	-1.94	-0.62	-0.15	0.55	2.62
abundanceIndex_commonTimePeriod	31	0.65	0.00	0.99	-1.91	-0.64	-0.17	0.61	2.52

10.2 About

Input `.qmd` file for this section was: **process_juv.qmd** and can be found in the main directory.

Input data file(s) - in `data/intermediate/juvenile_dfs.RData` (the single `.RData` file contains multiple data frames):

- `juv_sc_cpue`

- juv_sc_size
- juv_ol_count
- juv_ol_size

Modifications to data:

Generated data file(s): - fully processed files are in the data/processed directory:

- juvenile_bySurveyAndMonth.csv, in the subfolder moreDetailed, is summarized to species and month within year, by survey.

- juvenile_bySurveyAndShrimpYear.csv, in the subfolder moreDetailed, is summarized to species and shrimp year, by survey.
- juvenile_byShrimpYear.csv is summarized to species and shrimp year, as the average shrimp-year abundance across the two surveys. This is the file that will be used in further analyses.

Other notes about data:

- none

11 Subadults

There are two datasets that represent subadults: Georgia trawls and SC estuarine trawls.

Setup

GA Trawls - compile to monthly

SCDNR - compile to monthly

Combine monthly data frames

Make sure numbers are generally on the same scale before averaging.

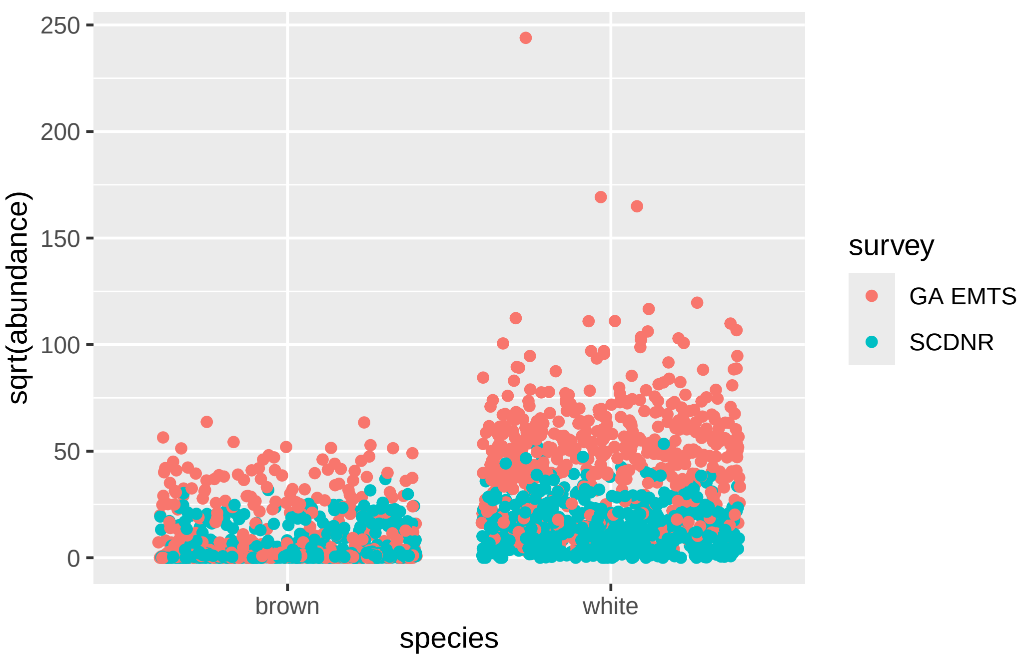


Figure 11.1: Jittered graphic of the square-root of abundance by species and survey. The different surveys used different measures of abundance and appear to be on different enough scales that values will be normalized before averaging across surveys.

Compile to Shrimp Year

Write out.

11.1 Tabular summaries of output datasets

11.1.1 Monthly

Table 11.1: Summary of monthly subadult abundance data, GA trawls

Table 11.1: Data summary

Name	subad_ga_monthly
Number of rows	1054
Number of columns	8
Column type frequency:	
character	4
numeric	4
Group variables	None

Variable type: character

Table 11.2: Summary of monthly subadult abundance data, GA trawls

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	7	7	0	1	0
species	0	1	5	5	0	2	0
life_stage	0	1	8	8	0	1	0
abundance_measure	0	1	12	12	0	1	0

Variable type: numeric

Table 11.3: Summary of monthly subadult abundance data, GA trawls

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
year	0	1	1998.82	13.00	1975	1988.00	1999.00	2010.00	2021.00
month	0	1	6.51	3.45	1	4.00	6.00	9.00	12.00
shrimp_year	0	1	1998.74	13.00	1975	1988.00	1999.00	2010.00	2022.00

Table 11.4: Summary of monthly juvenile abundance data, SC Estuarine Trawls

```
Error: object 'sub_sc_monthly' not found
```

Table 11.3: Summary of monthly subadult abundance data, GA trawls

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
abundance	0	1	1575.14	3021.84	0	2.09	426.06	2368.71	59513.63

11.1.2 Yearly

Table 11.5: Summary of subadult abundance data by shrimp year, after normalizing then averaging GA and SC surveys.

Table 11.5: Data summary

Name	subad_combined_byShrimpYe...
Number of rows	98
Number of columns	8
Column type frequency:	
character	4
numeric	4
Group variables	None

Variable type: character

Table 11.6: Summary of subadult abundance data by shrimp year, after normalizing then averaging GA and SC surveys.

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
survey	0	1	17	17	0	1	0
species	0	1	5	5	0	2	0
life_stage	0	1	8	8	0	1	0
abundance_measure	0	1	50	50	0	1	0

Variable type: numeric

Table 11.7: Summary of subadult abundance data by shrimp year, after normalizing then averaging GA and SC surveys.

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
shrimp_year	0	1.00	1999.50	14.22	1975.00	1987.25	1999.50	2011.75	2024.00
abundance	0	1.00	0.03	0.84	-2.86	-0.48	-0.01	0.47	3.87
abundanceIndex_fullTimePeriod	0	1.00	0.00	0.99	-3.47	-0.57	-0.04	0.54	4.39
abundanceIndex_commonTimePeriod	20	0.59	0.00	0.99	-1.90	-0.78	-0.01	0.61	2.47

11.2 About

Input `.qmd` file for this section was: **process_subad.qmd** and can be found in the main directory.

Input data file(s) - in `data/intermediate/subadult_dfs.RData` (the single `.RData` file contains multiple data frames):

- `subad_ga_count`
- `subad_ga_size`
- `subad_sc_cpue`
- `subad_sc_size`

Modifications to data:

Generated data file(s): - fully processed files are in the `data/processed` directory:

- `subadult_bySurveyAndMonth.csv`, in the subfolder `moreDetailed`, is summarized to species and month within year, by survey.

- `subadult_bySurveyAndShrimpYear.csv` , in the subfolder `moreDetailed`,is summarized to species and shrimp year, by survey.
- `subadult_byShrimpYear.csv` is summarized to species and shrimp year, as the average shrimp-year abundance across the two surveys. This is the file that will be used in further analyses.

Other notes about data:

- none

12 Adults - fisheries-independent

This file focuses on SEAMAP data.

Setup

Compiling to season

SEAMAP does seasonal sampling; not monthly.

So here, I average to sampling event; then average to season within state; and then calculate annual totals.

Yearly totals by state.

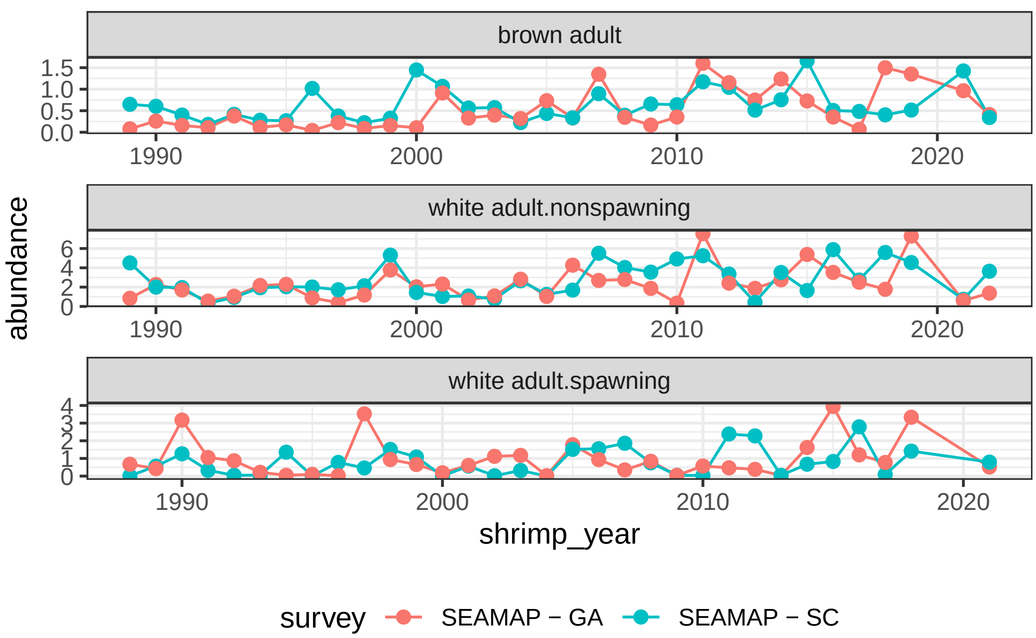


Figure 12.1: Time series of total abundance (weight) by species x life stage, colored by state, in SEAMAP surveys.

Average the states each year and calculate abundance index.

Write out the files.

12.1 About

Input `.qmd` file for this section was: **process_ad_seamap.qmd** and can be found in the main directory.

Input data file(s) - in `data/intermediate/adultSeamap_dfs.RData` (the single `.RData` file contains multiple data frames):

- `seamap` - the unfiltered data frame
- `adsm_wide`
- `adsm_long`

Modifications to data:

- filtered to only brown and white shrimp
- filtered to only SC and GA
- averaged to sampling event, then to season, by state. Summed seasonal averages by shrimp year, by state. Then averaged annual totals and calculated abundance index.

Generated data file(s): - fully processed files are in the `data/processed` directory:

- `adult.seamap_byStateAndSeason.csv` - in `moreDetailed` folder - abundance averaged to season within state, by species and life stage.
- `adult.seamap_byStateAndShrimpYear.csv` - in `moreDetailed` folder - contains annual totals by species, life stage, shrimp year, and state.
- `adult_byShrimpYear.csv` - contains average of annual totals by shrimp year and calculated abundance index, by species, life stage, and shrimp year.

Other notes about data:

- did not worry about order of rows in output files, but all life stages are represented equally.

13 Adults - landings and trip tickets

Setup

13.1 Landings Data

13.1.1 Brown Shrimp

Use annual data for brown shrimp, because it corresponds to the life cycle and shrimp year. Three states are represented in this dataset. NC is not represented elsewhere, so it is removed and we will focus only on GA and SC.

Figure 13.1

For white shrimp landings, we only have totals across the states of GA and SC. For Juvenile and Subadult stages (both species), we averaged the values from the two states.

Here, I calculated both ways, and only later realized they would be exactly the same once normalized - because averaging is just dividing the total by 2, for all samples; so they all stay the same relative to each other when you re-scale.

Because 'mean' will probably be primary, the variable names that match those of other files will contain values using the mean across GA and SC. "sum" will be appended to variable names using a total across the two states, in case such values are needed later.

Write out.

13.1.2 White Shrimp

Using seasonal data for white shrimp, to separate non-spawning adults (time period 2 for a calendar year, season Fall, Months Aug-Dec) from spawners (time period 1 of the next calendar year, season Winter-Spring-Summer, months Jan-July).

The input file contained heads-off landings from SC and GA.

Generate abundance index. Handling a little differently because there are two life stages in here (spawning and non-spawning adults).

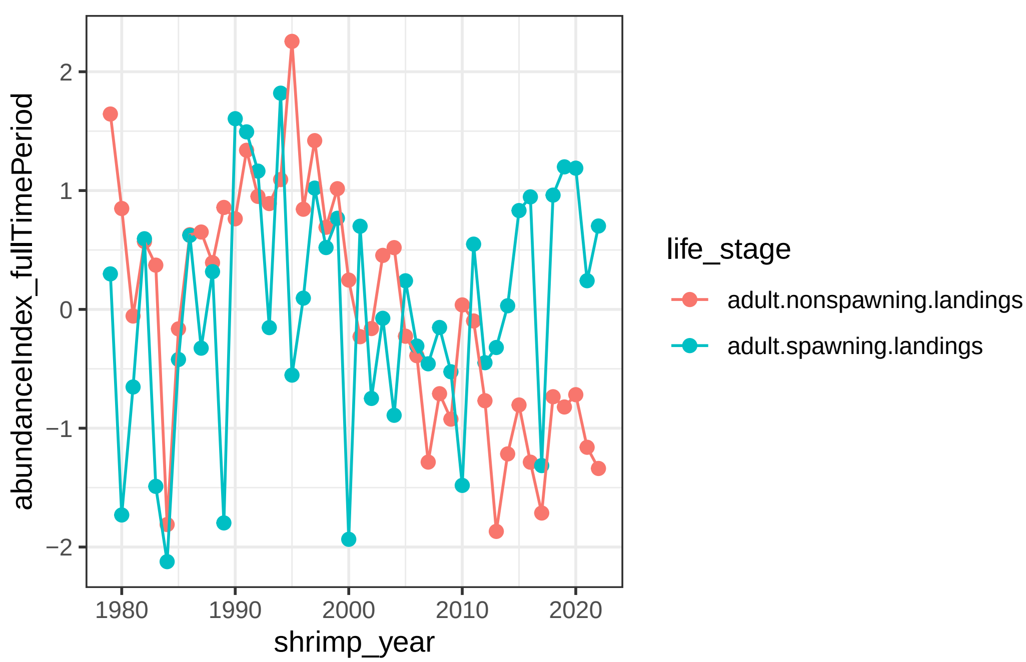


Figure 13.2: Abundance Index for white shrimp, by spawning vs. non-spawning adults

Write out white shrimp data. Both 'adult' life stages are in the single file.

13.2 Trip Ticket Data

I intended to compile to annual totals, but in early years there are big differences in reporting frequencies. So the only files I will write out will contain the 'Wave', which is the 2-month period of reporting. I will not compile to annual values or abundance indices.

Would probably be best to pull out a wave or two and use those as an index. Waves 3, 4, and 5 seem to be pretty consistently reported after about 2000.

13.2.1 Brown Shrimp

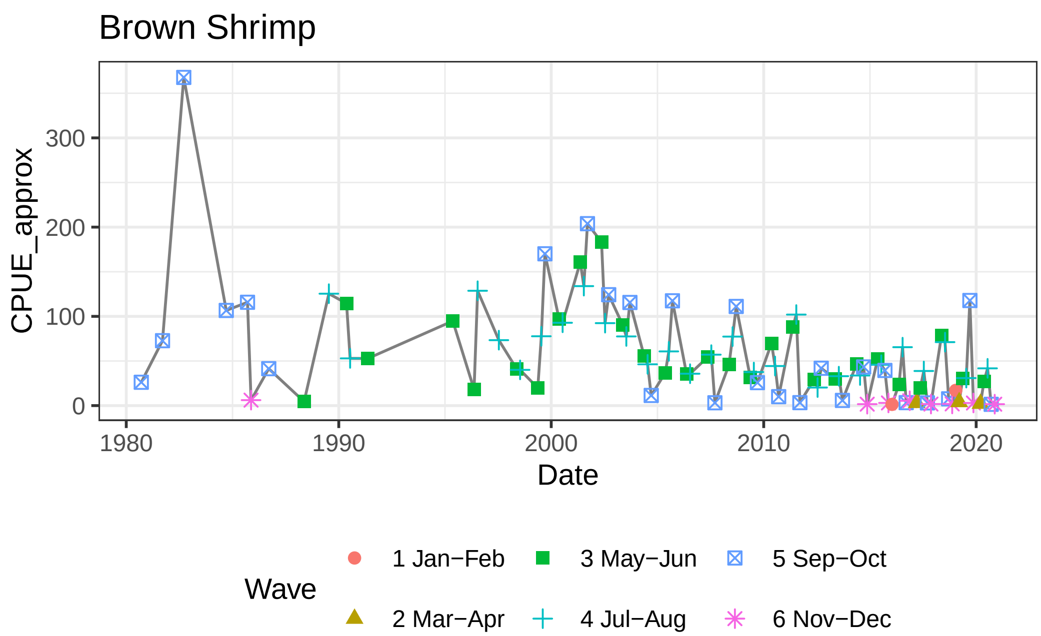


Figure 13.3: Approximate CPUE for GA & SC combined (mean CPUE per time period across states) for brown shrimp by time period. Notice the vastly different reporting frequencies before about 2000.

13.2.2 White Shrimp

For whites, ideally I would compile to the same seasons reported in landings: Jan-Jul and Aug-Dec. However the waves here split; Jul-Aug are combined. Per advice from a work group member, July landings are negligible and this wave should be dominated by August landings, so we will count them with Fall/non-spawning shrimp.

There are most likely differences in when reporting happened seasonally, as with brown shrimp, so I will again only compile to “Wave”.

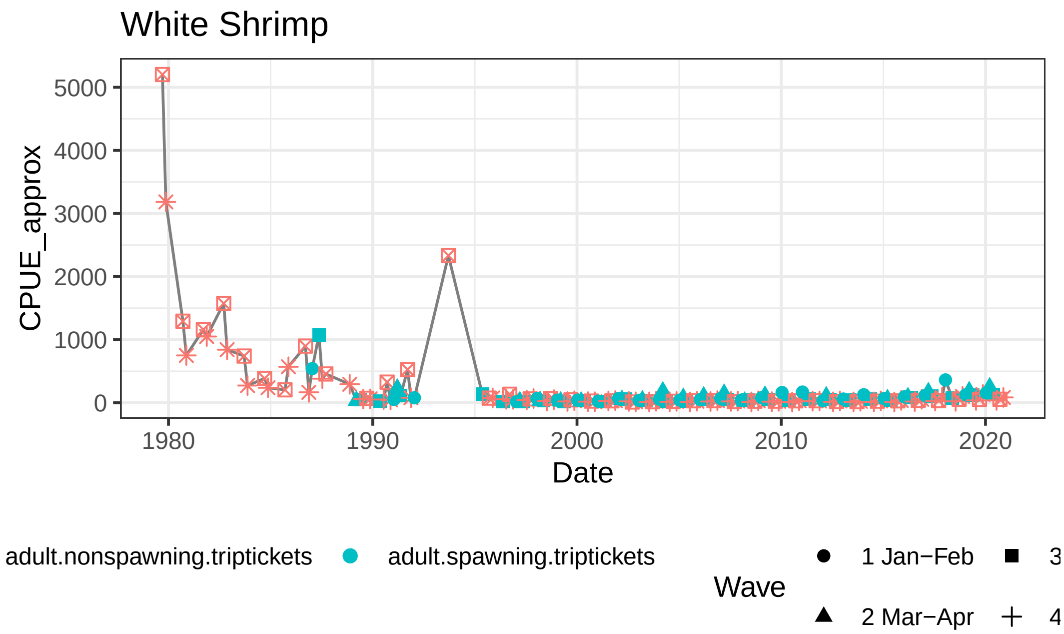


Figure 13.4: Approximate CPUE for GA & SC combined (added together) for white shrimp by time period. Notice the vastly different reporting frequencies before about 2000.

Looks like there was a lot going on before about 1995 so let’s filter that data.

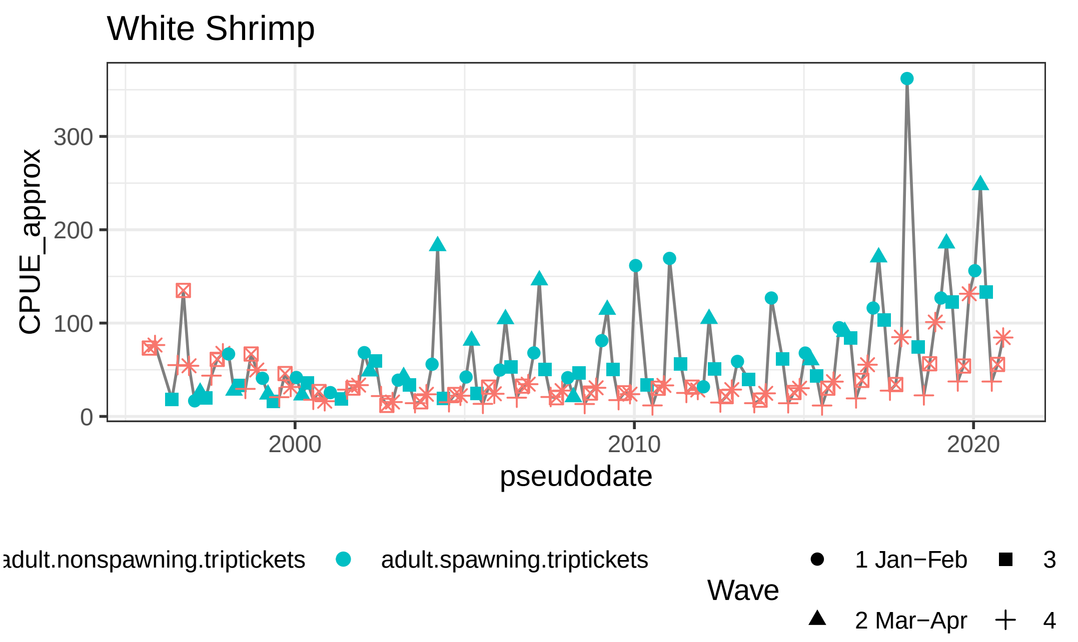


Figure 13.5: Approximate CPUE for GA & SC combined (added together) for white shrimp by time period, starting in 1995

There are several years where Wave 2 is missing (triangle), but 1, 3, 4, 5, and 6 seem pretty consistent in this time period (post-1995).

All years are included in these output files.

13.3 About

Input `.qmd` file for this section was: **process_ad_commercial.qmd** and can be found in the main directory.

Input data file(s) - in `data/intermediate/adultCommercial_dfs.RData` (the single `.RData` file contains multiple data frames):

- `landings_annual`
- `landings_seas`
- `tripTick_abund`

Modifications to data:

- pink shrimp removed.

- any areas other than GA and SC were removed.

- `pseudodate` column was created based on the year and first month of each 'Wave' of reporting, in order to make time series effectively.

Generated data file(s):

NOTE - for brown shrimp files, abundances and abundance indices were aggregated from the 2 states of GA and SC in two different ways. They turn out to give the same numbers when scaled, so the columns with "sum" appended can be ignored. They were retained in case someone else wants to verify in the future.

Landings Data

- fully processed files are in the `data/processed` directory.
- `adult.landings.brown_byStateAndShrimpYear.csv`, in the subfolder `moreDetailed`, shows the separate totals by year.

- `adult.landings.brown_byShrimpYear.csv` - annual landings were totalled across the two states, to be consistent with how white shrimp landings were reported in our input file.

- `adult.landings.white_byShrimpYear.csv` - has both spawning and nonspawning in the same file, as separate `life_stage`s.

Trip Ticket Data

Fully processed files are in the `moreDetailed` folder of the `data/processed` directory. Abundance indices for trip ticket data were not generated due to inconsistency in the consituents (which Waves had values through time):

- `adult.triptickets.brown_01byStateAndWave.csv` - abundances and number reporting for GA and SC separately by Wave and year.

- `adult.triptickets.brown_02byWave.csv` - abundances and reporters from the two states are added together for Wave and year.

- `adult.triptickets.white_01byStateAndWave.csv` - same as brown file

- `adult.triptickets.white_02byWave.csv` - same as brown file

Other notes about data:

- none

14 Combine abundance information

Pull together data from all life stages. There are a couple of graphs in here; life stages are not in order and I did not try to make them look nice, but they will give a very rough idea of what's in the data.

The files for each life stage have the same columns in them. Some have additional columns that are removed at this stage.

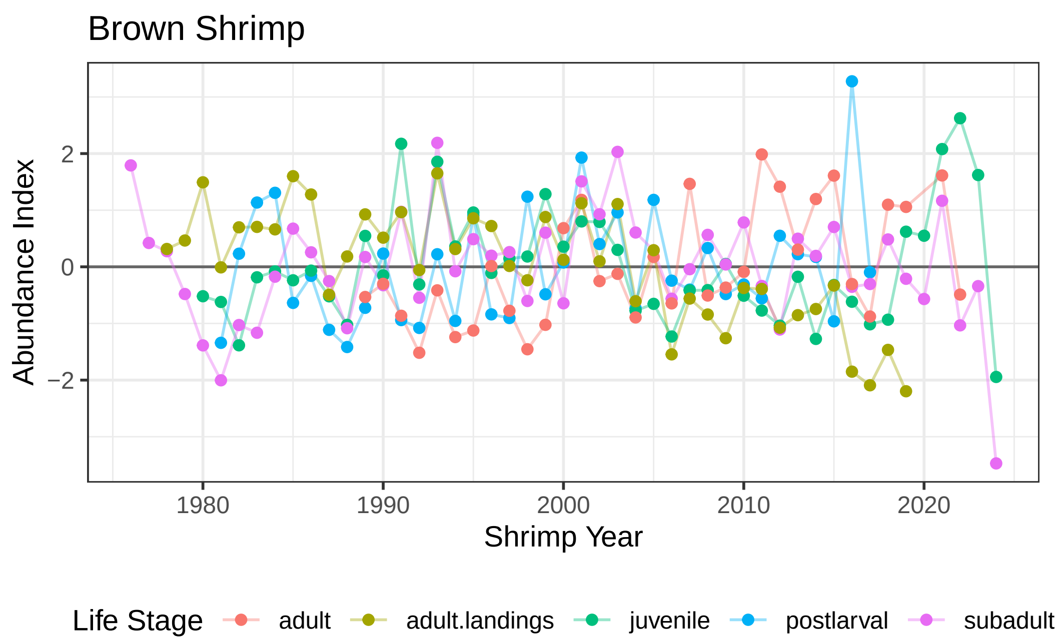
I will create a few files: one with all abundance information for both species, so future data users can compare whichever subsets they are interested in. That will be in a subfolder `data/processed/combinedLifeStages`.

Then I will split brown and white shrimp, and combine the environmental data with abundance data. These will be saved in `data/processed`.

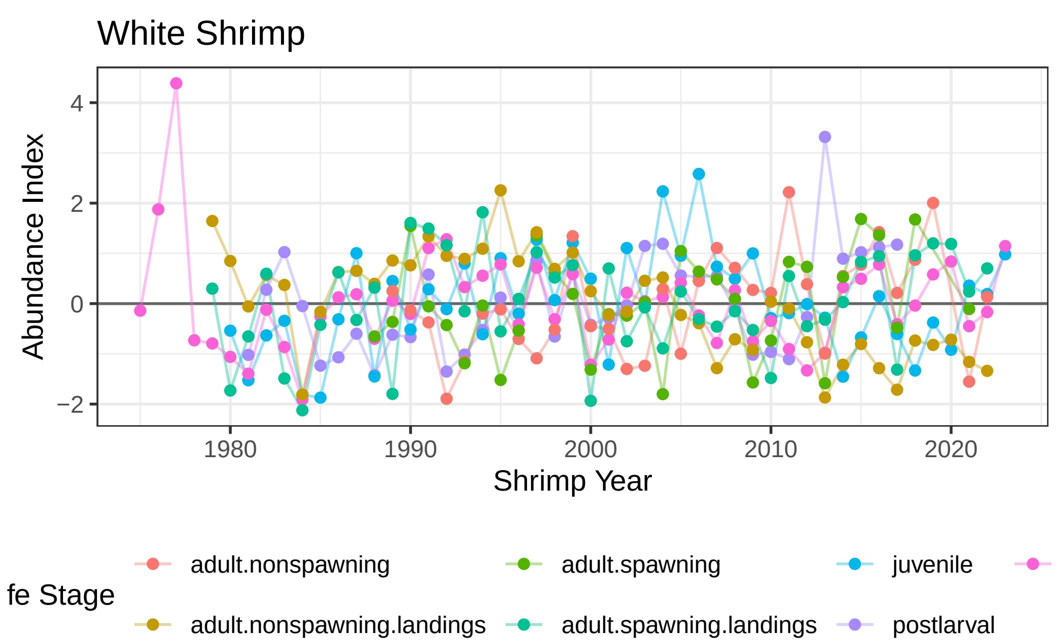
Landings data are retained in these files. Trip Ticket data has not been incorporated.

14.1 Brown Shrimp

Write it out.



14.2 White Shrimp



14.3 About

Input `.qmd` file for this section was: **combine__processed__abundances.qmd** and can be found in the main directory.

Input data file(s) - in `data/processed/byLifeStage`:

- `postlarval_byShrimpYear.csv`
- `juvenile_byShrimpYear.csv`
- `subadult_byShrimpYear.csv`
- `adult_byShrimpYear.csv`
- `adult.landings.brown_byShrimpYear.csv`
- `adult.landings.white_byShrimpYear.csv`

Modifications to data:

- reduced to only common shrimp abundance columns
- combined abundance and environmental data, based on shrimp year.
- **NOTE:** because the abundances are in a long format, meaning each life stage has a separate row for shrimp year, the same environmental data is repeated on multiple rows.

Generated data file(s):

- `allAbundanceIndices.csv`, in `data/processed/combinedLifeStages`. Contains abundance indices for each life stage and shrimp year, for both brown and white shrimp.
- `brn_abundAndEnv.csv`, in `data/processed`. Contains abundance index and environmental metrics (nursery period salinity; winter days below temperature thresholds) for brown shrimp only.
- `wht_abundAndEnv.csv`, in `data/processed`. Contains abundance index and environmental metrics (nursery period salinity; winter days below temperature thresholds) for white shrimp only.

Other notes about data:

- none

Part V

Relationships between datasets

15 Abundance Index Relationships

This chapter will look at time series of abundance indices and the relationships between abundance index of different life stages.

Relationships will only be examined in years where we have data from all life stages (using the variable `abundanceIndex_commonTimePeriod`).

Read in data; trim to only common years; put life stages in proper order.

Set up colors and shapes

15.1 Brown Shrimp

15.1.1 Time Series

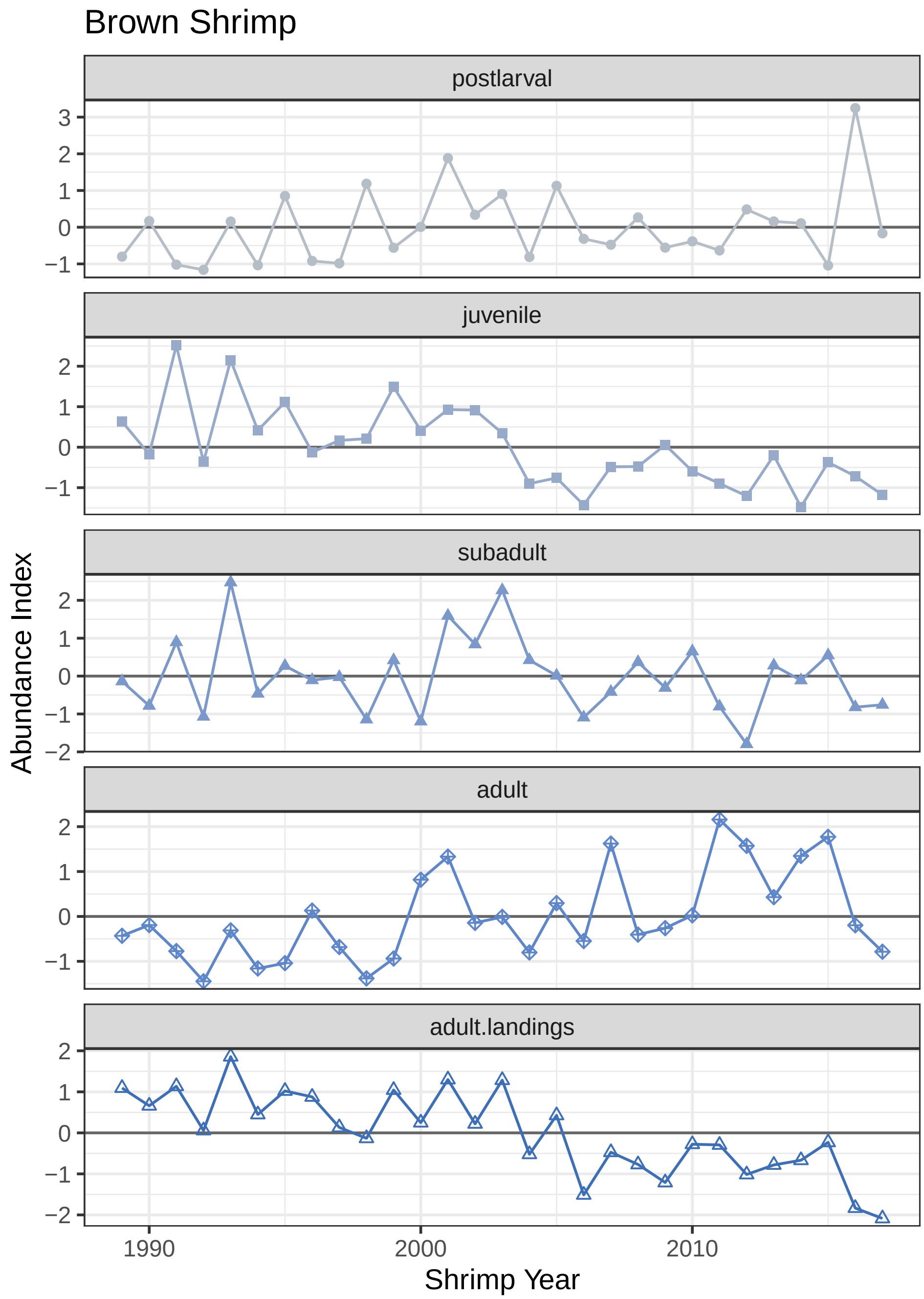


Figure 15.1: Time series of brown shrimp abundance index by life stage.

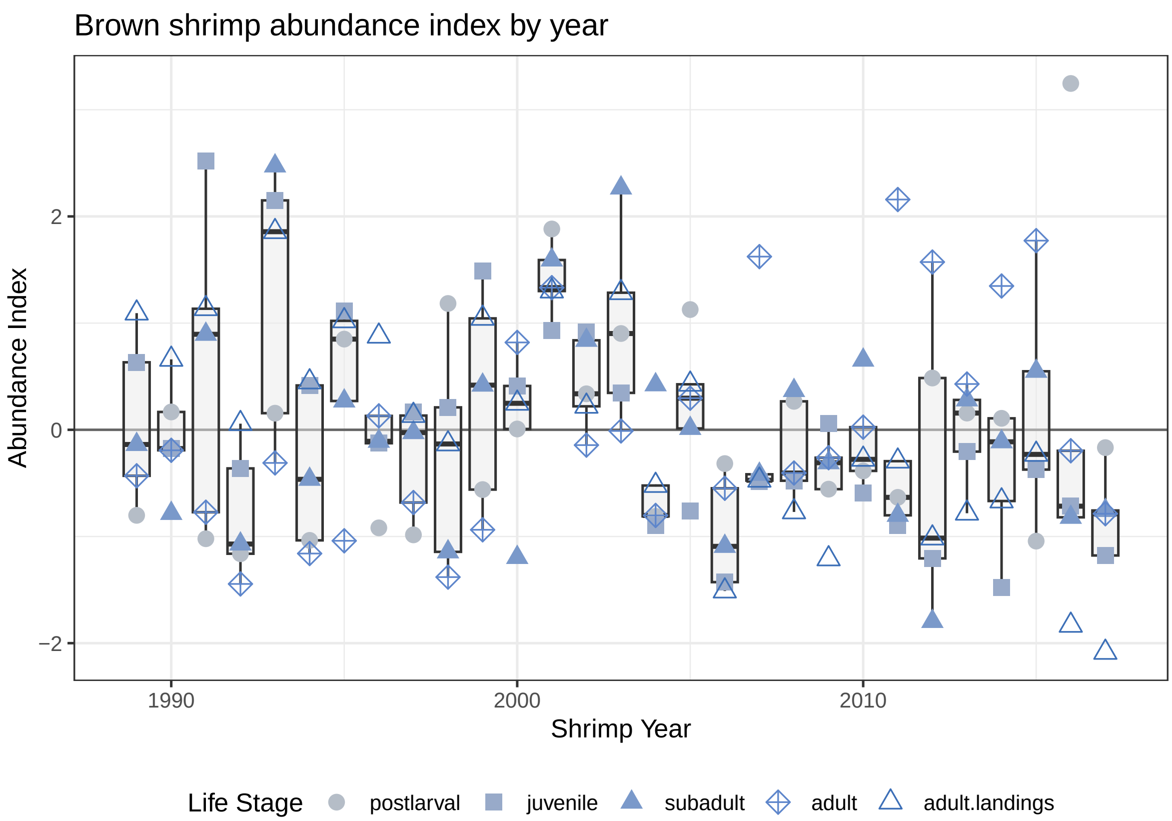


Figure 15.2: Time series of abundance index for brown shrimp, focused on the variability between life stages within each year.

15.1.2 PCA

Table 15.1: Variance explained by PC axes

	PC1	PC2	PC3	PC4	PC5
Eigenvalue	2.3613	1.1139	0.9166	0.4141	0.1941
Proportion Explained	0.4723	0.2228	0.1833	0.0828	0.0388
Cumulative Proportion	0.4723	0.6950	0.8784	0.9612	1.0000

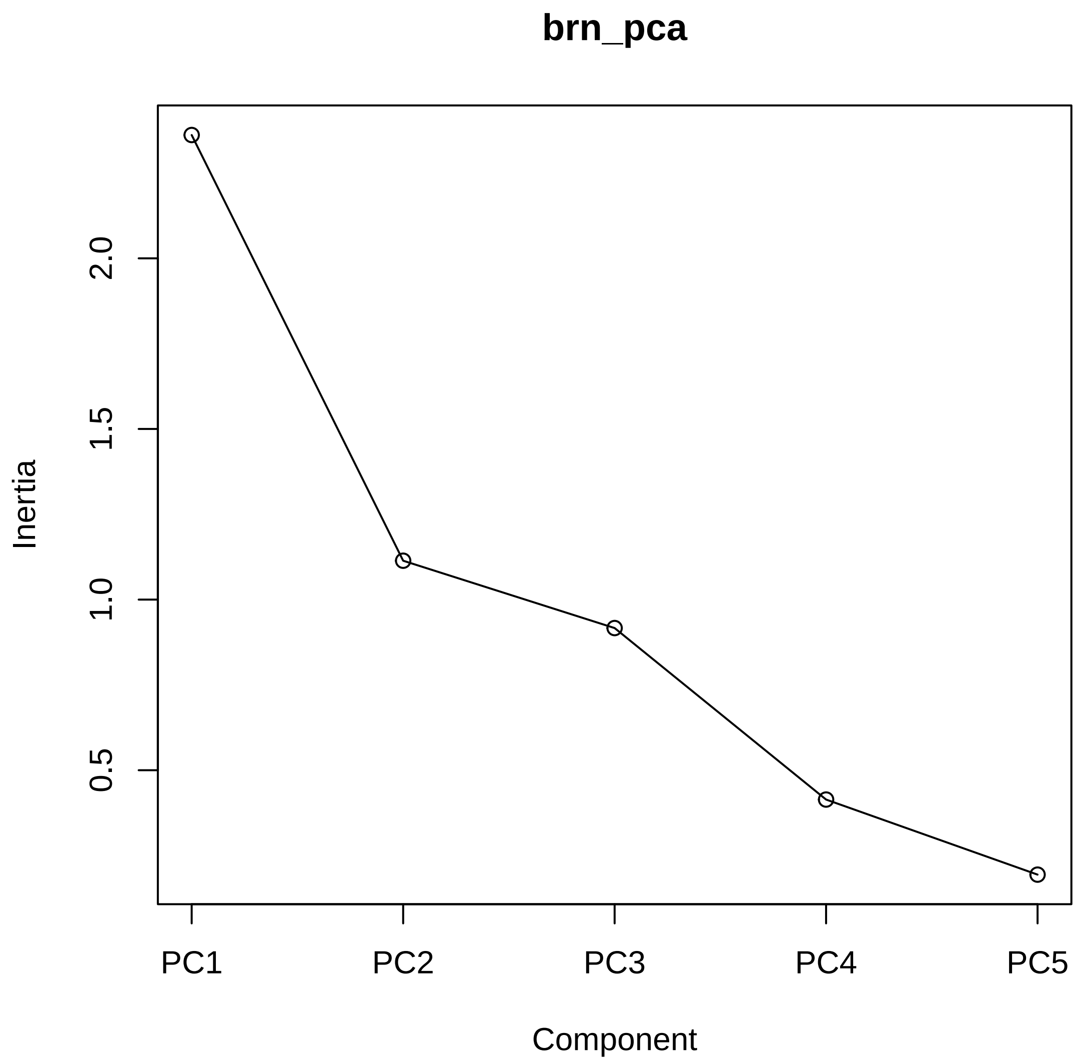


Figure 15.3: Screeplot for brown shrimp PCA.

Table 15.2: unscaled loadings onto PC axes

	PC1	PC2	PC3
postlarval	0.0545	-0.7008	0.6873
juvenile	-0.5970	0.0586	0.0906
subadult	-0.5084	-0.3326	-0.1034
adult	0.2051	-0.6259	-0.6899

Table 15.2: unscaled loadings onto PC axes

	PC1	PC2	PC3
adult.landings	-0.5832	-0.0557	-0.1810

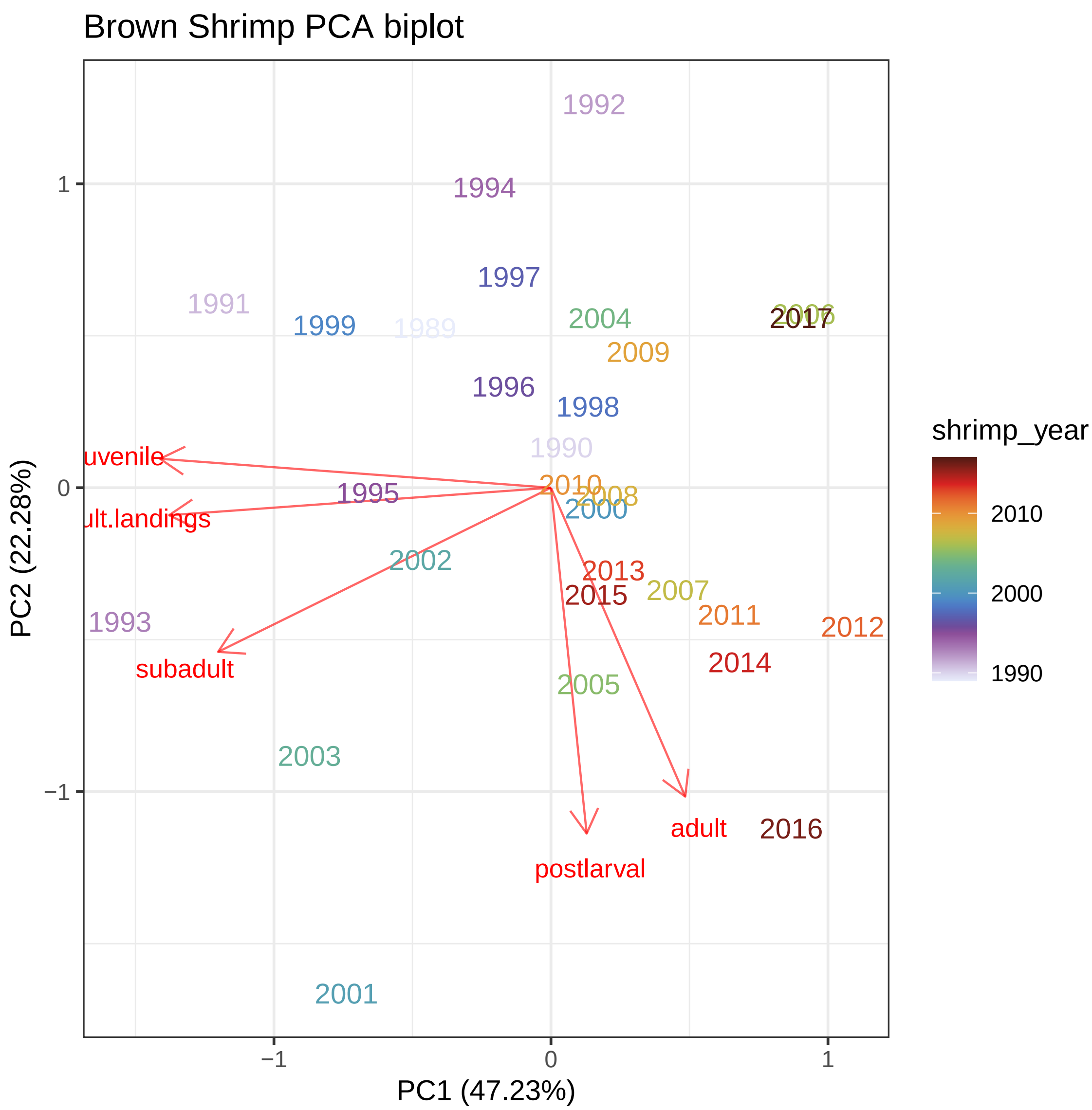


Figure 15.4: Biplot from brown shrimp PCA. Each point represents a year. Columns were life stages, rows were years, and cell values were abundance index.

15.1.3 Correlations between life stages

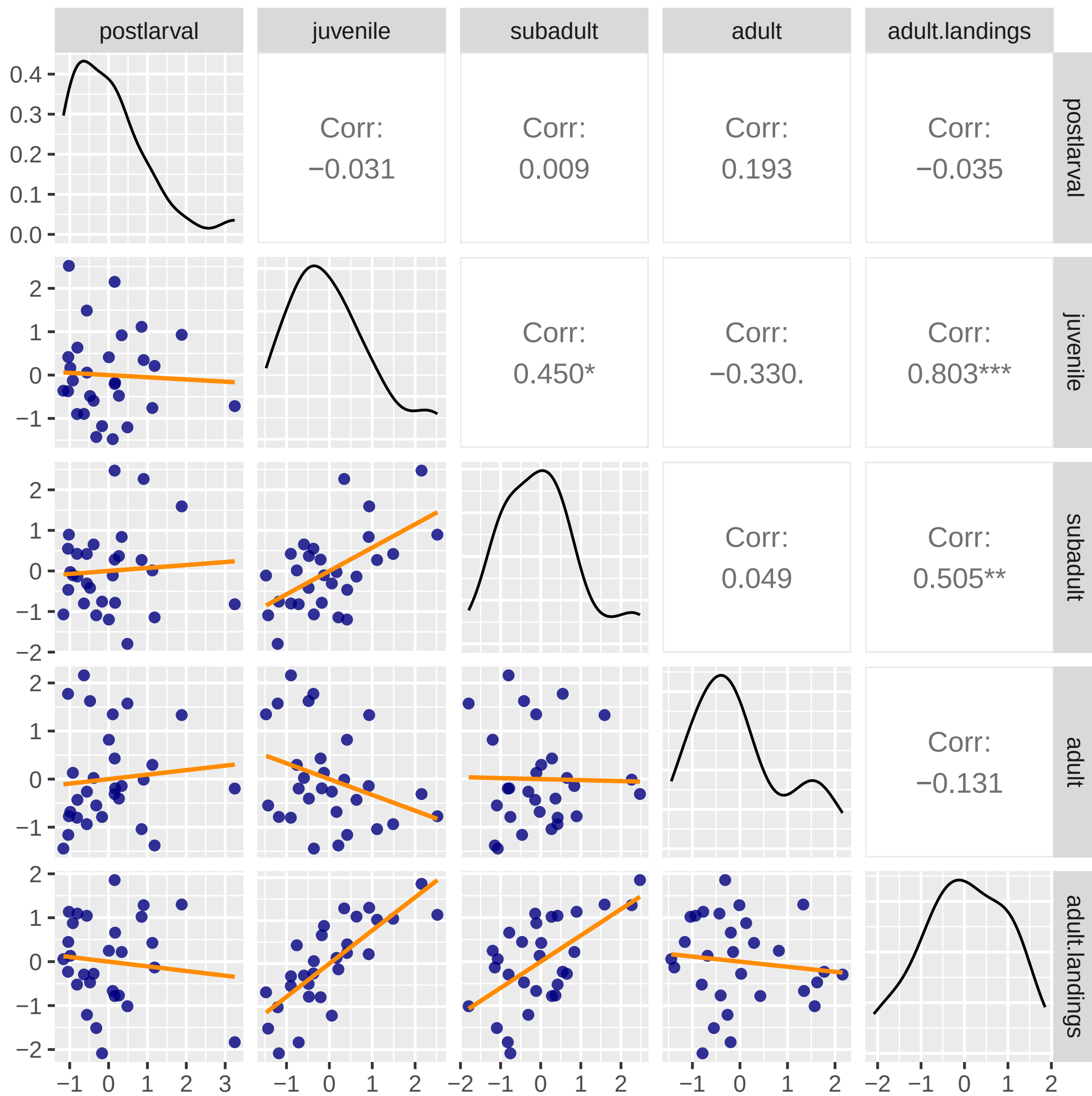


Figure 15.5: Pairs plot of brown shrimp life stages, with Spearman correlation coefficients in the upper triangle.

Juveniles positively correlated with subadults. Adult landings positively correlated with juveniles and subadults.

15.1.4 Cross-correlations (lags)

The lag k value returned by ccf(x, y) estimates the correlation between x[t+k] and y[t].

These are Pearson correlation coefficients, so lag-0 correlations will be a little different than the Spearman coefficients in the pairs plots.

Juveniles vs. subadults

```
Autocorrelations of series 'X', by lag

   -11    -10     -9     -8     -7     -6     -5     -4     -3     -2     -1
 0.014  0.290 -0.070  0.318 -0.140  0.221 -0.052  0.223 -0.020  0.562 -0.009
     0      1      2      3      4      5      6      7      8      9     10
 0.575 -0.197  0.086 -0.274 -0.052 -0.025  0.217 -0.077  0.132 -0.136 -0.015
    11
-0.217
```

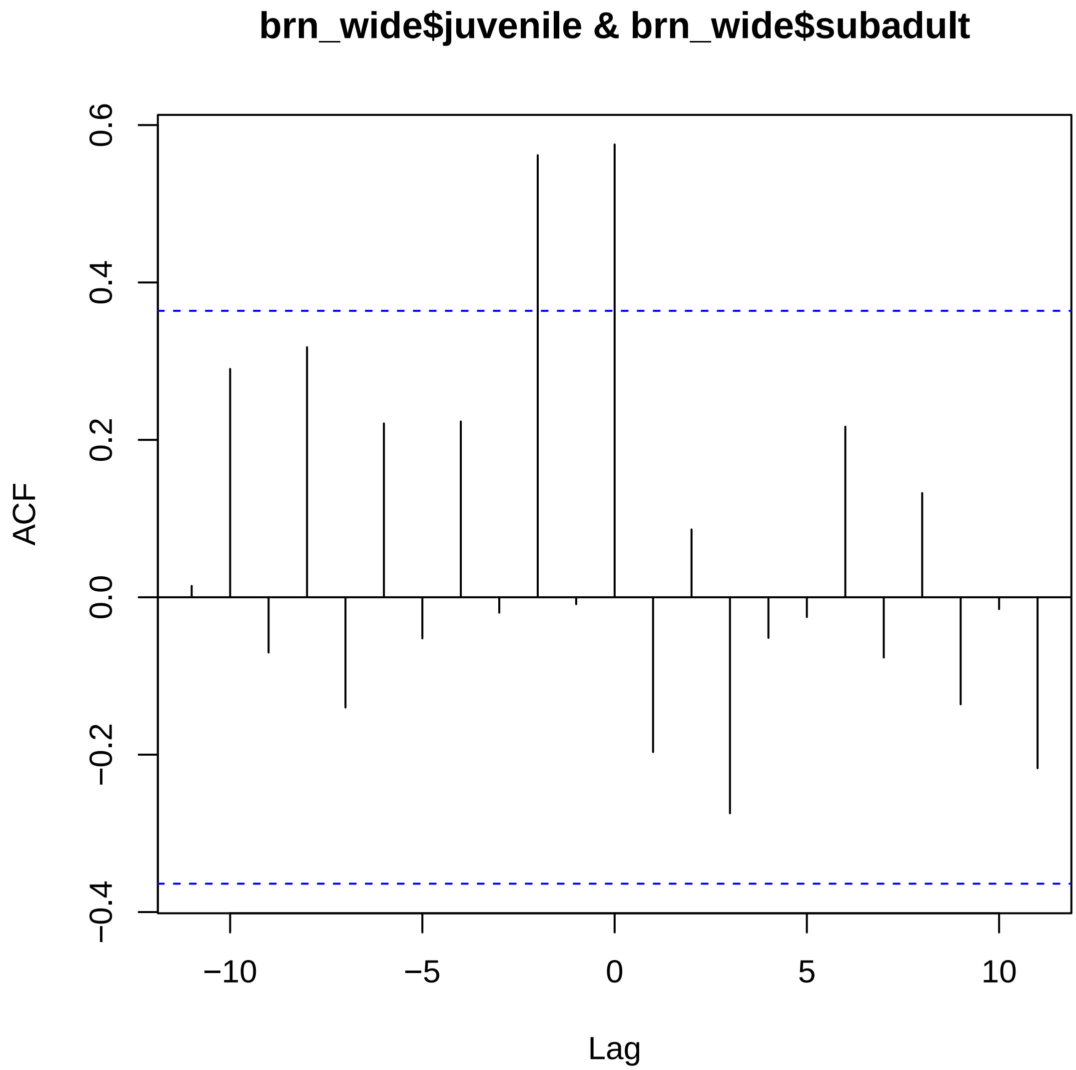


Figure 15.6: Cross-correlation between juvenile and adult stages of brown shrimp

The highest correlation is at lag 0: the time series are most correlated without lagging either of them. Lag -2 is almost as big, but doesn't seem to make much sense - juveniles 2 years ago correlated with this year's subadults?

Postlarvae vs. adults

Can we see if adults are correlated with the following year's postlarvae?

```
Autocorrelations of series 'X', by lag

   -11    -10     -9     -8     -7     -6     -5     -4     -3     -2     -1
 0.012  0.203  0.106 -0.024  0.060  0.238 -0.050 -0.057  0.021  0.205 -0.090
     0      1      2      3      4      5      6      7      8      9     10
 0.093  0.304  0.277 -0.115 -0.001  0.389 -0.154 -0.082 -0.183  0.081 -0.195
    11
 0.008
```

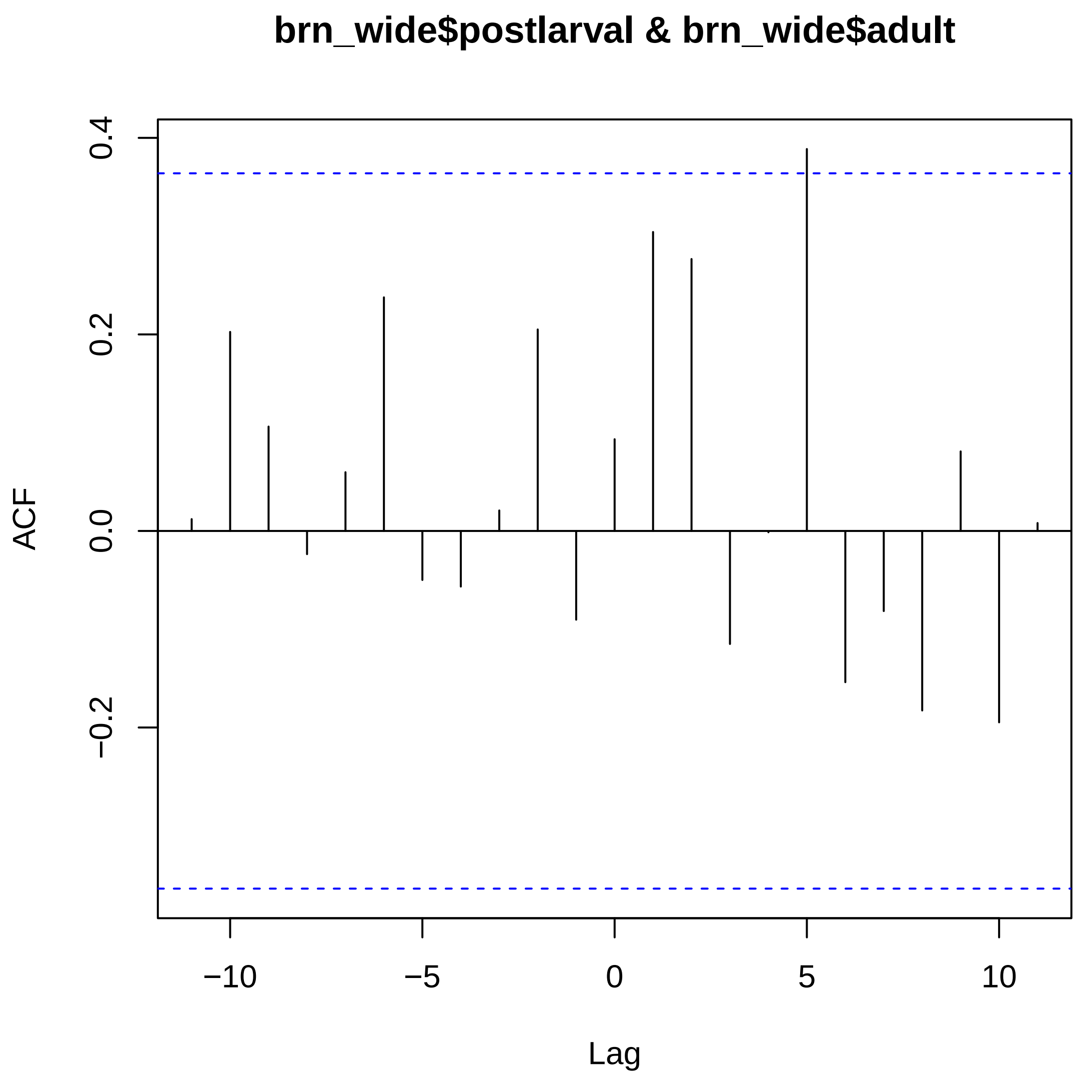


Figure 15.7: Cross-correlation between postlarval and adult stages of brown shrimp

Nothing shows up except something at 5 years, which could just be noise.

15.2 White Shrimp

15.2.1 Time Series

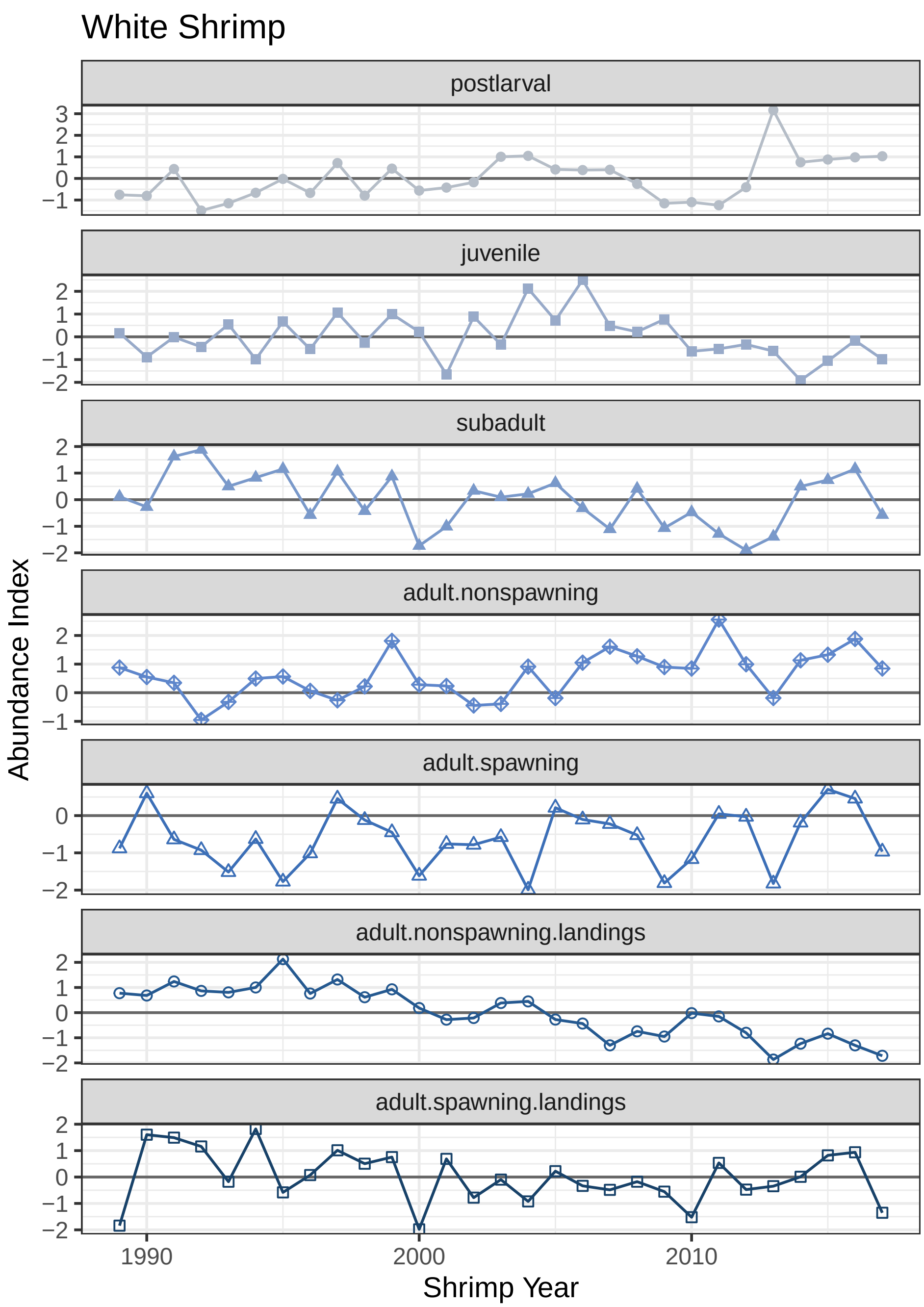


Figure 15.8: Time series of white shrimp abundance index by life stage.

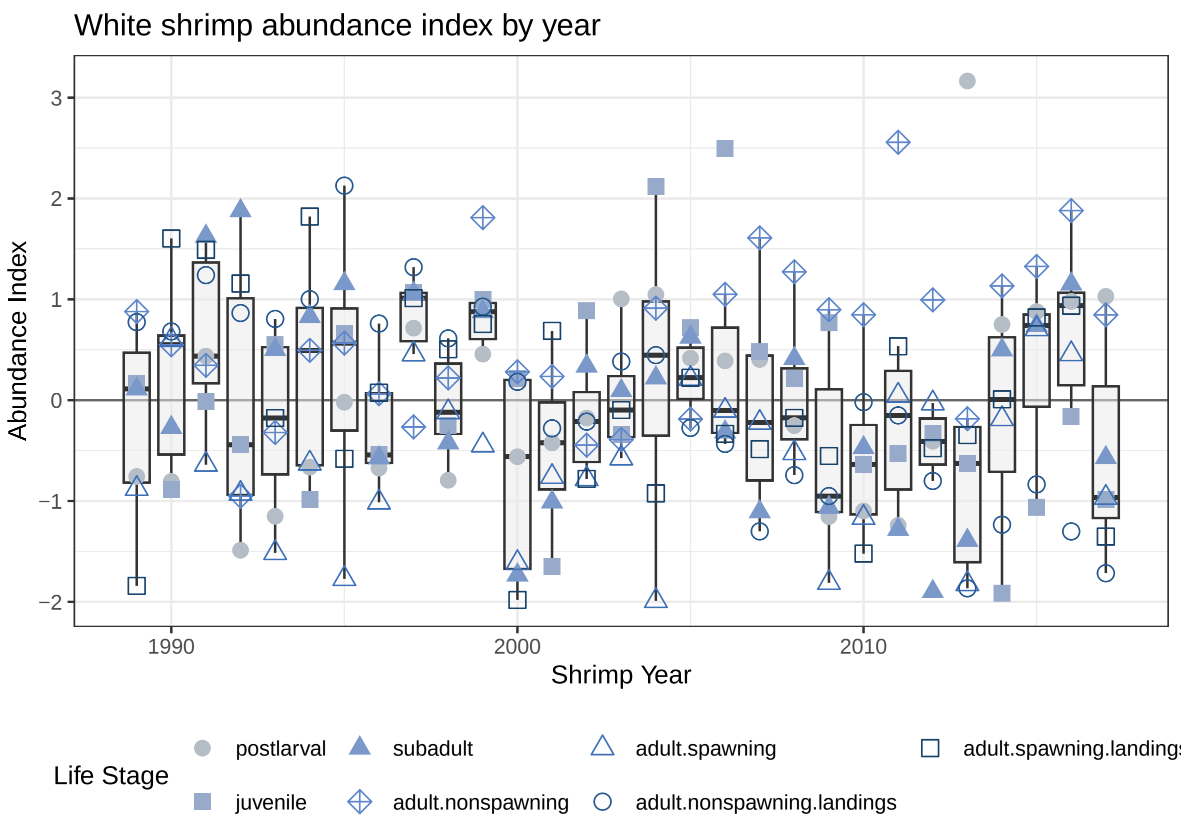


Figure 15.9: Time series of abundance index for white shrimp, focused on the variability between life stages within each year.

15.2.2 PCA

Table 15.3: Variance explained by PC axes

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Eigenvalue	1.9731	1.4765	1.1318	0.7690	0.3830	0.2897	0.2127
Proportion Explained	0.3164	0.2368	0.1815	0.1233	0.0614	0.0465	0.0341
Cumulative Proportion	0.3164	0.5532	0.7347	0.8580	0.9194	0.9659	1.0000

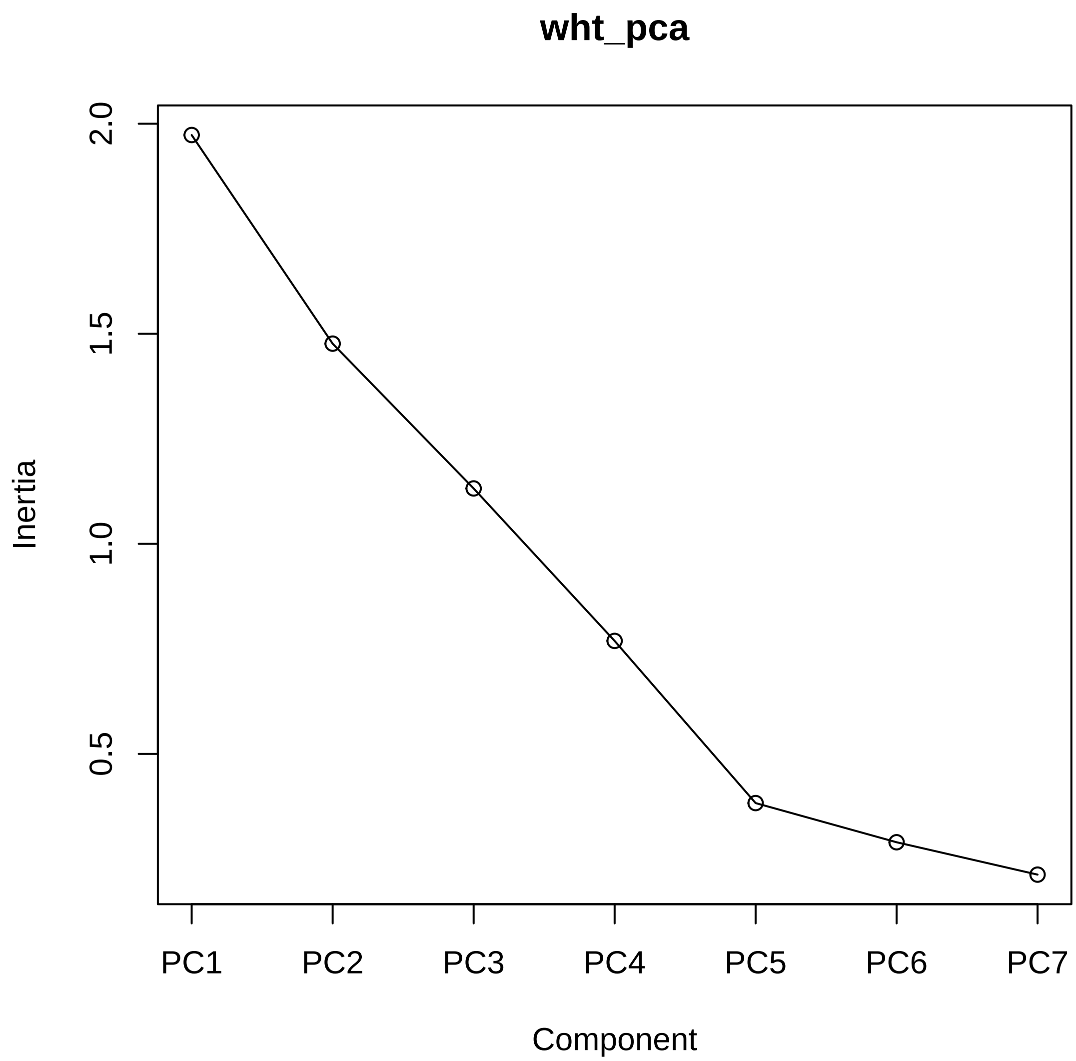


Figure 15.10: Screeplot for white shrimp PCA.

Table 15.4: unscaled loadings onto PC axes

	PC1	PC2	PC3
postlarval	0.2264	-0.2786	-0.7669
juvenile	-0.0509	0.5427	-0.5116
subadult	-0.5736	-0.0614	-0.3670
adult.nonspawning	0.1827	-0.1965	0.0279

Table 15.4: unscaled loadings onto PC axes

	PC1	PC2	PC3
adult.spawning	-0.1388	-0.4319	0.0016
adult.nonspawning.landings	-0.5643	0.3847	0.1216
adult.spawning.landings	-0.4961	-0.5009	-0.0018

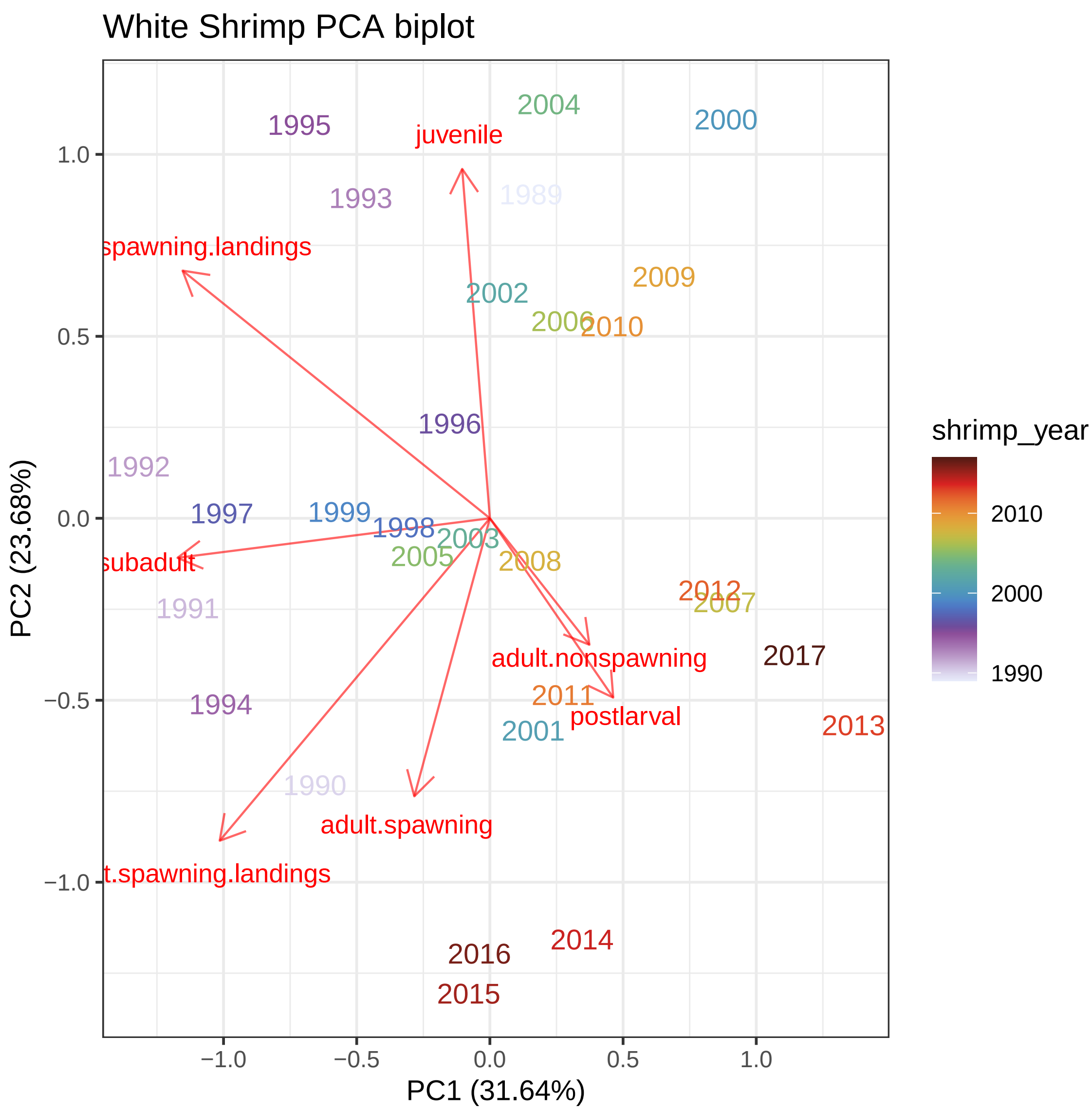


Figure 15.11: Biplot from white shrimp PCA. Each point represents a year. Columns were life stages, rows were years, and cell values were abundance index.

15.2.3 Correlations between life stages

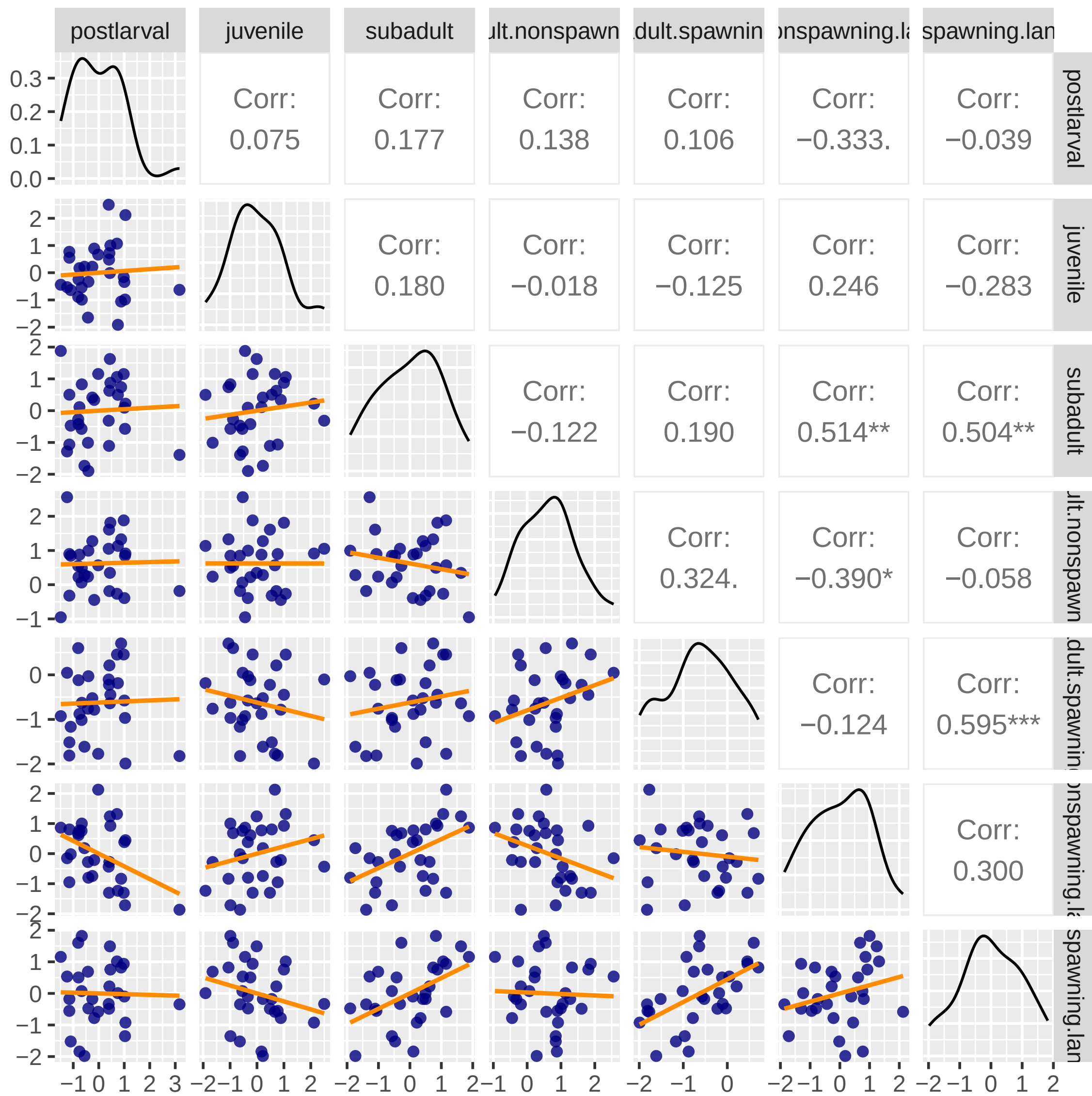


Figure 15.12: Pairs plot of white shrimp life stages, with Spearman correlation coefficients in the upper triangle.

Subadults positively correlated with landings (both spawning and non-spawning); interesting. Adult spawning from SEAMAP is positively correlated with adult spawning landings. Small negative correlation between non-spawning adults in SEAMAP vs. landings, though the graph mostly looks like scatter.

15.2.4 Cross-correlations (lags)

The lag k value returned by ccf(x, y) estimates the correlation between x[t+k] and y[t].

These are Pearson correlation coefficients, so lag-0 correlations will be a little different than the Spearman coefficients in the pairs plots.

Juveniles vs. subadults

```
Autocorrelations of series 'X', by lag

   -11    -10     -9     -8     -7     -6     -5     -4     -3     -2     -1
 0.111  0.165 -0.014 -0.062 -0.323 -0.061 -0.375 -0.134 -0.345 -0.061 -0.328
     0      1      2      3      4      5      6      7      8      9     10
 0.129  0.202  0.179  0.185  0.057  0.241 -0.126  0.234  0.054 -0.012  0.094
    11
 0.205
```

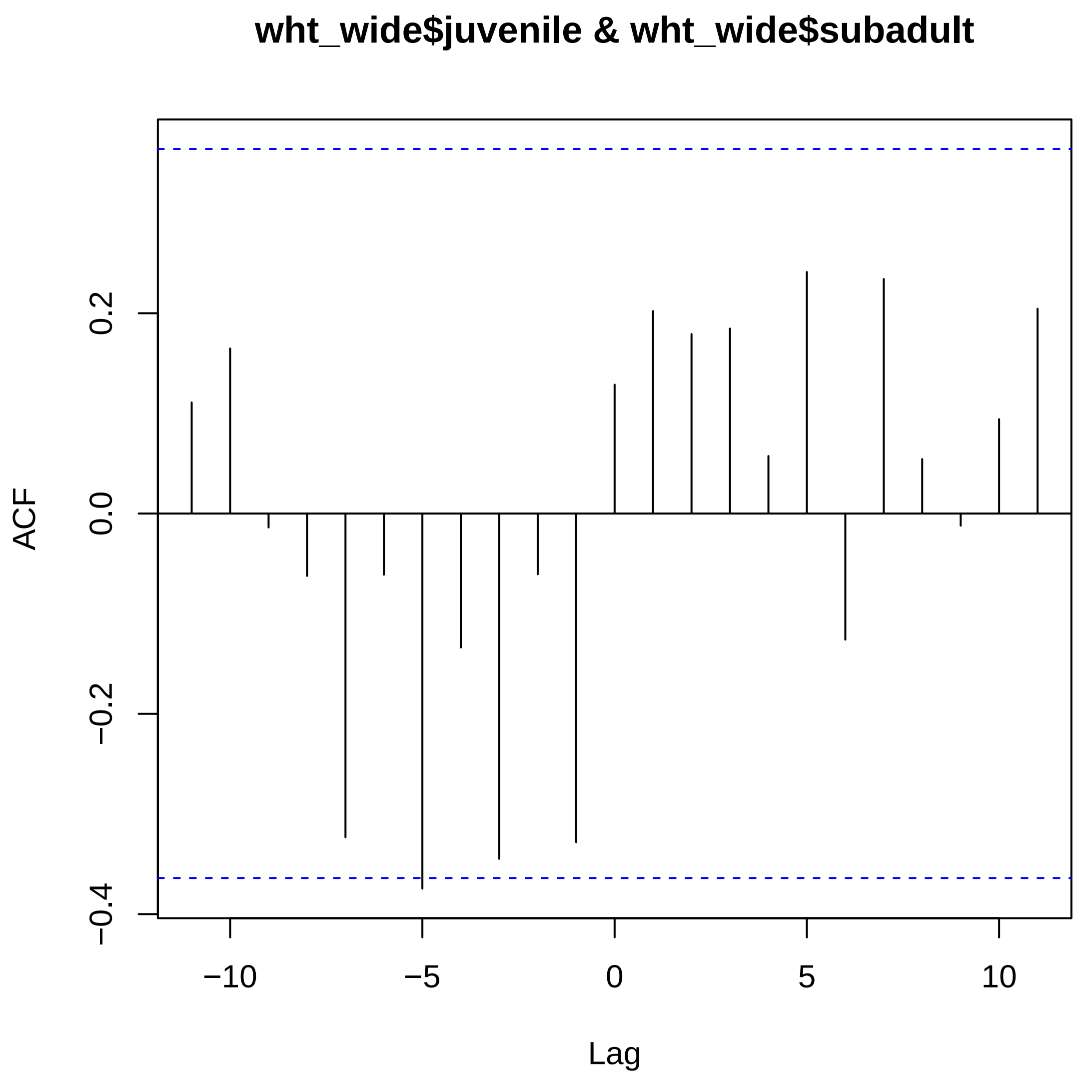


Figure 15.13: Cross-correlation between juvenile and adult stages of white shrimp

Nope, nothing to see.

Postlarvae vs. spawning adults

Can we see if adults are correlated with the following year's postlarvae?

```
Autocorrelations of series 'X', by lag

   -11    -10     -9     -8     -7     -6     -5     -4     -3     -2     -1 
 0.184  0.046  0.014  0.188 -0.091 -0.076 -0.310 -0.099  0.225  0.376  0.143 
     0      1      2      3      4      5      6      7      8      9     10 
 0.031  0.326  0.089 -0.212 -0.405  0.068  0.069  0.189  0.087 -0.047 -0.004 
    11 
-0.107 
```

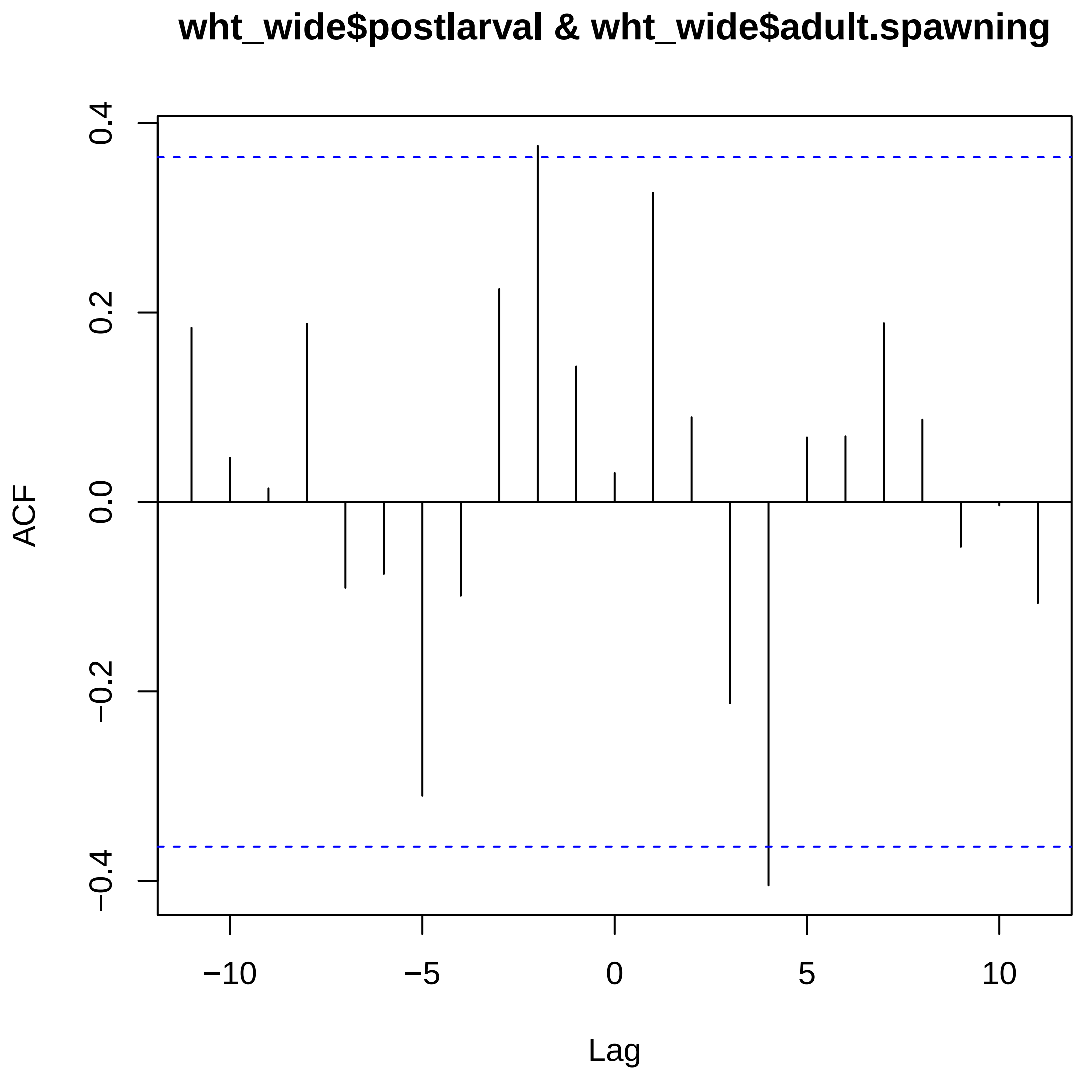


Figure 15.14: Cross-correlation between postlarval and adult (spawning) stages of white shrimp

Again, not much to see.

15.3 About

Input `.qmd` file for this section was: **rel_abundances.qmd** and can be found in the main directory.

Input data file(s) - in `data/processed`:

- `brn_abundAndEnv.csv`
- `wht_abundAndEnv.csv`

16 Temperature thresholds

This chapter will look at temperature thresholds and relationships to abundance indices.

Relationships will only be examined in years where we have data from all life stages (using the variable `abundanceIndex_commonTimePeriod`).

Read in data; trim to only common years; put life stages in proper order.

16.1 Brown Shrimp

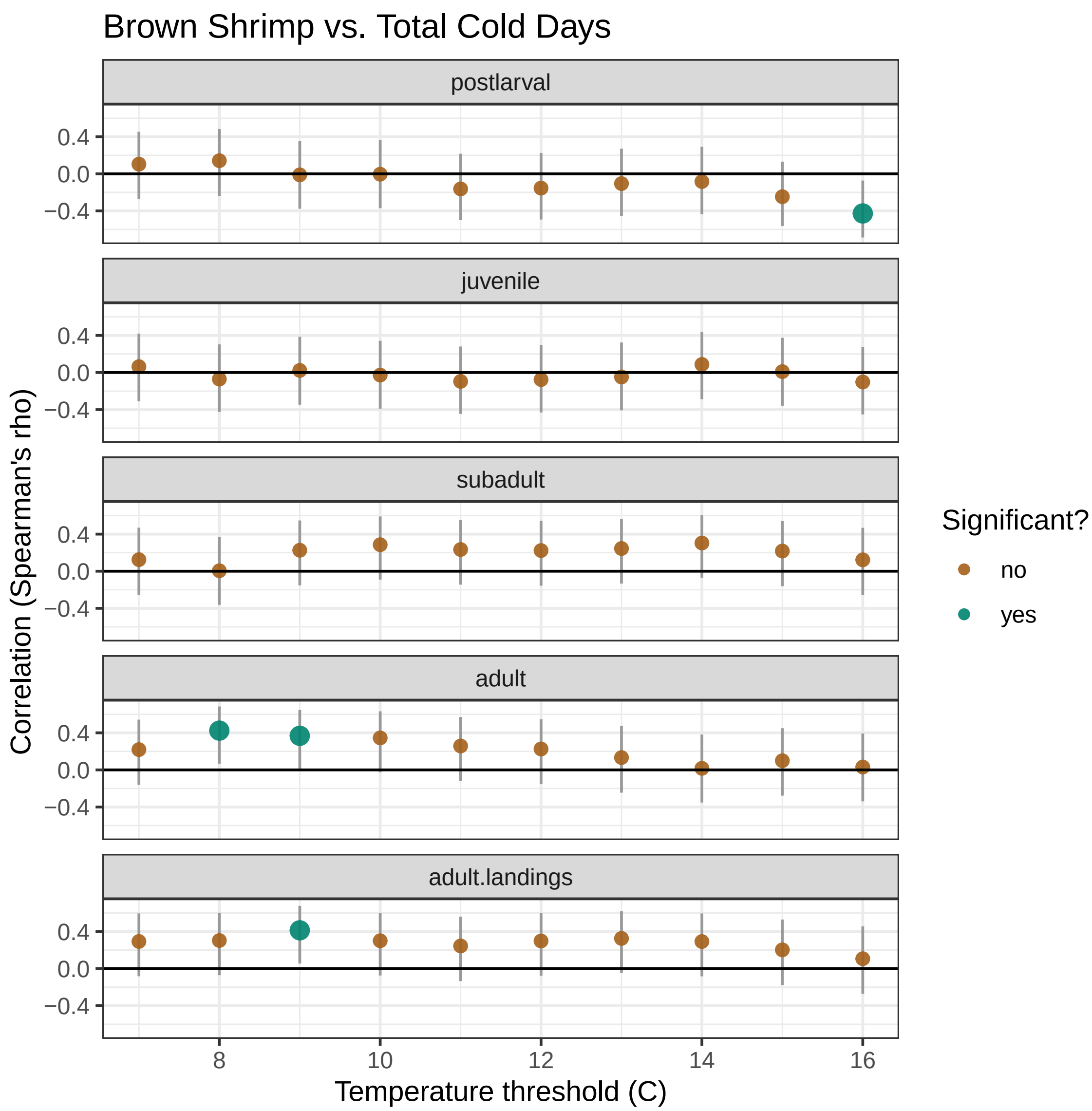


Figure 16.1: Estimated Spearman correlations between each life stage and the total number of winter days below a selected temperature threshold for brown shrimp. Points represent the estimated Spearman correlation coefficient, and vertical gray lines represnt the 95% confidence interval for the coefficient. Points are colored brown if the correlation is not significantly different from 0 ($p > 0.05$) and teal if the correlation is significant ($p < 0.05$). Significant points are also larger. These p-values are unadjusted, so expect around 1 'false' significant result for every 20 tests at this level.

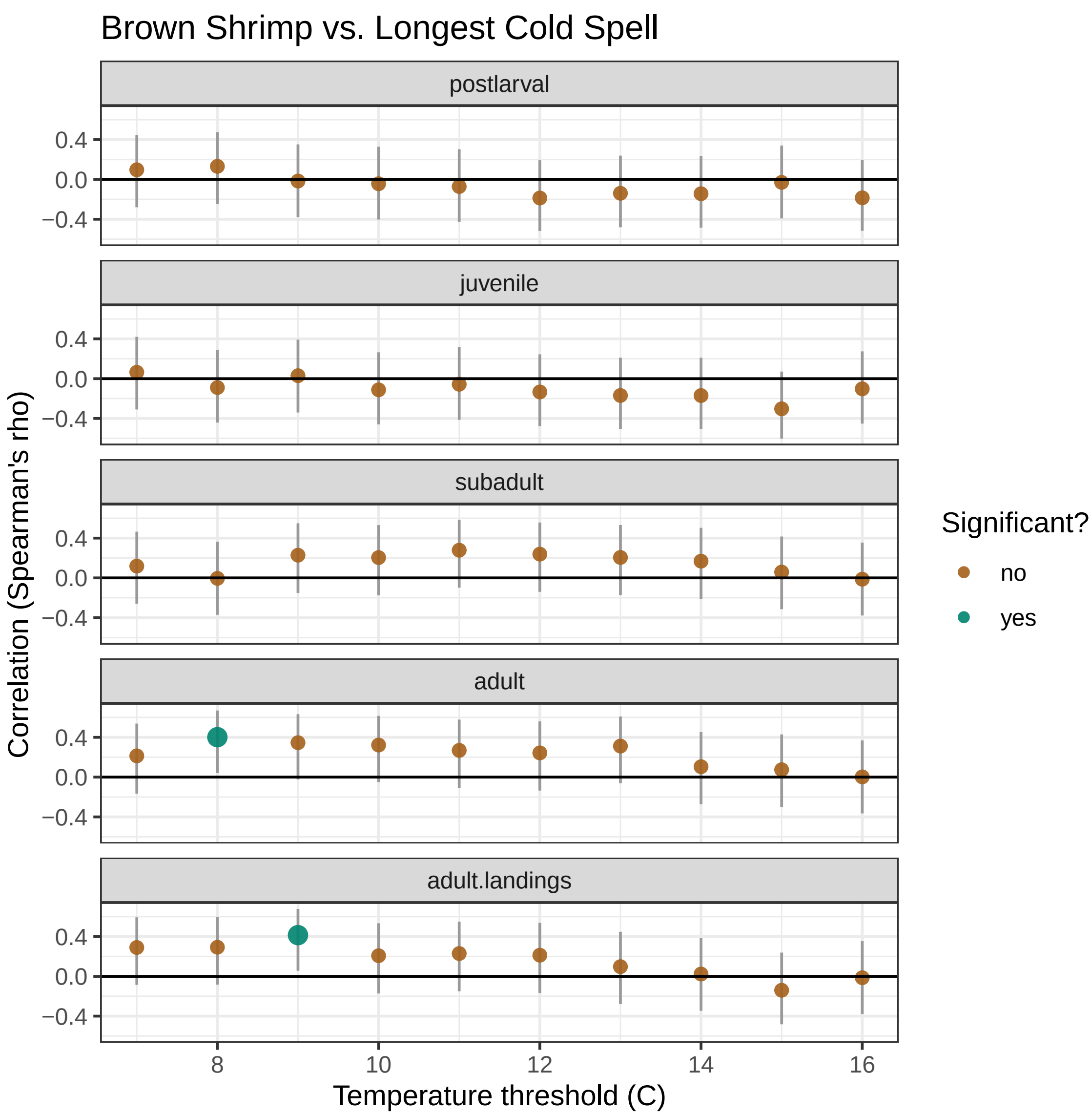
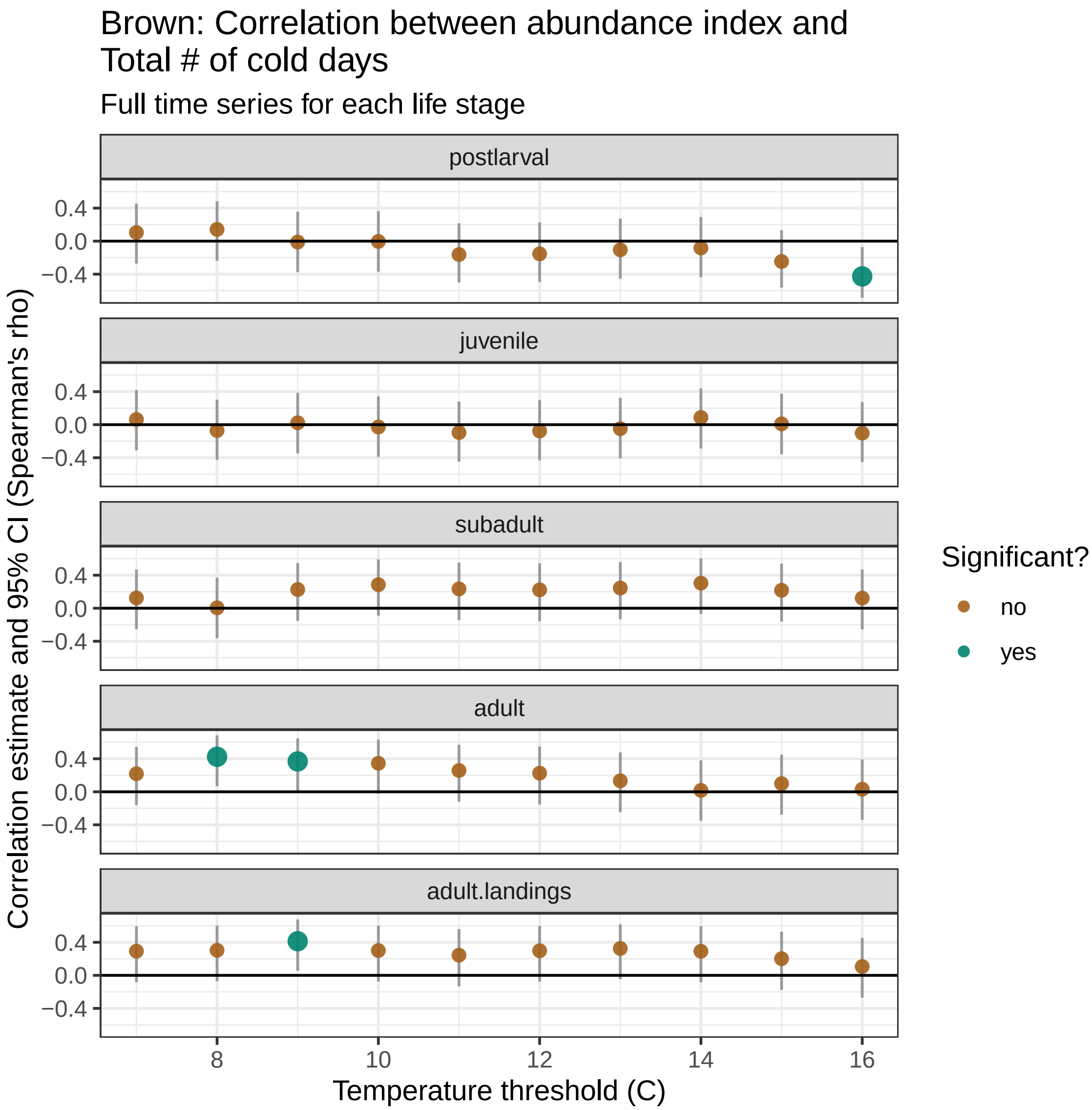


Figure 16.2: Estimated Spearman correlations between each life stage and the largest number of consecutive winter days below a selected temperature threshold for brown shrimp. Points represent the estimated Spearman correlation coefficient, and vertical gray lines represnt the 95% confidence interval for the coefficient. Points are colored brown if the correlation is not significantly different from 0 ($p > 0.05$) and teal if the correlation is significant ($p < 0.05$). These p-values are unadjusted, so expect around 1 'false' significant result for every 20 tests at this level.



16.2 White Shrimp

For white shrimp, need to break up the columns reasonably to correlate postlarval and juvenile with one winter, and adults with the next.

16.3 About

Input `.qmd` file for this section was: **rel_temp_thresholds.qmd** and can be found in the main directory.

Input data file(s) - in `data/processed`:

- `brn_abundAndEnv.csv`
- `wht_abundAndEnv.csv`

17 Nursery Period Salinity

stuff

18 Regression Models

Start to explore what happens when we do more than just correlations.